

# **Smooth Structures on PL-manifolds of Dimensions between 8 and 10**

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by

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*Dedicated to my beloved parents.*



## Thesis Approval

This thesis entitled **Smooth Structures on PL-manifolds of Dimensions between 8 and 10** by **Priyanka Tanaji Magar** is approved for the degree of **Doctor of Philosophy**.

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# Abstract

In this thesis, various results related to classification of smooth structures on manifolds are discussed. Primarily, the concordance classes of smooth structures on  $PL$ -manifolds with dimensions between 8 and 10 are presented in terms of cohomologies and Steenrod operations, which also leads to the computation of homotopy inertia groups. Additionally, it is demonstrated that the concordance inertia group of a connected sum of manifolds is the sum of the individual manifold's concordance inertia group. Furthermore, explicit values of the concordance structure sets are provided for  $k$ -fold connected sum of projective spaces,  $\#^k \mathbb{P}^n$  for  $\mathbb{P} = \mathbb{C}, \mathbb{H}, \mathbb{O}$  in real dimensions between 8 to 16. Moreover, the smooth normal invariant set of  $\#^k \mathbb{C}P^n$  is obtained as a direct sum of two groups – the reduced 0<sup>th</sup> stable cohomotopy group of  $\#^k \mathbb{C}P^n$  which constitutes the torsion part of this group, and the reduced  $KO$ -groups of  $\#^k \mathbb{C}P^n$  for  $k, n \in \mathbb{N}$  and  $k \geq 2$  which constitutes the free part of this group.





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# List of Abbreviations

|             |                                               |
|-------------|-----------------------------------------------|
| <i>Top</i>  | Category of closed topological manifolds      |
| <i>PL</i>   | Category of closed piecewise-linear manifolds |
| <i>Diff</i> | Category of closed smooth manifolds           |

# List of Symbols

|                  |                                                                      |
|------------------|----------------------------------------------------------------------|
| $\emptyset$      | Empty set                                                            |
| $\mathbb{N}$     | Set of natural numbers                                               |
| $\mathbb{Z}$     | Set of integers                                                      |
| $\mathbb{C}$     | Set of complex numbers                                               |
| $\mathbb{H}$     | Set of quaternion numbers                                            |
| $\mathbb{O}$     | Set of Cayley numbers                                                |
| $\mathbb{R}^n$   | $n$ -dimensional Euclidean space                                     |
| $I$              | Closed interval $[0, 1]$                                             |
| $\mathbb{D}^n$   | Closed unit disk in $\mathbb{R}^n$                                   |
| $\mathbb{CP}^n$  | Complex projective $n$ -space                                        |
| $\mathbb{HP}^n$  | Quaternionic projective $n$ -space                                   |
| $\mathbb{OP}^n$  | Octonionic projective $n$ -space                                     |
| $\Delta^n$       | $n$ -simplex                                                         |
| $\mathbb{S}^n$   | $n$ -dimensional sphere                                              |
| $\Sigma^n$       | Homotopy $n$ -sphere                                                 |
| $\Theta_n$       | Group of homotopy $n$ -spheres                                       |
| $[\Sigma^n]$     | Element of $\Theta_n$                                                |
| $[A, B]$         | Set of all homotopy classes of maps from $A$ to $B$                  |
| $\pi_n(X)$       | $n^{\text{th}}$ homotopy group of $X$                                |
| $[f]$            | Set of all maps homotopic to the map $f : A \rightarrow B$           |
| $\mathcal{H}(X)$ | $\{[f] \mid f : X \rightarrow X \text{ is homeomorphism}\}$          |
| $\mathcal{E}(X)$ | $\{[f] \mid f : X \rightarrow X \text{ is a homotopy equivalence}\}$ |
| $\Omega^n(X)$    | $n$ -fold loop space of $X$                                          |

|              |                          |
|--------------|--------------------------|
| $\Sigma^n X$ | n-fold suspension of $X$ |
| $K(G, n)$    | Eilenberg–MacLane space  |



# Chapter 1

## Introduction

### 1.1 Background

In 1904, Henri Poincaré conjectured that any simply-connected, closed 3-dimensional manifold is homeomorphic to  $\mathbb{S}^3$ . This conjecture remained unresolved for a significant duration and was later recognized as one of the *millennium prize problems*. In the meantime, the conjecture was extended to all dimensions:

**Conjecture 1.1.1** (Generalized Poincaré Conjecture). *Any simply connected closed  $n$ -manifold homotopy equivalent to the standard  $n$ -sphere  $\mathbb{S}^n$  is homeomorphic to  $\mathbb{S}^n$ .*

The conjecture for  $n \geq 5$  was proven by Stephen Smale for smooth manifolds in 1962, and later for topological manifolds by Newman [44] and Connell [9], by Michael Freedman for  $n = 4$  in 1982, and by Grigori Perelman for  $n = 3$  in 2003.

In the early twentieth century, the concepts of piecewise linear (*PL*) manifolds and smooth manifolds were well-defined. Subsequently, the Poincaré conjecture was formulated more generally in *PL* and smooth categories of manifolds, denoted by *Cat=PL* and *Diff*, respectively.

In 1956, John Milnor disproved the smooth version of the Poincaré conjecture for  $n = 7$  by showing the existence of 7-dimensional manifolds that are homeomorphic to the standard 7-sphere but exhibit non-standard smooth structures. In fact, he proved the existence of at least 7 distinct smooth structures up to diffeomorphism on the 7-sphere. Soon after, in 1960, Michel Kervaire constructed a 10-dimensional *PL*-manifold which does not admit any smooth structure on it. In 1940, J.H.C. Whitehead [55] proved a fundamental result: any smooth manifold can be equipped with a *PL*-structure, which is unique up to *PL*-homeomorphism. Aim of the smooth theory is to explore the converse of this result, primarily focusing on the classification of smooth manifolds based on their underlying *PL*-manifold structure.

These results from the mid-twentieth century encourage interesting questions in the field of classification of *PL* or smooth structures on manifolds, including whether a given topological manifold admits a *PL* or smooth structure. If so, the subsequent challenge lies in determining the number of distinct *PL* or smooth structures on the given topological or *PL*-manifold. Since then, the study of classifying *PL* and smooth structures has remained an active area of research for topologists. This has led to the development of homotopy theory, bundle theory, cobordism theory, surgery theory, obstruction theory, and more, opening new vistas in the classification of *PL* or smooth structures of manifolds in higher dimensions.

The classification of *PL* or smooth structures on the underlying topological manifolds of dimension at most 3 was completed in the early twentieth century. However, in dimension 4, this problem is still considered to be particularly challenging, even for the case of the standard sphere, and so for other 4-manifolds. Using aforementioned theories, it has been proven that for a given underlying topological manifold  $M$ , the existence and number of distinct *PL* structures on  $M$  depend only on the third and fourth cohomologies of  $M$  with  $\mathbb{Z}_2$ -coefficients. Furthermore, it is also known that for a given underlying *PL*-manifold  $M$  of dimension 5 or 6, there exists a unique smooth structure up to diffeomorphism. Additionally, extensive findings are available for 7-dimensional manifolds.

In this thesis, we discuss the classification of smooth structures on the underlying *PL*-manifold of dimensions 8, 9 and 10.

## 1.2 Literature review

In geometric topology, one primarily addresses the following question.

**Question 1.2.1.** *Given an underlying topological (resp. PL) manifold, does it admit a PL (resp. smooth) structure? If yes, then how many distinct PL (resp. smooth) structures are there up to PL-homeomorphism (resp. diffeomorphism)?*

The above question was resolved in the early twentieth century for lower dimensions. Every topological manifold of dimension at most 3 admits a unique PL (resp. smooth) structure up to a PL-homeomorphism (resp. diffeomorphism) [40, 43, 56]. In case of Euclidean spaces, Stallings in 1962 proved that up to isomorphism there is a unique PL (resp. smooth) structure on  $\mathbb{R}^n$  with  $n \geq 5$  [50]. In case of  $\mathbb{R}^4$ , surprisingly, there exist uncountably many non-isomorphic PL or smooth structures [14, 15].

As discussed in Section 1.1, the topological version of the Poincaré conjecture is true in all dimensions. For PL manifolds, the generalized Poincaré conjecture holds true for all dimensions except for  $n = 4$ . In 1962, Smale proved that a PL homotopy  $n$ -sphere (a closed PL-manifold homotopy equivalent to  $n$ -sphere) is PL-homeomorphic to the standard  $n$ -sphere for all  $n \geq 5$ . In 2003 Grigori Perelman proved the conjecture for the case  $n = 3$ , where the topological, PL, and smooth cases all coincide. For the case  $n = 4$ , the conjecture remains open and is equivalent to the smooth version. The smooth Poincaré conjecture was first disproved for  $n = 7$  by John Milnor [38]. Later, he along with Michel Kervaire, showed that the smooth Poincaré conjecture is false for infinitely many dimensions. Using cobordism theory and surgery theory they found the number of distinct smooth structures on  $\mathbb{S}^n$  for  $5 \leq n \leq 18$  [29]. It is conjectured that the smooth Poincaré conjecture is true for only finitely many dimensions.

As directly dealing with the PL or smooth classification problem is highly difficult, there are various approaches to this classification problem. These include using cobordism theory, surgery theory, bundle theory, and homotopy theory, among others. Below, we provide an overview of these theories.

Lev Pontryagin was the one to introduce *cobordism theory* explicitly. The cobordism relation (refer to Chapter 2) is significantly easier to study and compute than the PL-homeomorphism

or diffeomorphism relation of manifolds. The proof of topological or  $PL$  version of generalized Poincaré conjecture, given by Stephen Smale for  $n \geq 5$ , uses cobordism theory. The surprising result that there are only a finite number of distinct smooth structures on  $\mathbb{S}^n$  in all dimensions  $\geq 5$  also relies on cobordism theory [29]. Furthermore, the  $h$ -cobordism theorem is a powerful result in the theory of high-dimensional manifolds, which gives sufficient conditions to prove that two spaces are diffeomorphic.

*Surgery theory* is also one of the important and major tools used for the classification of manifolds in high dimensions. One of the early significant milestones in surgery theory was the classification of smooth structures on  $\mathbb{S}^n$ , which motivated further developments in the field. Essentially, surgery theory addresses questions such as:

- Question 1.2.2.**    1. *Let  $X$  be a Poincaré complex of dimension  $n$ . Does  $X$  have the homotopy type of a smooth manifold?*
2. *Is a homotopy equivalence  $f : M \rightarrow N$  between two smooth  $n$ -manifolds homotopic to a diffeomorphism?*

Answering these questions mainly uses surgery exact sequence which is discussed in Chapter 2 (2.2).

Additional approaches to the classification problem include bundle theory and homotopy theory. In these approaches, researchers investigate topological,  $PL$ , or smooth tangent bundles over a given topological or  $PL$  manifold. Certain results offer necessary and sufficient conditions for an underlying topological manifold to accommodate a  $PL$  or smooth structure, along with quantifying the number of distinct  $PL$  or smooth structures on it up to concordance relation [48]. Determining the concordance structure set for a given manifold is comparatively easier than computing the set of diffeomorphism classes of smooth structures on it; primarily due to its relationship with homotopy theory. This relation is detailed in Chapter 2.



### 1.3 Thesis outline

The thesis presents various results related to the classification problem of smooth structures on  $PL$ -manifolds, divided into two parts. Part I includes Chapters 3, 4, and primarily focuses on the smooth structures on 8, 9, and 10-dimensional  $PL$ -manifolds up to concordance relation and their homotopy inertia groups. In Part II, which includes chapters 5, 6, and 7, the focus shifts to the connected sum of manifolds and particularly to the connected sum of complex projective spaces, motivated by the concept of T-stabilization in geometric topology.

- In Chapter 3, we present the concordance structure set  $\mathcal{C}(M)$  for a smooth manifold  $M$  with  $8 \leq \dim(M) \leq 10$  which is explicitly determined in terms of the action of Steenrod operations on the cohomology of  $M$ . For this we utilize the structure of the 10<sup>th</sup>-Postnikov section of  $PL/O$ .
- In Chapter 4, we compute the concordance and homotopy inertia groups of smooth  $n$ -manifold for  $8 \leq n \leq 10$ , and inertia group of particular manifolds.
- In Chapter 5, we present a short exact sequence containing the group  $[\#^{i \in I} M_i, Y]$  where  $\#^{i \in I} M_i$  denotes the connected sum of manifolds and  $Y$  an  $H$ -space. In Section 5.2, it is demonstrated that the concordance inertia group of a connected sum is the sum of the individual concordance inertia groups. As an application, explicit values of the  $\mathcal{C}(\#^k \mathbb{C}P^n)$  and  $\pi_s^0(\#^k \mathbb{C}P^n)$  are presented in Sections 5.3 and 5.4 for  $4 \leq n \leq 8$ .
- In Chapter 6, the reduced complex  $K$ -groups  $\widetilde{K}^i(\#^k \mathbb{C}P^n)$  for  $i = -1, 0$ , and the reduced real  $K$ -groups  $\widetilde{KO}^i(\#^k \mathbb{C}P^n)$  for  $-7 \leq i \leq 0$  are determined along with their generators for  $k, n \in \mathbb{N}$  with  $k \geq 2$ . (see Theorems 6.1.1, 6.1.2 and 6.2.2)
- In Chapter 7, we express the smooth normal invariant set  $\mathcal{N}(\#^k \mathbb{C}P^n)$  in terms of  $\pi_s^0(\#^k \mathbb{C}P^n)$  and  $\widetilde{KO}^0(\#^k \mathbb{C}P^n)$ . In addition to this, we compute the smooth and  $PL$  tangential structure set of  $\#^k \mathbb{C}P^n$  for  $3 \leq n \leq 7$ , and compare the tangential homotopy smoothings and  $PL$ -tangential smoothings of  $\#^k \mathbb{C}P^n$ .



# Chapter 2

## Preliminaries

A topological space  $M$  is said to be a *manifold of dimension  $n$*  or an  *$n$ -manifold* if it is a Hausdorff, second countable and locally Euclidean  $n$ -space, i.e., each point of  $M$  has a neighborhood homeomorphic to an open subset of  $\mathbb{R}^n$ . Let  $Top$  denote the category whose objects are topological manifolds and morphisms are continuous maps.

We now recall the category of piecewise linear manifolds. Let  $K$  and  $L$  be simplicial complexes. Then a continuous map  $f: K \rightarrow L$  is said to be *piecewise linear* (or *PL*) if for every simplex  $\sigma \in K$ , the restricted map  $f|_{\sigma}$  is a linear map. Let  $v$  be a vertex of  $K$ . Then  $Star(v)$  is the union of simplices in  $K$  which contain  $v$ ,  $\{\sigma \in K \mid v \in \sigma\}$ , and  $Link(v)$  is the union of those simplices of  $Star(v)$  which are disjoint from  $v$ ,  $\{\tau \in Star(v) \mid v \notin \tau\}$ . Thus,  $|Star(v)|$  is a topological join of  $v$  and  $|Link(v)|$ .

A *piecewise linear manifold* (*PL-manifold*) of dimension  $n$  is a topological  $n$ -manifold  $M$  equipped with a homeomorphism  $h: |K| \rightarrow M$ , where  $K$  is a simplicial complex with  $|K|$  as a topological  $n$ -manifold, and for any vertex  $v$  of  $K$ ,  $Star(v)$  is *PL-homeomorphic* to the standard  $n$ -simplex  $\Delta^n$ ; equivalently,  $Link(v)$  is *PL-homeomorphic* to the boundary of  $\Delta^n$ . We denote the category *PL* whose objects are *PL-manifolds* and morphisms are *PL-continuous maps*.

A smooth  $n$ -manifold  $M$  is a topological manifold equipped with a smooth atlas, where a smooth atlas is defined such that any transition map between any two charts is a diffeomorphism. Consider the category  $Diff$  with objects being the smooth manifolds and morphisms being the smooth maps.

Note that there is a proper containment  $Top \supsetneq PL \supsetneq Diff$ . Here are some briefly discussed examples illustrating the proper containment:

**Example 2.0.1.** In 1956, John Milnor [38] constructed an exotic 7-sphere as follows: Consider two copies of  $\mathbb{D}^4 \times \mathbb{S}^3$  along with their boundary  $\mathbb{S}^3 \times \mathbb{S}^3$ . Identify  $\mathbb{S}^3$  with the group of unit quaternion, glue  $(a, b)$  on the boundary of former with  $(a, a^2ba^{-1})$  on the boundary of latter. He showed that the resulting manifold has a smooth structure and is homeomorphic to the standard 7-sphere but not diffeomorphic to it.

**Example 2.0.2.** The first example of a  $PL$ -manifold which does not admit smooth structure was given by Michel Kervaire in 1960 [28]. A Kervaire manifold  $K^{4k+2}$  is a  $(4k+2)$ -dimensional  $PL$  manifold constructed by plumbing together the tangent bundles of two  $(2n+1)$ -spheres, and then gluing a ball to the result. Moreover,  $K^{10}$  does not have a homotopy type of any smooth manifold.

**Example 2.0.3.** Another example of a topological manifold with no  $PL$  or smooth structures is a Poincaré homology sphere [45]: Recall the Poincaré homology sphere is a closed manifold given by  $SO(3)/A_5$ , where  $A_5$  denotes the group isomorphic to the alternating group on 5 elements. It is non-simply connected manifold with the fundamental group of order 120.

## 2.1 Classification of Cat-structures

Let us fix  $Cat = PL$  or  $Diff$ , where the  $Cat$ -isomorphism denotes either a  $PL$ -homeomorphism or a diffeomorphism, respectively.

Let  $X$  be an underlying topological manifold with two  $Cat$ -structures  $\mathcal{U}_1$  and  $\mathcal{U}_2$ . Then  $\mathcal{U}_1$  and  $\mathcal{U}_2$  are said to be equivalent *up to Cat-isomorphism* if there exists a homeomorphism

$$f : (X, \mathcal{U}_1) \rightarrow (X, \mathcal{U}_2),$$

such that the pullback of  $(X, \mathcal{U}_2)$  under  $f$  is  $(X, \mathcal{U}_1)$ , i.e.,  $f^*(X, \mathcal{U}_2) = (X, \mathcal{U}_1)$ .

This gives an equivalence relation on the set of all *Cat*-structures on the underlying topological manifold  $X$ . Given a topological manifold  $X$  the ultimate aim is to determine the set of all equivalence classes of *Cat*-structures on  $X$  up to *Cat*-isomorphisms, denoted by  $\mathcal{S}_{Cat}(X)$  [34].

Every topological manifold  $M$  of dimension at most 3 admits a *PL* as well as a smooth structure, and essentially it is unique, i.e.,  $|\mathcal{S}_{Cat}(M)| = 1$  for both *Cat* = *PL* or *Diff* [46, 40]. In case of Euclidean spaces, we have  $|\mathcal{S}_{Cat}(\mathbb{R}^n)| = 1$  for  $n \neq 4$  proved in [50, 43, 19]. For  $\mathbb{R}^4$  there are uncountable many non-isomorphic smooth or *PL*-structures [14]. Every *PL*-manifold homeomorphic to  $\mathbb{S}^n$  is actually *PL*-homeomorphic to  $\mathbb{S}^n$ , i.e.  $|\mathcal{S}_{PL}(\mathbb{S}^n)| = 1$ , for all  $n \neq 4$ . This implies that the *PL* version of the generalized Poincaré conjecture holds true for all  $n \neq 4$ . However, for the case  $n = 4$ , the conjecture remains open and is equivalent to the smooth version. The smooth Poincaré conjecture is false in most of the dimensions, i.e.,  $|\mathcal{S}_{Diff}(\mathbb{S}^n)| > 1$  for infinitely many  $n$ .

For topological manifolds, the ultimate goal is to determine the set of all equivalence classes of *Cat*-structures up to *Cat*-isomorphism, where *Cat* can be *Diff* or *PL*, i.e., to determine  $\mathcal{S}_{Cat}(M)$ . This is equivalent to asking the following classification problem:

**Problem 2.1.1.** *Classify PL (resp. smooth)-structures on the underlying topological (resp. PL) manifold  $M$  up to PL-homeomorphism (resp. diffeomorphism).*

Recognizing the inherent challenge in directly addressing this problem, we acknowledge that its difficulty arises from the absence of any algebraic structure, which typically plays a crucial role in computational task. Therefore, researchers often focus on exploring other equivalence relations closely related to the aforementioned relations described in the next section.

## 2.2 Structure sets under various equivalences

In this section, we revisit several equivalence relations within the category of smooth manifolds, that are essential for the upcoming chapters.

**Note:** The spaces  $B\text{Top}$ ,  $B\text{PL}$ ,  $\text{BO}$ ,  $\text{BF}$ ,  $\text{Top/O}$ ,  $\text{PL/O}$ ,  $\text{F/O}$  occurring in this and the forthcoming section play a very important role in the existence of Cat-structures and related classification problem. For further details and description of these spaces, we refer the reader to Section 2.4.

### 2.2.1 Surgery structure set

We recall the smooth surgery structure set exhibited by Sullivan and Wall. This set helps in providing answers to Question 1.2.2, for  $n \geq 5$ .

**Definition 2.2.1** ([6]). *Let  $X$  be a space. A smooth manifold structure on  $X$  is a pair  $(M, f)$  where  $M$  is a smooth  $n$ -dimensional manifold and  $f : M \rightarrow X$  is a homotopy equivalence. The smooth structure set  $S^{\text{Diff}}(X)$  is the set of smooth manifold structures up to the following equivalence relation:  $(M, f) \sim (N, g)$  if there is a diffeomorphism  $h : M \rightarrow N$  such that  $f \simeq g \circ h$ .*

In other words,  $S^{\text{Diff}}(X)$  measures the number of distinct manifolds up to diffeomorphisms within the homotopy class of  $X$ .

**Example 2.2.2.** The generalized Poincaré conjecture in the topological category says that  $S^{\text{Diff}}(\mathbb{S}^n)$  is equal to  $\mathcal{S}_{\text{Diff}}(\mathbb{S}^n)$ . The  $S^{\text{Diff}}(\mathbb{S}^p \times \mathbb{S}^q)$  is discussed for  $p + q \geq 5$  in [10].

**Question 2.2.3.** *What is the relationship between the space of smooth structures  $S^{\text{Diff}}(X)$  and the classification of smooth manifolds up to diffeomorphism?*

To examine the relationship between the smooth structure set and the set of smooth structures up to diffeomorphism, observe that there is a natural forgetful surjective map;

$$\begin{aligned} S^{\text{Diff}}(X) &\twoheadrightarrow \mathcal{S}_{\text{Diff}}(X) \\ [N, f] &\mapsto [N]. \end{aligned}$$

Let  $\mathcal{E}(X)$  be the group of self-homotopy equivalences up to homotopy of  $X$ . Then, note

that, we have the following group action

$$\begin{aligned}\mathcal{E}(X) \times S^{\text{Diff}}(X) &\rightarrow S^{\text{Diff}}(X) \\ ([h], [N, f]) &\mapsto [N, h \circ f].\end{aligned}$$

It can be checked easily that the orbit space of this action yields the following bijection with the set  $\mathcal{S}_{\text{Diff}}(X)$ .

$$\left( S^{\text{Diff}}(X) / \mathcal{E}(X) \right) \rightarrow \mathcal{S}_{\text{Diff}}(X). \quad (2.1)$$

The surjectivity is fairly straightforward. For injectivity, if elements  $[M, f]$  and  $[N, g]$  in  $S^{\text{Diff}}(X)$  represent the same element in  $\mathcal{S}_{\text{Diff}}(X)$  then we can write  $f = h \circ g$  for  $h = f^{-1} \circ g \in \mathcal{E}(X)$ , where  $f^{-1}$  is a homotopy inverse of  $f$ .

**Remark 2.2.4.** In general,  $S^{\text{Diff}}(X)$  does not admit a group structure [10].

### 2.2.1.1 Surgery exact sequence

Let us recall some key terminologies essential in understanding the surgery exact sequence. A map  $f : M \rightarrow X$  is called a *degree one map* if the fundamental class of a closed, oriented connected  $n$ -manifold  $M$  under  $f$  is mapped to the fundamental class of  $X$ , i.e.,

$$f_*([M]) = [X] \in H_n(X),$$

where  $[M]$  and  $[X]$  denote the fundamental classes of  $M$  and  $X$  respectively, and  $H_n(X)$  denotes the  $n$ -th homology group of  $X$ .

A *normal map* on a finite connected  $n$ -dimensional Poincaré complex  $X$  (i.e. a CW-complex whose cellular chain complex satisfies Poincaré duality) consists of an  $n$ -manifold  $M$  with a map  $f : M \rightarrow X$ , a stable vector bundle  $\xi : E \rightarrow X$ , and a bundle map from the stable

normal bundle covering  $f$ .

$$\begin{array}{ccc} v(M) & \xrightarrow{b} & E \\ \downarrow & & \downarrow \xi \\ M & \xrightarrow{f} & X \end{array}$$

**Definition 2.2.5.** An  $h$ -cobordism is a triple  $(W; M, N)$  consisting of an  $(n+1)$ -dimensional compact, oriented, smooth manifold  $W$  with boundary  $\partial W$ , and closed, oriented, smooth  $n$ -manifolds  $M, N$  such that

$$\partial W = M \sqcup -N,$$

and the inclusions  $M \hookrightarrow W$  and  $N \hookrightarrow W$  are homotopy equivalences, and here  $-N$  denotes the manifold  $N$  with opposite orientation.

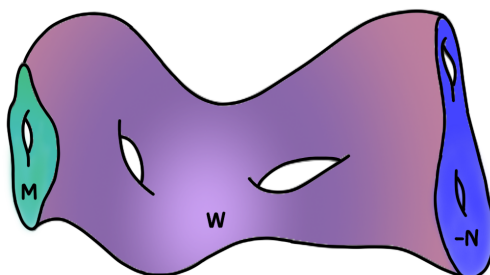


Figure 2.1: Cobordism  $(W; M, N)$

This gives an equivalence relation on the set of closed, oriented, smooth  $n$ -manifolds. All manifolds cobordant to a fixed given manifold  $M$  form the cobordism class of  $M$ .

Then, consider the collection of all degree one normal maps on a Poincaré complex  $X$  under the cobordism relation. Denote this set as  $\mathcal{N}(X)$ , representing the normal cobordism classes of degree one normal maps, often referred to as the *normal invariant set*.

Now, we have the following surgery exact sequence associated with a simply connected Poincaré complex  $X$ ,

$$\cdots \rightarrow L_{n+1}(\mathbb{Z}) \xrightarrow{\omega} S^{\text{Diff}}(X) \xrightarrow{\theta} \mathcal{N}(X) \xrightarrow{\sigma} L_n(\mathbb{Z}). \quad (2.2)$$



The spaces  $L_n(\mathbb{Z})$  and  $L_{n+1}(\mathbb{Z})$  represent surgery obstruction groups. The map  $\sigma$  is the surgery obstruction map that measures whether  $S^{\text{Diff}}(X)$  is nonempty or not. The map  $\omega$  represents the action of  $L_{n+1}(\mathbb{Z})$  on  $S^{\text{Diff}}(X)$ . Elements  $a$  and  $b$  in  $S^{\text{Diff}}(X)$  map to the same element in  $\mathcal{N}(X)$  if and only if there exists  $x \in L_{n+1}(\mathbb{Z})$  such that  $\omega(x, a) = b$ . It is important to note that  $S^{\text{Diff}}(X)$  is a set and not a group. Therefore, the exact sequence (2.2) conveys that for  $y \in \text{Im}(\theta)$ ,  $\sigma(y) = 0$  if and only if  $y \in \text{Im}(S^{\text{Diff}}(X))$ .

If  $X$  is simply connected then the values of surgery obstruction group are

$$L_n(\mathbb{Z}) = \begin{cases} \mathbb{Z} & n \equiv 0 \pmod{4} \\ 0 & n \equiv 1 \pmod{4} \\ \mathbb{Z}_2 & n \equiv 2 \pmod{4} \\ 0 & n \equiv 3 \pmod{4}. \end{cases}$$

Furthermore, if  $\mathcal{N}(X) \neq \emptyset$  then we have the bijection

$$\mathcal{N}(X) \cong [X, F/O],$$

where  $F/O$  is a homotopy fiber of the map  $BO \rightarrow BF$  (see Section 2.4.1). For computational purposes concerning  $\mathcal{N}(X)$ , we examine the following exact sequence induced by the aforementioned fibration sequence

$$\cdots \rightarrow [X, SO] \rightarrow [X, SF] \rightarrow [X, F/O] \rightarrow [X, BSO] \rightarrow \cdots \quad (2.3)$$

Note that the set  $[X, SF]$  coincides with the reduced  $0^{\text{th}}$  stable cohomotopy group of  $X$ , the set  $[X, SO]$  is isomorphic to the reduced  $-1^{\text{th}}$   $KO$ -group, and  $[X, BSO]$  is isomorphic to the reduced  $0^{\text{th}}$   $KO$ -group of  $X$ .

In particular, below table provides reduced  $n^{\text{th}}$  stable cohomotopy group of sphere denoted by  $\pi_n^s$  (for generators refer [53]):

Next table presents reduced  $0^{\text{th}}$   $KO$ -group of  $\mathbb{S}^n$ :

In general  $S^{\text{Diff}}(M)$  may not admit a group structure (refer [10]). Therefore, researchers are encouraged to explore other equivalence relations in bundle theory and homotopy theory,

| $n$ | $\pi_n^s$                          | $n$ | $\pi_n^s$                                                 | $n$ | $\pi_n^s$                                                                     |
|-----|------------------------------------|-----|-----------------------------------------------------------|-----|-------------------------------------------------------------------------------|
| 0   | $\mathbb{Z}$                       | 7   | $\mathbb{Z}_{16} \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5$ | 14  | $\mathbb{Z}_2 \oplus \mathbb{Z}_2$                                            |
| 1   | $\mathbb{Z}_2$                     | 8   | $\mathbb{Z}_2 \oplus \mathbb{Z}_2$                        | 15  | $\mathbb{Z}_{32} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5$ |
| 2   | $\mathbb{Z}_2$                     | 9   | $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$    | 16  | $\mathbb{Z}_2 \oplus \mathbb{Z}_2$                                            |
| 3   | $\mathbb{Z}_8 \oplus \mathbb{Z}_3$ | 10  | $\mathbb{Z}_2 \oplus \mathbb{Z}_3$                        | 17  | $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$    |
| 4   | 0                                  | 11  | $\mathbb{Z}_8 \oplus \mathbb{Z}_9 \oplus \mathbb{Z}_7$    | 18  | $\mathbb{Z}_2 \oplus \mathbb{Z}_8$                                            |
| 5   | 0                                  | 12  | 0                                                         | 19  | $\mathbb{Z}_8 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_{11}$ |
| 6   | $\mathbb{Z}_2$                     | 13  | $\mathbb{Z}_3$                                            |     |                                                                               |

Table 2.1: Stable homotopy groups of sphere for  $1 \leq n \leq 19$ .

| $q \pmod{8}$                     | 0            | 1              | 2              | 3 | 4            | 5 | 6 | 7 |
|----------------------------------|--------------|----------------|----------------|---|--------------|---|---|---|
| $\widetilde{KO}^0(\mathbb{S}^q)$ | $\mathbb{Z}$ | $\mathbb{Z}_2$ | $\mathbb{Z}_2$ | 0 | $\mathbb{Z}$ | 0 | 0 | 0 |

Table 2.2: Reduced  $KO$ -groups of sphere

discussed below.

## 2.2.2 Concordance structure set

**Definition 2.2.6.** Let  $X$  be a closed topological  $n$ -manifold. Let  $(N, f)$  be a pair consisting of a closed, smooth  $n$ -manifold  $N$  together with a homeomorphism  $f : N \rightarrow X$ . Then  $(N_1, f_1)$  and  $(N_2, f_2)$  are concordant if there exists a diffeomorphism  $g : N_1 \rightarrow N_2$  and a homeomorphism

$$F : N_1 \times \mathbb{I} \rightarrow X \times \mathbb{I}$$

such that  $F|_{N_1 \times 0} = (f_1, 0)$  and  $F|_{N_1 \times 1} = (f_2 \circ g, 1)$ .

The concordance class of  $(N, f)$  is denoted by  $[N, f]$ , and the set of all such concordance classes is denoted by  $\mathcal{C}(X)$ . The class  $[X, \text{Id}]$  of the identity  $\text{Id} : X \rightarrow X$  can be considered as the base point of  $\mathcal{C}(X)$ . Recall that, in Definition 2.2.6, if we add the level-wise homeomorphism condition on  $F$  then it defines an *isotopy relation*. Note that, from [19, Part I], concordance

implies isotopy for smooth manifolds of dimensions  $\geq 5$ .

**Question 2.2.7.** *What is the relationship between the space of smooth structures  $\mathcal{C}(X)$  and the classification of smooth manifolds up to diffeomorphism?*

Let us examine the relationship between the concordance structure set and the set of smooth structures up to diffeomorphism. Any element  $[N]$  in  $\mathcal{S}_{\text{Diff}}(X)$  represents a smooth structure on  $X$  up to diffeomorphism, i.e. there exists a homeomorphism  $f : N \rightarrow X$  which induces a smooth structure in  $X$  representing the class  $[N]$ . Therefore, observe that there is a natural forgetful surjective map;

$$\begin{aligned}\mathcal{C}(X) &\twoheadrightarrow \mathcal{S}_{\text{Diff}}(X) \\ [N, f] &\mapsto [N]\end{aligned}$$

Let  $\mathcal{H}(X)$  be the group of self-homeomorphisms up to homotopy on  $X$ . Then we have the following group action

$$\begin{aligned}\mathcal{H}(X) \times \mathcal{C}(X) &\rightarrow \mathcal{C}(X) \\ ([h], [N, f]) &\mapsto [N, h \circ f].\end{aligned}$$

It can be checked easily that the orbit space of this action yields the following a bijection with the set  $\mathcal{S}_{\text{Diff}}(X)$ .

$$\left( \mathcal{C}(X) / \mathcal{H}(X) \right) \rightarrow \mathcal{S}_{\text{Diff}}(X) \tag{2.4}$$

The surjectivity is fairly straightforward. For injectivity, if elements  $[N_1, f]$  and  $[N_2, g]$  in  $\mathcal{C}(X)$  represents same element in  $\mathcal{S}_{\text{Diff}}(X)$  then we can write  $f = h \circ g$  for  $h = f^{-1} \circ g \in \mathcal{H}(X)$ .

**Question:** Why is it advantageous to study the concordance class space  $\mathcal{C}(X)$  instead of the space  $S^{\text{Diff}}(X)$  or the space  $\mathcal{S}_{\text{Diff}}(X)$ ?

The most important reason for studying the concordance structure set is its correspondence in homotopy theory, as demonstrated by Cairns-Hirsch-Mazur, which is not true for the set  $\mathcal{S}_{\text{Diff}}(X)$ ;

**Theorem 2.2.8.** [34, Theorem 7.2] *Let  $X$  be a closed, smooth manifold of dimension  $m \geq 5$ . Then there exists a connected  $H$ -space  $Top/O$  such that there is a bijection  $\psi$  between  $\mathcal{C}(X)$  and the set  $[X, Top/O]$  of homotopy classes of maps  $X$  into  $Top/O$ .*

Shortly, if  $X$  admits a smooth structure then there is a set bijection

$$\mathcal{C}(X) \cong [X, Top/O].$$

**Remark 2.2.9.** *Let  $X$  be a smooth  $n$ -manifold for  $n \geq 5$ . Then cardinality of the set  $[X, Top/O]$  is equal to the number of distinct smooth structures on the underlying topological manifold  $X$  up to concordance/isotopy relation.*

Analogously, there is a notion of *PL-concordance structure set* which is the classification of smooth structures up to concordance relation on the underlying *PL*-manifold.

**Definition 2.2.10.** *Let  $X$  be a closed *PL*-manifold. Let  $(N, f)$  be a pair consisting of a closed, smooth  $n$ -manifold  $N$  together with a *PL*-homeomorphism  $f : N \rightarrow X$ . Then  $(N_1, f_1)$  and  $(N_2, f_2)$  are concordant if there exists a diffeomorphism  $g : N_1 \rightarrow N_2$  and a *PL*-homeomorphism  $F : N_1 \times \mathbb{I} \rightarrow X \times \mathbb{I}$  such that  $F|_{N_1 \times 0} = (f_1, 0)$  and  $F|_{N_1 \times 1} = (f_2 \circ g, 1)$ .*

The set of all such equivalence classes is denoted by  $\mathcal{C}_{PL}(X)$ , the set of all smooth structures up to *PL*-concordance on the underlying *PL*-manifold  $X$ .

**Theorem 2.2.11.** [34, Theorem 7.2] *Let  $X$  be a closed, smooth manifold of dimension  $m \geq 5$ . Then there exists a connected  $H$ -space  $PL/O$  such that there is a bijection  $\psi$  between  $\mathcal{C}_{PL}(X)$  and the set  $[X, PL/O]$  of homotopy classes of maps  $X$  into  $PL/O$ .*

Equivalently, given a smooth  $n$ -manifold  $X$  for  $n \geq 5$ , the cardinality of the set  $[X, PL/O]$  is equal to the number of distinct smooth structures on the underlying *PL* manifold  $X$  up to concordance relation [19, Theorem 4.2],

$$\mathcal{C}_{PL}(M) \cong [M, PL/O]. \quad (2.5)$$

In a particular case of standard sphere  $\mathbb{S}^n$ ,  $\mathcal{C}(\mathbb{S}^n)$  is nothing but the set of all (oriented)

diffeomorphism classes of smooth structures on  $\mathbb{S}^n$ , for  $n \geq 5$

$$|\mathcal{C}(\mathbb{S}^n)| = |\mathcal{C}_{PL}(\mathbb{S}^n)| = |\mathcal{S}_{\text{Diff}}(\mathbb{S}^n)|.$$

Note that  $\mathcal{C}(\mathbb{S}^p \times \mathbb{S}^q) \cong \mathcal{C}(\mathbb{S}^p) \oplus \mathcal{C}(\mathbb{S}^q) \oplus \mathcal{C}(\mathbb{S}^{p+q})$  for  $p + q \geq 5$ . More generally, for manifolds  $M^p, N^q$ , we have  $\mathcal{C}(M^p \times N^q) \cong \mathcal{C}(M^p) \oplus \mathcal{C}(N^q) \oplus \mathcal{C}(M^p \wedge N^q)$  for  $p + q \geq 5$  (for proof refer to [21, Chapter 2]). In case of complex projective space  $\mathbb{CP}^n$ , there is a unique smooth structure up to diffeomorphism for  $n = 1, 2, 3$  (for  $n = 3$  case refer to [22]). However,  $\mathbb{CP}^4$  has two smooth structures up to concordance as well as up to diffeomorphism [22]. The quaternionic projective space  $\mathbb{HP}^2$  has two smooth structures up to concordance but a unique smooth structure up to diffeomorphism [3].

## 2.3 Group of homotopy $n$ -spheres - $\Theta_n$

In 1963, John W. Milnor, along with Michel A. Kervaire, constructed a group of h-cobordism classes of homotopy  $n$ -spheres, denoted by  $\Theta_n$ , for  $n \geq 5$  in [29]. A *homotopy  $n$ -sphere* is a closed, oriented, smooth  $n$ -manifold homotopy equivalent to  $\mathbb{S}^n$ .

The *connected sum* of two closed, oriented,  $n$ -manifolds is obtained by removing a small  $n$ -cell from each and then pasting together the resulting boundaries homeomorphically in such a way that the resulting  $n$ -manifold is oriented. Furthermore, Milnor and Kervaire showed that the connected sum operation is well defined, associative, and commutative up to orientation preserving diffeomorphisms. In particular, for spheres they proved the following:

**Theorem 2.3.1.** [29] *The h-cobordism classes of homotopy  $n$ -spheres form an abelian group  $\Theta_n$ ,  $n \geq 5$ , under the connected sum operation.*

Note that, for  $n \geq 5$  we have

$$\Theta_n \cong S^{\text{Diff}}(\mathbb{S}^n) \cong \mathcal{C}(\mathbb{S}^n) \cong \mathcal{C}_{PL}(\mathbb{S}^n) \cong \mathcal{C}_{\text{Diff}}(\mathbb{S}^n).$$

In the same paper, it is proved that there is a finite number of distinct smooth structures on

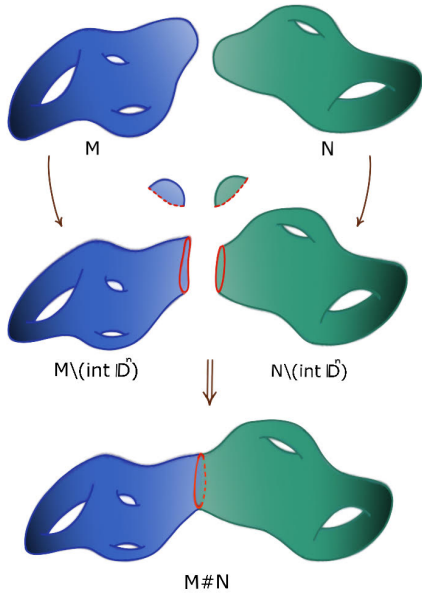


Figure 2.2: Connected sum of two manifolds

spheres in all dimensions other than 3.

**Theorem 2.3.2.** *For  $n \neq 3$ , the group  $\Theta_n$  is finite.*

Furthermore, they also computed a few values of  $\Theta_n$  (see Table 2.3).

| $n$      | $\Theta_n$                                             | $n$ | $\Theta_n$                              |
|----------|--------------------------------------------------------|-----|-----------------------------------------|
| $\leq 6$ | $\{1\}$                                                | 13  | $\mathbb{Z}_3$                          |
| 7        | $\mathbb{Z}_4 \oplus \mathbb{Z}_7$                     | 14  | $\mathbb{Z}_2$                          |
| 8        | $\mathbb{Z}_2$                                         | 15  | $\mathbb{Z}_2 \oplus \mathbb{Z}_{8128}$ |
| 9        | $\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2$ | 16  | $\mathbb{Z}_2$                          |
| 10       | $\mathbb{Z}_2 \oplus \mathbb{Z}_3$                     | 17  | $\mathbb{Z}_2 \oplus \mathbb{Z}_8$      |
| 11       | $\mathbb{Z}_{31} \oplus \mathbb{Z}_{32}$               | 18  | $\mathbb{Z}_2 \oplus \mathbb{Z}_8$      |
| 12       | $\{1\}$                                                |     |                                         |

Table 2.3: Group  $\Theta_n$  for  $1 \leq n \leq 18$

### 2.3.1 Subgroups of $\Theta_n$

In the classification of smooth structures of manifolds, the group of homotopy  $n$ -spheres  $\Theta_n$  plays an important role. Recall that, an exotic sphere  $\Sigma^n$  is an element of  $\Theta_n$  homeomorphic to  $\mathbb{S}^n$  but not diffeomorphic. A connected sum of a smooth manifold with an exotic sphere results in a manifold homeomorphic to the underlying topological manifold; however, it may not be diffeomorphic to it. Therefore, the determination of subgroups of  $\Theta_n$ , such as, the inertia group, the homotopy inertia group, and the concordance inertia group correlates with the classification problem.

**Definition 2.3.3.** *Let  $M^n$  be a closed, oriented, smooth manifold.*

- (i) *The inertia group of  $M^n$  is the subgroup  $I(M^n) \subseteq \Theta_n$  of homotopy spheres  $\Sigma^n$  such that  $M^n \# \Sigma^n$  is (orientation-preserving) diffeomorphic to  $M^n$ .*
- (ii) *The homotopy inertia group  $I_h(M^n)$  consists of  $\Sigma^n \in \Theta_n$  such that there exists an orientation-preserving diffeomorphism from  $M^n \# \Sigma^n \rightarrow M^n$  that is homotopic to the canonical homeomorphism  $h_{\Sigma^n} : M^n \# \Sigma^n \rightarrow M^n$ .*
- (iii) *The concordance inertia group  $I_c(M^n)$  consists of  $\Sigma^n \in \Theta_n$  such that there exists an orientation-preserving diffeomorphism from  $M^n \# \Sigma^n \rightarrow M^n$  that is concordant to the canonical homeomorphism  $h_{\Sigma^n} : M^n \# \Sigma^n \rightarrow M^n$ .*

Here, observe that

$$I_c(M) \subseteq I_h(M) \subseteq I(M).$$

Note that,  $I(M)$  is not a PL-homeomorphism or topological invariant of  $M$  [27]. Further,  $I_h(M)$  is not a homotopy invariant of  $M$  but is an h-cobordism invariant when dimension of  $M$  is at least 8, and  $I_c(M)$  is homotopy invariant of  $M$  for dimension of  $M$  at least 5 [8].

**Remark 2.3.4.** *The notions in Definition 2.3.3 can also be defined for closed, smooth manifolds, where the requirement for the diffeomorphism to be orientation-preserving is relaxed. However, for simply connected manifolds, the two notions coincide [13, Page 6].*

Recall that, the group  $\Theta_n \cong S^{\text{Diff}}(\mathbb{S}^n)$  acts on the smooth structure set  $S^{\text{Diff}}(M)$  [6, 54], given by

$$\begin{aligned}\Theta_n \times S^{\text{Diff}}(M) &\rightarrow S^{\text{Diff}}(M) \\ ([\Sigma, f], [N, g]) &\mapsto [\Sigma \# N, f \# g]\end{aligned}\tag{2.6}$$

The stabilizer of this action at the base point  $[M, \text{Id}]$ , can be identified with the homotopy inertia group, denoted by  $I_h(M)$ .

Similarly, the group  $\Theta_n \cong \mathcal{C}(\mathbb{S}^n)$  acts on the concordance structure set  $\mathcal{C}(M)$  [6, 54], given by

$$\begin{aligned}\Theta_n \times \mathcal{C}(M) &\rightarrow \mathcal{C}(M) \\ ([\Sigma, f], [N, g]) &\mapsto [\Sigma \# N, f \# g]\end{aligned}\tag{2.7}$$

The stabilizer of this action at the base point  $[M, \text{Id}]$ , can be identified with the homotopy inertia group, denoted by  $I_c(M)$ .

### 2.3.2 Exact sequence involving $\Theta_n$

It follows from [29] that, the group  $\Theta_n$  for  $n \geq 5$  fits into the following exact sequence, which we are using in Chapter 4,

$$0 \rightarrow bP_{n+1} \rightarrow \Theta_n \rightarrow \pi_n^s / \text{Im}(J) \rightarrow \mathbb{Z}_2 \text{ or } 0,$$

where  $bP_{n+1} \subseteq \Theta_n$  such that a homotopy  $n$ -sphere  $\Sigma^n$  represents an element in  $bP_{n+1}$  if and only if  $\Sigma^n$  is the boundary of a parallelizable manifold (manifold whose tangent bundle is trivial). Here,  $\pi_n^s$  is the  $n^{\text{th}}$  stable homotopy group of  $\mathbb{S}^0$ ,  $J : \pi_n(SO) \rightarrow \pi_n^s$  is the J-homomorphism, and  $\pi_n^s / \text{Im}(J) \subseteq \pi_n(F/O)$  where the space  $F/O$  is discussed in Section 2.4.

Let us briefly examine how this exact sequence arises (for more details, refer to [21], Section 3). Using spaces from Section 2.4.1, one can obtain the homotopy fibration  $PL/O \rightarrow$



$F/O \rightarrow F/PL$ , which induces the following long exact sequence of homotopy groups

$$\cdots \rightarrow \pi_{n+1}(F/PL) \xrightarrow{p_*} \pi_n(PL/O) \xrightarrow{i_*} \pi_n(F/O) \rightarrow \pi_n(F/PL) \rightarrow \cdots.$$

In this sequence we use the identification of the space  $\pi_n(PL/O)$  with  $\Theta_n$  for  $n \geq 5$ , and in [29], it is proved that the  $\text{Im}(p_*) = bP_{n+1}$ , therefore we obtain the following exact sequence

$$0 \rightarrow bP_{n+1} \rightarrow \Theta_n \xrightarrow{i_*} \pi_n(F/O) \rightarrow \pi_n(F/PL) \rightarrow \cdots.$$

The group  $bP_{n+1}$  is discussed for some values of  $n$  in the same paper [29].

Next, the inclusion of  $\text{Im}(i_*) \subseteq \pi_n^s / \text{Im}(J)$  is readily observed from the following commutative diagram. This diagram is constructed using the homotopy fibrations  $F/O \rightarrow BO \rightarrow BF$  and  $PL/O \rightarrow BO \rightarrow BPL$ .

$$\begin{array}{ccccccc} \cdots & \rightarrow & \pi_{n+1}(BO) & \rightarrow & \pi_{n+1}(BPL) & \longrightarrow & \pi_n(PL/O) \longrightarrow \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \rightarrow & \pi_{n+1}(BO) & \rightarrow & \pi_{n+1}(BF) & \longrightarrow & \pi_n(F/O) \longrightarrow \cdots \\ & & \parallel & & \parallel & & \uparrow \\ & & \pi_n(O) & \xrightarrow{J} & \pi_n(F) & \longrightarrow & \pi_n(F) / \text{Im}(J) = \pi_n^s / \text{Im}(J) \end{array}$$

Finally, in the exact sequence 2.3.2 for  $n \geq 5$  the right side homomorphism  $\pi_n^s / \text{Im}(J) \rightarrow \pi_n(F/PL)$  has the image equal to 0 if  $n \neq 2^m - 2$  and is  $\mathbb{Z}_2$  for  $n = 2^m - 2$ .

## 2.4 Stable classifying spaces

The study of classification of *Cat*-structures involves stable classifying spaces. These spaces determine whether a given topological manifold admits a *Cat*-structure and also quantify these structures. Following this, we discuss particular examples of stable classifying spaces essential for the subsequent results.

In this section, we use the fact that for a topological group  $G$ , there exists a classifying space  $BG$  (under mild hypotheses) such that isomorphism classes of principal  $G$ -bundles over

$X$  are in natural bijective correspondence with  $[X, BG]$  [30]. The correspondence is given by pulling back a universal principal  $G$ -bundle over  $BG$ . Here, we discuss specific examples of classifying spaces for topological groups that are important for our further discussions.

Let  $Top_n$  denote the topological group of self-homeomorphisms of  $\mathbb{R}^n$ ,  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  satisfying  $f(0) = 0$ , equipped with the compact-open topology.

A **Top<sub>n</sub>-bundle** over a space  $B$  is a fiber bundle  $\xi = \{p : E \rightarrow B\}$  with fiber  $\mathbb{R}^n$ , and equipped with the zero section  $s : B \rightarrow E$  which is a cofibration. (For the definition of fiber bundle, cofibration see Appendix A.)

Given two Top<sub>n</sub>-bundles  $\xi = \{p : E \rightarrow B\}$  and  $\eta = \{q : Y \rightarrow X\}$ , a Top<sub>n</sub>-morphism  $\varphi : \xi \rightarrow \eta$  is defined as a commutative diagram, representing a bundle morphism, as follows:

$$\begin{array}{ccc} E & \xrightarrow{g} & Y \\ p \downarrow & & \downarrow q \\ B & \xrightarrow{f} & X \end{array} \quad (2.8)$$

where  $\varphi$  respects the sections ( $gs_\xi = s_\eta f$ ) and induces a homeomorphism on each of the fibers.

**Theorem 2.4.1.** [19] *There exists a Top<sub>n</sub>-bundle  $\gamma_{Top}^n$  with the following universal property: For every Top<sub>n</sub>-bundle  $\xi$  over a CW-space  $B$ , for every CW-subspace  $A$  of  $B$  and every morphism  $\psi : \xi_A \rightarrow \gamma_{Top}^n$  of Top<sub>n</sub>-bundles, there exists a morphism  $\varphi : \xi \rightarrow \gamma_{Top}^n$  which is an extension of  $\psi$ . The base space of the bundle  $\gamma_{Top}^n$  is called the **classifying space for Top<sub>n</sub>-bundle** and denoted by **BTop<sub>n</sub>***

Let  $PL_n$  denote the topological group of PL-homeomorphisms  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  satisfying  $f(0) = 0$ , equipped with the compact-open topology.

**Definition 2.4.2.** A **PL<sub>n</sub>-bundle** is a principal PL<sub>n</sub>-bundle  $\xi = \{p : E \rightarrow B\}$  with  $E$  and  $B$  as PL-manifolds, and maps  $p : E \rightarrow B$  and a section  $s : B \rightarrow E$  are PL-morphisms. Further, for every simplex  $\Delta \subset B$ , there is a PL-homeomorphism  $h : p^{-1}(\Delta) \rightarrow \Delta \times \mathbb{R}^n$  satisfying  $h(s(\Delta)) = \Delta \times \{0\}$ .

**Theorem 2.4.3** ([19]). *There exists a universal PL<sub>n</sub>-bundle  $\gamma_{PL}^n$  with the following property: for every PL<sub>n</sub>-bundle  $\xi$  over a CW space  $B$ , for every CW subspace  $A$  of  $B$  and every morphism  $\psi : \xi_A \rightarrow \gamma_{PL}^n$  of PL<sub>n</sub>-bundle, there exists a morphism  $\varphi : \xi \rightarrow \gamma_{PL}^n$  which is an extension of  $\psi$ . The base space of  $\gamma_{PL}^n$  is called the **classifying space for PL<sub>n</sub>-bundles** and denoted by **BPL<sub>n</sub>**.*

Similarly, let  $O_n$  be a topological group of vector space isomorphisms  $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$  equipped with the usual subspace topology on  $\mathbb{R}^{n^2}$ .

**Theorem 2.4.4** ([19]). *There exists an  $O_n$ -bundle  $\gamma_O^n$  with the following universal property: For every  $O_n$ -bundle  $\xi$  over a CW-space  $B$ , for every CW-subspace  $A$  of  $B$  and every morphism  $\psi : \xi_A \rightarrow \gamma_O^n$  of  $O_n$ -bundle, there exists a morphism  $\phi : \xi \rightarrow \gamma_O^n$  which is an extension of  $\psi$ . The base sapce of  $\gamma_O^n$  is called the classifying space for  $O_n$ -bundle and denoted by  $BO_n$*

For  $G_n = Top_n, PL_n$  or  $O_n$ , let  $\gamma_G^n \oplus \theta_{BG_n}^1$  denote the  $G_{n+1}$ -bundle where  $\theta_{BG_n}^1$  is a trivial line bundle. Then, there exists a classifying map,

$$\beta_n^G : BG_n \rightarrow BG_{n+1}.$$

This map can be regarded as a standard inclusion of  $G_n \subset G_{n+1}$ , and hence  $BG_n \subset BG_{n+1}$ . Consider the following direct limit of these spaces over  $n$

$$G := \bigcup_n G_n \quad \text{and} \quad BG := \bigcup_n BG_n,$$

equipped with direct limit topology. Then **BG** is the *stable classifying space of G-bundle*.

Hence, the stable classifying spaces for  $G$ -bundles, where  $G$  denotes  $Top$ ,  $PL$ , and  $O$  respectively, are  $BTop$ ,  $BPL$ , and  $BO$ .

Another example of interest arises from a topological group  $F_n$  of self-homotopy equivalences of  $\mathbb{S}^{n-1}$ , equipped with the compact-open topology. The associated  $F_n$ -bundle is also referred as homotopy  $\mathbb{S}^{n-1}$ -fibration. The **stable classifying space for spherical fibration** is obtained as the direct limit of  $BF_n$ , denoted by **BF**, equipped with the direct limit topology.

### 2.4.1 Homotopy fibers Top/O, PL/O, F/O

Consider the map  $p_n : BO_n \rightarrow BTop_n$  for classifying spaces induced from a forgetful map  $O_n \rightarrow Top_n$ . If necessary, replacing  $p_n$  with a fibration map gives the following homotopy fibration

sequence.

$$\dots \rightarrow \Omega BO_n \rightarrow \Omega BTop_n \rightarrow F \rightarrow BO_n \xrightarrow{p_n} BTop_n \rightarrow \dots$$

Note that,  $\Omega BG \simeq G$  for any classifying space associated with a group  $G$ . Therefore, in the above sequence,  $F$  is homotopy equivalent to  $Top_n/O_n$  (see [30] for details). Thus, we obtain a homotopy fibration  $Top_n/O_n \rightarrow BO_n \rightarrow BTop_n$  for each  $n$ , which fits into the following homotopy commutative diagram:

$$\begin{array}{ccccccc} \dots & \rightarrow & Top_{n-1}/O_{n-1} & \rightarrow & Top_n/O_n & \rightarrow & Top_{n+1}/O_{n+1} \rightarrow \dots \rightarrow Top/O \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & BO_{n-1} & \longrightarrow & BO_n & \longrightarrow & BO_{n+1} \longrightarrow \dots \longrightarrow BO \\ & & \downarrow p_{n-1} & & \downarrow p_n & & \downarrow p_{n+1} \\ \dots & \longrightarrow & BTop_{n-1} & \longrightarrow & BTop_n & \longrightarrow & BTop_{n+1} \longrightarrow \dots \longrightarrow BTop \end{array}$$

If necessary, then replace the map  $p$  by its homotopy fibration, in particular we have

$$Top/O \rightarrow BO \rightarrow BTop.$$

Another way is to interpret the space  $Top/O$  as a direct limit space  $\bigcup_n (Top_n/O_n)$  with the direct limit topology.

Similarly, we have a forgetful map  $g_n : BO_n \rightarrow BPL_n$  with a homotopy fiber  $PL_n/O_n$ . Then for each  $n \in \mathbb{N}$ , we have the following homotopy commutative diagram

$$\begin{array}{ccccccc} \dots & \rightarrow & PL_{n-1}/O_{n-1} & \rightarrow & PL_n/O_n & \rightarrow & PL_{n+1}/O_{n+1} \rightarrow \dots \rightarrow PL/O \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & BO_{n-1} & \longrightarrow & BO_n & \longrightarrow & BO_{n+1} \longrightarrow \dots \longrightarrow BO \\ & & \downarrow p_{n-1} & & \downarrow p_n & & \downarrow p_{n+1} \\ \dots & \longrightarrow & BPL_{n-1} & \longrightarrow & BPL_n & \longrightarrow & BPL_{n+1} \longrightarrow \dots \longrightarrow BPL \end{array}$$

inducing the following homotopy fibration

$$PL/O \rightarrow BO \rightarrow BPL.$$

Now, from any  $O_n$ -bundle  $\xi = \{p : E \rightarrow B\}$ , we delete the zero section to obtain the homotopy  $\mathbb{S}^{n-1}$ -fibration  $\xi' = \{p' : E \setminus s(B) \rightarrow B\}$ . This gives a map  $BO_n \rightarrow BF_n$  with homotopy fiber  $F_n/O_n$ , and its colimit over  $n$  results in space  $F/O$ . So we have a homotopy fibration

$$F/O \rightarrow BO \rightarrow BF.$$

Due to the well-known result of Milnor [39], these spaces have the homotopy type of CW-spaces. Additionally, these spaces are infinite-loop spaces, therefore if we consider the homotopy classes of maps from a space  $X$  to any of these spaces, say  $[X, Top/O]$ , it admits an abelian group structure on it (see Appendix A).

Homotopy groups of  $Top/O$  are

$$\pi_n(Top/O) = \begin{cases} 0 & \text{if } n = 1, 2, 4, \\ \mathbb{Z}_2 & \text{if } n = 3, \\ \Theta_n & \text{if } n \geq 5; \end{cases} \quad (2.9)$$

and the homotopy groups of  $PL/O$  are

$$\pi_n(PL/O) = \begin{cases} 0 & \text{if } n \leq 4, \\ \Theta_n & \text{if } n \geq 5. \end{cases} \quad (2.10)$$

In particular we have

$$\pi_n(Top/O) \cong \pi_n(PL/O) \text{ for all } n \geq 5.$$

## 2.4.2 Spectrum and its properties

In stable homotopy theory, a **spectrum** is a sequence of topological spaces (or pointed spaces)  $X_0, X_1, X_2, \dots$  equipped with structure maps that stabilize the homotopy-theoretic information in some sense. Formally, a spectrum  $X$  consists of: a sequence of pointed spaces  $X_n$  for  $n =$

$0, 1, 2, \dots$ , and structure maps  $\Sigma X_n \rightarrow X_{n+1}$ , where  $\Sigma X_n$  denotes the suspension of  $X_n$ .

The homotopy groups of a spectrum  $X$  are defined as:

$$\pi_n(X) = \lim_{k \rightarrow \infty} \pi_{n+k}(X_k),$$

where the limit is taken over the structure maps of the spectrum. These homotopy groups are stable under suspension.

Note that, every spectrum gives rise to a generalized cohomology theory. Specifically, for a space  $Y$ , the cohomology groups are defined as:

$$E^n(Y) = [Y, X_n],$$

where  $[Y, X_n]$  denotes the set of homotopy classes of maps from  $Y$  to  $X_n$ .

Examples of spectra include suspension spectrum, sphere spectrum, Eilenberg-MacLane spectrum, K-theory spectrum, Thom spectrum, and complex cobordism spectrum, etc.

In Chapter 3, for an abelian group  $A$ , we have used an Eilenberg-MacLane spectrum denoted by  $HA$ . This spectrum is associated with ordinary cohomology. The  $n$ -th space in the spectrum is an Eilenberg-MacLane space  $K(A, n)$ , and the structure maps are induced by the canonical suspension maps. For example, the spectrum  $H\mathbb{Z}_2$  gives rise to ordinary cohomology with integer coefficients. Its homotopy groups are  $\pi_n(H\mathbb{Z}_2) = \mathbb{Z}_2$  for  $n = 0$  and  $0$  for  $n \neq 0$ . Other spectra used in Chapter 3 include  $PL/O$  and  $G/PL$ , which correspond to the spaces  $PL/O$  and  $G/PL$ , respectively.

## 2.5 Topological K-theory

Topological K-theory studies vector bundles and their classifications through homotopical methods. It provides a way to classify real or complex vector bundles over topological spaces and is particularly useful in understanding stable phenomena in topology.

The K-theory of a topological space  $X$ , denoted  $K(X)$ , is defined using the Grothendieck

group of the monoid of isomorphism classes of vector bundles over  $X$ . Formally, if  $V(X)$  represents the monoid of vector bundles over  $X$ , the K-theory group is given by:

$$K(X) = K^0(X) = V(X) / \sim$$

where the relation  $\sim$  identifies two bundles  $E$  and  $F$  if there exists a third bundle  $G$  such that  $E \oplus G \cong F \oplus G$

This definition captures the idea that two vector bundles can be considered equivalent if they can be “added” to another bundle to form isomorphic bundles.

### 2.5.1 Real K-Theory

Real K-theory, denoted  $\widetilde{KO}^0(X)$ , classifies vector bundles over a topological space  $X$  using real vector spaces. This group can be identified with the homotopy classes of maps from  $X$  into the classifying space for stable real vector bundles, denoted  $BO$ :

$$\widetilde{KO}^0(X) \cong [X, BO]$$

The space  $BO$  (the infinite real Grassmannian) classifies stable isomorphism classes of real vector bundles.

**Example 2.5.1.** 1. For a point  $pt$ ,  $\widetilde{KO}^0(pt) \cong \mathbb{Z}$  because the only bundles are trivial bundles, counted by their ranks.

2. K-theory of the Circle:

$$\widetilde{KO}^0(S^1) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$$

The  $\mathbb{Z}$  part corresponds to the rank of vector bundles, and  $\mathbb{Z}/2\mathbb{Z}$  accounts for the classification of line bundles (the only nontrivial line bundle over  $S^1$  being the Möbius bundle).

3.  $\widetilde{KO}^0(\mathbb{RP}^n)$  varies depending on  $n$ . For instance:  $\widetilde{KO}^0(\mathbb{RP}^1) \cong \mathbb{Z}/2\mathbb{Z}$ ,  $\widetilde{KO}^0(\mathbb{RP}^2) \cong \mathbb{Z} \oplus$

$$\mathbb{Z}/2\mathbb{Z}, K(\mathbb{RP}^3) \cong \mathbb{Z}.$$

Recall that the Bott periodicity theorem states that  $\widetilde{KO}^s(X)$  are 8-periodic, i.e.,

$$\widetilde{KO}^{-s}(X) \cong \widetilde{KO}^{-s+8}(X) = [\Sigma^s X, BSO]$$

for  $s \in \mathbb{Z}$ . Therefore, in case of real vector bundles, it is enough to calculate  $\widetilde{KO}^s(X)$  for any eight consecutive integers  $s \in \mathbb{Z}$ .

## 2.5.2 Complex K-Theory

Complex K-theory, denoted  $\widetilde{K}^0$ , classifies complex vector bundles over a topological space  $X$ . This group can be identified with the homotopy classes of maps from  $X$  into the classifying space for stable complex vector bundles, denoted  $BU$ :

$$\widetilde{K}^0(X) \cong [X, BU].$$

**Example 2.5.2.** 1. For a point  $pt$ ,  $\widetilde{K}^0(pt) \cong \mathbb{Z}$ .

$$2. \widetilde{K}^0(S^1) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$$

3.  $\widetilde{K}^0(\mathbb{CP}^n)$  can be expressed as  $\mathbb{Z}[x]/(x^{n+1})$ , where  $x$  corresponds to the generator of the K-group corresponding to the hyperplane bundle.

Recall that the Bott periodicity theorem states that  $\widetilde{K}^s(X)$  are 2-periodic, i.e.,

$$\widetilde{K}^{-s}(X) \cong \widetilde{K}^{-s+2}(X) = [\Sigma^s X, BSU],$$

for  $s \in \mathbb{Z}$ . Therefore, in case of complex vector bundles, it is enough to calculate  $\widetilde{K}^s(X)$  for any two consecutive integers  $s \in \mathbb{Z}$ .



## 2.6 Localization of topological spaces

Localization of topological spaces is a crucial concept in both algebraic and geometric topology. It is often used to study spaces by simplifying certain aspects of their structure without losing essential information. The idea originates from the concept of localization in algebra, where one focuses on particular elements or properties of an object, filtering out others.

Let  $p$  be an arbitrary prime integer which will be fixed throughout the section and all localization will be made with respect to it. Consider a space  $X$  whose fundamental group is abelian. For the prime  $p$ , let  $X_{(p)}$  denote the space satisfying

$$\pi_*(X_{(p)}) \cong \pi_*(X)_{(p)}.$$

The space  $X_{(p)}$  is called a  $p$ -local space of  $X$  and the construction of such a space  $X_{(p)}$  is detailed in [52]. A map  $\ell : X \rightarrow X_{(p)}$  is said to be a localization of  $X$  if it is universal for maps of  $X$  in to spaces whose homotopy groups are  $\mathbb{Z}_{(p)}$ -module, i.e., given  $f : X \rightarrow Y$  for another space  $Y$  whose homotopy groups are  $\mathbb{Z}_{(p)}$ -module, there is a unique map  $f_\ell$  making the following diagram commutative:

$$\begin{array}{ccc} X & \xrightarrow{\ell} & X_{(p)} \\ & \searrow f & \swarrow f_\ell \\ & Y & \end{array}$$

A *simple space* is a connected topological space having the homotopy type of a CW complex and an abelian fundamental group which acts trivially on the homotopy and homology of the universal covering spaces.

Below theorem characterize local spaces and localization.

**Theorem 2.6.1** (Theorem 2.1 [52]). *Let  $\ell : X \rightarrow X'$  be a map of arbitrary simple spaces. Then the following are equivalent.*

1.  $\ell$  is a localization

2.  $\ell$  localizes integral homology

$$\begin{array}{ccc} \tilde{H}_*X & \xrightarrow{\ell} & \tilde{H}_*X' \\ & \searrow \otimes \mathbb{Z}_{(p)} & \swarrow \cong \\ & \tilde{H}_*X \otimes \mathbb{Z}_{(p)} & \end{array}$$

3.  $\ell$  localizes homotopy ( $* > 0$ )

$$\begin{array}{ccc} \pi_*X & \xrightarrow{\ell} & \pi_*X' \\ & \searrow \otimes \mathbb{Z}_{(p)} & \swarrow \cong \\ & \pi_*X \otimes \mathbb{Z}_{(p)} & \end{array}$$

**Remark 2.6.2.** *Any simply connected space has a localization.*

The crux of this theory is to understand the homotopy or homology of spaces. Localization allows one to focus on properties related to a specific prime number or family of primes, making computations more manageable. For example, instead of dealing with the full homotopy or homology groups, which may be complicated, one can study their behavior relative to a prime  $p$  or a set of primes, thereby simplifying the study. The construction and properties of certain local spaces are thoroughly discussed in [52].

One of the important property of the local spaces that would be used frequently in this thesis is the following.

**Remark 2.6.3.** *For a manifold  $X$  and an infinite loop space  $Y$ , we have*

$$[X, Y]_{(p)} = [X, Y] \otimes \mathbb{Z}_{(p)} = [X, Y_{(p)}].$$

**Convention:** In this thesis, manifolds are assumed to be smooth, closed, connected of dimension  $\geq 5$ , and oriented, and all maps are orientation-preserving unless otherwise stated.

# Part I



## Chapter 3

# Smooth Structures on PL-manifolds of Dimensions Between 8 and 10

In this chapter, we compute the set of smooth structures up to concordance on 8, 9, and 10-dimensional  $PL$ -manifolds. Recall that, the concordance structure set  $\mathcal{C}(M)$  for any smooth manifold  $M$  corresponds to the set  $[M, PL/O]$  (refer to Chapter 2), and our central objective is to explicitly compute the latter set.

As the space  $PL/O$  is 6-connected, through obstruction theory one can establish that the group  $\mathcal{C}(M) \cong H^7(M; \Theta_7)$  for a closed, smooth 7-manifold  $M$ , where  $\Theta_7 \cong \mathbb{Z}_{28}$  is the group of homotopy 7-spheres. Now, we extend this result to manifolds of dimensions 8, 9 and 10, utilizing the structure of the 10th Postnikov section of  $PL/O$ .

The underlying idea is to use the cohomology operations to gain sufficient knowledge of the cell attachments of  $M$  in degrees 7 to 10, leading to the computation of  $[M, PL/O]$  through the initial stages of the Postnikov tower of  $PL/O$ .

### 3.1 Smooth structures on 8-manifolds

Let us begin with the construction of the 8<sup>th</sup> Postnikov section of  $PL/O$ , denoted by  $\tau_{\leq 8} PL/O$ , and compute  $[M, PL/O]$  for 8-dimensional manifolds. This deduces the following theorem. Theorem 3.1.1 also follows from the computations done by Jahren in [21, Page 41]; however, here we independently confirm this result through a direct calculation.

**Theorem 3.1.1.** *Let  $M^8$  be a closed, smooth manifold. Then*

$$[M^8, PL/O] \cong H^7(M^8; \mathbb{Z}_{28}) \oplus H^8(M^8; \mathbb{Z}_2)$$

*Proof.* We prove that the 8<sup>th</sup> Postnikov section of  $PL/O$  splits as a product  $K(\mathbb{Z}_{28}, 7) \times K(\mathbb{Z}_2, 8)$ , implying the required isomorphism. It suffices to prove splitting in a  $p$ -local category for every prime  $p$ . As the  $k$ -invariants lie in the stable range, we work stably using Eilenberg-MacLane spectra, denoted by lowercase letters, rather than their underlying spaces. For instance, for the space  $PL/O$ , we use its associated spectra denoted by  $pl/o$ . Note that, the homotopy groups of  $pl/o$  in degrees at most 8 have  $p$ -torsion only for  $p = 2$  and 7. For the case  $p = 7$ , the homotopy group is non-zero only in degree 7. Therefore, it suffices to work 2-locally.

The stable 8<sup>th</sup> Postnikov section of  $pl/o$  is the fiber of a map  $\Sigma^7 H\mathbb{Z}_4 \rightarrow \Sigma^9 H\mathbb{Z}_2$ . Up to homotopy, this map is either 0 or  $(\Sigma^7 Sq^2) \circ q$ , that is

$$\Sigma^7 H\mathbb{Z}_4 \xrightarrow{q} \Sigma^7 H\mathbb{Z}_2 \xrightarrow{\Sigma^7 Sq^2} \Sigma^9 H\mathbb{Z}_2.$$

This can be readily seen from the following argument. Consider the following long exact sequence induced from the cofibration  $\Sigma^7 H\mathbb{Z}_2 \rightarrow \Sigma^7 H\mathbb{Z}_4 \xrightarrow{q} \Sigma^7 H\mathbb{Z}_2$  of spectra

$$\{\Sigma^7 H\mathbb{Z}_2, \Sigma^9 H\mathbb{Z}_2\} \xrightarrow{q^*} \{\Sigma^7 H\mathbb{Z}_4, \Sigma^9 H\mathbb{Z}_2\} \rightarrow \{\Sigma^7 H\mathbb{Z}_2, \Sigma^9 H\mathbb{Z}_2\} \xrightarrow{\beta^*} \{\Sigma^6 H\mathbb{Z}_2, \Sigma^9 H\mathbb{Z}_2\},$$

where  $\beta = Sq^1$ , and  $\{\Sigma^7 H\mathbb{Z}_2, \Sigma^9 H\mathbb{Z}_2\} \cong \{0, \Sigma^7 Sq^2\}$  implies any  $\psi \in \{\Sigma^7 H\mathbb{Z}_2, \Sigma^9 H\mathbb{Z}_2\}$  is equal to  $\Sigma^7 Sq^2$  or 0. Since  $Sq^2 \circ \beta \neq 0$ , the map  $q^*$  is surjective. This implies an existence of  $\zeta$  in the

below diagram.

$$\begin{array}{ccccccc}
\Sigma^6 H\mathbb{Z}_2 & \xrightarrow{\beta} & \Sigma^7 H\mathbb{Z}_2 & \longrightarrow & \Sigma^7 H\mathbb{Z}_4 & \xrightarrow{q} & \Sigma^7 H\mathbb{Z}_2 \\
& & \searrow \psi & & \downarrow \phi & \swarrow \zeta & \\
& & & & \Sigma^9 H\mathbb{Z}_2 & & 
\end{array}$$

We intend to show that  $(\Sigma^7 Sq^2) \circ q$  does not occur as the  $k$ -invariant. The idea is to exhibit a map  $\Sigma^7 H\mathbb{Z}_4 \rightarrow \tau_{\leq 8} pl/o_{(2)}$  such that its composition with the fibration map  $b : \tau_{\leq 8} pl/o_{(2)} \rightarrow \Sigma^7 H\mathbb{Z}_4$  is an equivalence. This implies that the  $k$ -invariant is 0.

Consider the natural fibration map  $\Sigma^{-1}g/pl_{(2)} \rightarrow pl/o_{(2)}$  and restrict it to the 6-connected cover  ${}^{>6}\tau\Sigma^{-1}g/pl_{(2)}$  of  $\Sigma^{-1}g/pl_{(2)}$ . Since  $\pi_7({}^{>6}\tau\Sigma^{-1}g/pl) \cong \mathbb{Z}$  and  $\pi_8({}^{>6}\tau\Sigma^{-1}g/pl) = 0$ , we have

$${}^{>6}\tau_{\leq 8}(\Sigma^{-1}g/pl_{(2)}) \cong \Sigma^7 H\mathbb{Z}_{(2)} \xrightarrow{g} \tau_{\leq 8} pl/o_{(2)}.$$

Consequently, we have the following diagram

$$\begin{array}{ccccc}
\Sigma^7 H\mathbb{Z}_{(2)} & \xrightarrow{\times 4} & \Sigma^7 H\mathbb{Z}_{(2)} & \xrightarrow{p} & \Sigma^7 H\mathbb{Z}_4 \\
& & \downarrow g & & \downarrow \\
\Sigma^8 H\mathbb{Z}_2 & \longrightarrow & \tau_{\leq 8} pl/o_{(2)} & \xrightarrow{b} & \Sigma^7 H\mathbb{Z}_4
\end{array}$$

where  $p$  is the natural projection from  $\mathbb{Z}_{(2)}$  onto  $\mathbb{Z}_4$ . The right square commutes up to homotopy because maps of degree zero between Eilenberg-MacLane spectra are determined by the induced group homomorphism on homotopy groups. Observe that the composition  $b \circ g \circ (\times 4)$  is 0, since the composition  $p \circ (\times 4)$  is 0. Therefore we get a homotopy commutative diagram

$$\begin{array}{ccccc}
\Sigma^7 H\mathbb{Z}_{(2)} & \xrightarrow{\times 4} & \Sigma^7 H\mathbb{Z}_{(2)} & \xrightarrow{p} & \Sigma^7 H\mathbb{Z}_4 \\
\downarrow \cong & & \downarrow g & & \downarrow \cong \\
\Sigma^8 H\mathbb{Z}_2 & \longrightarrow & \tau_{\leq 8} pl/o_{(2)} & \xrightarrow{b} & \Sigma^7 H\mathbb{Z}_4
\end{array}$$

Since every map from  $\Sigma^7 H\mathbb{Z}_{(2)} \rightarrow \Sigma^8 H\mathbb{Z}_2$  is null homotopic, we have the following dia-

gram

$$\begin{array}{ccccc}
 \Sigma^7 H\mathbb{Z}_{(2)} & \xrightarrow{\times 4} & \Sigma^7 H\mathbb{Z}_{(2)} & \xrightarrow{p} & \Sigma^7 H\mathbb{Z}_4 \\
 & \searrow 0 & \downarrow g & \swarrow \tilde{b} & \\
 & & \tau_{\leq 8} pl/o_{(2)} & & 
 \end{array}$$

where  $g \circ (\times 4) = 0$ . The existence of a map  $\tilde{b}$  having the desired property can be verified by considering the following induced exact sequence

$$\{\Sigma^7 H\mathbb{Z}_4, \tau_{\leq 8} pl/o\}_{(2)} \xrightarrow{p^*} \{\Sigma^7 H\mathbb{Z}_{(2)}, \tau_{\leq 8} pl/o\}_{(2)} \xrightarrow{4^{\times*}} \{\Sigma^7 H\mathbb{Z}_{(2)}, \tau_{\leq 8} pl/o\}_{(2)}$$

Thus, the map  $\tilde{b}$  is a homotopy section for the map  $b$ , implying the following decomposition

$$\tau_{\leq 8} pl/o_{(2)} \simeq \Sigma^7 H\mathbb{Z}_4 \vee \Sigma^8 H\mathbb{Z}_2. \quad (3.1)$$

This completes the proof. □

Theorem 3.1.1 shows how a decomposition result for the 8<sup>th</sup> Postnikov section facilitates the computation of  $\mathcal{C}(M) = [M, PL/O]$  for 8-manifold  $M$ .

## 3.2 Smooth structures on 9-manifolds

We now recall from [21, Page 41, 42] the formula for  ${}^{>6}\tau_{\leq 9} pl/o$ . Note that,  $\text{Ext}(\mathbb{Z}_4, \mathbb{Z}_2) \cong \mathbb{Z}_2$  along with the universal coefficient theorem gives  $H^{n+1}(K(\mathbb{Z}_4, n); \mathbb{Z}_2) \cong \mathbb{Z}_2$ . Let us fix the notation  $d_2$  for the corresponding stable map  $H\mathbb{Z}_4 \rightarrow \Sigma H\mathbb{Z}_2$ , and define

$$\begin{aligned}
 \mathcal{F} &:= \text{Fibre}(H\mathbb{Z}_2 \xrightarrow{Sq^2} \Sigma^2 H\mathbb{Z}_2) \\
 \mathcal{F}_2 &:= \text{Fibre}(H\mathbb{Z}_4 \xrightarrow{Sq^2 \circ d_2} \Sigma^3 H\mathbb{Z}_2).
 \end{aligned} \quad (3.2)$$



With this, the 9<sup>th</sup> Postnikov section of  $pl/o$  (detailed description is given in [21, Page 42]) is

$$\tau_{\leq 9} pl/o \simeq \Sigma^8 \mathcal{F} \vee \Sigma^7 \mathcal{F}_2 \vee \Sigma^7 H\mathbb{Z}_7 \vee \Sigma^9 H\mathbb{Z}_2. \quad (3.3)$$

Let  $M^9$  be a closed, oriented, smooth manifold. Then the minimal cell structure [17, §4.C] on  $M^9/M^{(6)}$  is of the following form

$$M^9/M^{(6)} \simeq \left( \bigvee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \vee (\mathbb{S}^7)^{\vee_l} \vee (\mathbb{S}^8)^{\vee_k} \right) \bigcup e^9,$$

where  $M(\mathbb{Z}_{p^r}, n)$  stands for the Moore space for the group  $\mathbb{Z}_{p^r}$  in degree  $n$ , and  $J$  is some finite indexing set. In this case, the attaching map of the 9-cell onto 8-cells is null homotopic. Therefore,

$$M^9/M^{(6)} \simeq \left( \left( \bigvee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \vee (\mathbb{S}^7)^{\vee_l} \right) \bigcup_f e^9 \right) \vee (\mathbb{S}^8)^{\vee_k}, \quad (3.4)$$

and hence the attaching map  $f$  lies in  $\pi_8 \left( \bigvee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \vee (\mathbb{S}^7)^{\vee_l} \right)$ . By the connectivity argument,

$$\pi_8 \left( \bigvee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \vee (\mathbb{S}^7)^{\vee_l} \right) \cong \left( \bigoplus_{(p,r) \in J} \pi_8(M(\mathbb{Z}_{p^r}, 7)) \right) \oplus \left( \bigoplus_l \pi_8(\mathbb{S}^7) \right)$$

with  $\pi_8(\mathbb{S}^7) \cong \mathbb{Z}_2\{\eta\}$  and

$$\pi_8(M(\mathbb{Z}_{p^r}, 7)) \cong \begin{cases} 0 & \text{if } p \text{ is odd,} \\ \mathbb{Z}_2\{\iota \circ \eta\} & \text{if } p = 2, \end{cases}$$

where  $\iota \circ \eta$  is the composite  $S^8 \xrightarrow{\eta} S^7 \xrightarrow{\iota} M(\mathbb{Z}_2^r, 7)$ .

Consider the Steenrod square operation  $Sq^2 : H^7(M^9; \mathbb{Z}_2) \rightarrow H^9(M^9; \mathbb{Z}_2)$ , and for each  $r \geq 1$ , there is a higher order Bockstein operation  $\beta_r : H^*(M^9; \mathbb{Z}_2) \rightarrow H^{*+1}(M^9; \mathbb{Z}_2)$ . Now, depending on the attaching map we have following possibilities (for more details see [35]):

1. If  $M^9$  is a spin manifold then the attaching map  $\mathbb{S}^8 \rightarrow \bigvee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \vee (\mathbb{S}^7)^{\vee_l}$  is null

homotopic, since  $Sq^2 : H^7(M^9; \mathbb{Z}_2) \rightarrow H^9(M^9; \mathbb{Z}_2)$  is zero. Thus

$$M^9 / M^{(6)} \simeq (\mathbb{S}^7)^{\vee_l} \vee (\mathbb{S}^8)^{\vee_k} \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \vee \mathbb{S}^9, \quad (3.5)$$

2. If  $M^9$  is a non-spin manifold then  $Sq^2 : H^7(M^9; \mathbb{Z}_2) \rightarrow H^9(M^9; \mathbb{Z}_2)$  is non-zero. In addition,

- (a) Suppose that for any  $u \in H^7(M^9; \mathbb{Z}_2)$  with  $Sq^2(u) \neq 0$  and any  $v \in \text{Ker}(Sq^2)$ , we have  $\beta_r(u+v) = 0$  and  $u+v \notin \text{Im}(\beta_s)$ ,  $\forall r, s \geq 1$ . Then, the non-trivial factor of the attaching map  $f$  in (3.4) is  $\eta$ , which implies

$$M^9 / M^{(6)} \simeq C(\eta) \vee M', \quad (3.6)$$

where  $M' \simeq (\mathbb{S}^7)^{\vee_{l-1}} \vee (\mathbb{S}^8)^{\vee_k} \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7)$ .

- (b) Suppose that for any  $u \in H^7(M^9; \mathbb{Z}_2)$  with  $Sq^2(u) \neq 0$  and any  $v \in \text{Ker}(Sq^2)$ , we have  $u+v \notin \text{Im}(\beta_s)$ ,  $\forall s \geq 1$ , while there exist  $u' \in H^7(M^9; \mathbb{Z}_2)$  with  $Sq^2(u') \neq 0$  and  $v' \in \text{Ker}(Sq^2)$  such that  $\beta_r(u'+v') \neq 0$  for some  $r \geq 1$ . Then the only non-trivial factor of the attaching map  $f$  in (3.4) is  $\iota \circ \eta$ , and

$$M^9 / M^{(6)} \simeq C(\iota \circ \eta) \vee M'' \quad (3.7)$$

where  $M'' \simeq (\mathbb{S}^7)^{\vee_l} \vee (\mathbb{S}^8)^{\vee_k} \vee_{(p,r) \in J'} M(\mathbb{Z}_{p^r}, 7)$ .

The following theorem applies the splitting of the Postnikov section (3.3) to the case of 9-manifolds.

**Theorem 3.2.1.** *Let  $M^9$  be a closed, oriented smooth manifold and let  $Sq^2 \circ d_2 : H^6(M^9; \mathbb{Z}_4) \rightarrow H^9(M^9; \mathbb{Z}_2)$ .*

1. *If  $M^9$  is a spin manifold then*

$$[M^9, PL/O] \cong H^7(M^9; \mathbb{Z}_{28}) \oplus H^8(M^9; \mathbb{Z}_2) \oplus H^9(M^9; \mathbb{Z}_2^3).$$

2. *Let  $M^9$  be a non-spin manifold.*

(a) If  $Sq^2 \circ d_2$  is non-trivial then

$$[M^9, PL/O] \cong H^7(M^9; \mathbb{Z}_{28}) \oplus H^8(M^9; \mathbb{Z}_2) \oplus H^9(M^9; \mathbb{Z}_2).$$

(b) If  $Sq^2 \circ d_2$  is trivial then

$$[M^9, PL/O] \cong H^7(M^9; \mathbb{Z}_7) \oplus H^8(M^9; \mathbb{Z}_2) \oplus H^9(M^9; \mathbb{Z}_2) \oplus [M^9, \Sigma^7 \mathcal{F}_2],$$

where  $[M^9, \Sigma^7 \mathcal{F}_2] \cong K \oplus \tilde{A}$ , with  $K$  is a part of the short exact sequence

$$0 \rightarrow \text{Ker}(Sq^2) \rightarrow K \rightarrow \text{Ker}(Sq^2 \oplus Sq^1) \rightarrow 0,$$

$Sq^1 : H^7(M^9; \mathbb{Z}_2) \rightarrow H^8(M^9; \mathbb{Z}_2)$  and  $Sq^2 : H^7(M^9; \mathbb{Z}_2) \rightarrow H^9(M^9; \mathbb{Z}_2)$ , and  $\tilde{A}$  is the non-trivial extension satisfying the short exact sequence

$$0 \rightarrow \mathbb{Z}_2 \rightarrow \tilde{A} \rightarrow A \rightarrow 0,$$

where  $A = \mathbb{Z}_4$  or  $\mathbb{Z}_2$ .

*Proof.* Using (3.3), we have  $[M^9, PL/O] \cong [M^9, \tau_{\leq 9} pl/o] \cong H^7(M^9; \mathbb{Z}_7) \oplus H^9(M^9; \mathbb{Z}_2) \oplus [M^9, \Sigma^8 \mathcal{F}] \oplus [M^9, \Sigma^7 \mathcal{F}_2]$ . Thus, it is enough to compute  $[M^9, \Sigma^7 \mathcal{F}_2]$  and  $[M^9, \Sigma^8 \mathcal{F}]$ .

Note that,  $\Sigma^8 \mathcal{F}$  is 7-connected, thus

$$[M^9, \Sigma^8 \mathcal{F}] \cong [M^9 / M^{(6)}, \Sigma^8 \mathcal{F}]. \quad (3.8)$$

The space  $\Sigma^7 \mathcal{F}_2$  is 6-connected and the group  $[M^9, \Sigma^7 \mathcal{F}_2]$  can be computed using the following commutative diagram:

$$\begin{array}{ccccccc} [M^9, \Sigma^6 H\mathbb{Z}_4] & \xrightarrow{Sq^2 \circ d_2} & [M^9, \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [M^9, \Sigma^7 \mathcal{F}_2] & \longrightarrow & [M^9, \Sigma^7 H\mathbb{Z}_4] \\ \uparrow & & \uparrow \cong & & \uparrow & & \uparrow \cong \\ 0 & \longrightarrow & [M^9 / M^{(5)}, \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [M^9 / M^{(5)}, \Sigma^7 \mathcal{F}_2] & \twoheadrightarrow & [M^9 / M^{(5)}, \Sigma^7 H\mathbb{Z}_4] \end{array} \quad (3.9)$$

1. Let  $M^9$  be a spin manifold. Then (3.5) and (3.8) together gives

$$[M^9, \Sigma^8 \mathcal{F}] \cong [M^9 / M^{(6)}, \Sigma^8 \mathcal{F}] \cong H^8(M^9 / M^{(6)}; \mathbb{Z}_2) \oplus H^9(M^9 / M^{(6)}; \mathbb{Z}_2).$$

For  $[M^9, \Sigma^7 \mathcal{F}_2]$ , since  $Sq^2 = 0$ , using (3.9) we get the following commutative diagram of short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & [M^9, \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [M^9, \Sigma^7 \mathcal{F}_2] & \longrightarrow & [M^9, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0 \\ & & \uparrow \cong & & \uparrow & & \uparrow \cong \\ 0 & \longrightarrow & [M^9 / M^{(5)}, \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [M^9 / M^{(5)}, \Sigma^7 \mathcal{F}_2] & \longrightarrow & [M^9 / M^{(5)}, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0. \end{array} \quad (3.10)$$

This implies

$$[M^9, \Sigma^7 \mathcal{F}_2] \cong [M^9 / M^{(5)}, \Sigma^7 \mathcal{F}_2].$$

In (3.10), we demonstrate the splitting of the short exact sequence in the second row, thereby implying the splitting of the sequence in the first row. For that purpose, consider the cofiber sequence  $M^9 / M^{(5)} \rightarrow M^9 / M^{(6)} \xrightarrow{\Psi} \Sigma(M^{(6)} / M^{(5)})$ , and the following commutative diagram induced from it

$$\begin{array}{ccccccc} 0 & \longrightarrow & [M^9 / M^{(5)}, \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [M^9 / M^{(5)}, \Sigma^7 \mathcal{F}_2] & \longrightarrow & [M^9 / M^{(5)}, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0 \\ & & \uparrow \cong & & \uparrow & & \uparrow \\ 0 & \longrightarrow & [M^9 / M^{(6)}, \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [M^9 / M^{(6)}, \Sigma^7 \mathcal{F}_2] & \longrightarrow & [M^9 / M^{(6)}, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0 \\ & & \uparrow \phi^* & & \uparrow \Psi^* & & \uparrow \gamma^* \\ 0 & \rightarrow & [\Sigma(M^{(6)} / M^{(5)}), \Sigma^9 H\mathbb{Z}_2] & \rightarrow & [\Sigma(M^{(6)} / M^{(5)}), \Sigma^7 \mathcal{F}_2] & \rightarrow & [\Sigma(M^{(6)} / M^{(5)}), \Sigma^7 H\mathbb{Z}_4] \rightarrow 0. \end{array} \quad (3.11)$$

Observe that due to (3.5), the map  $\Psi : M^9 / M^{(6)} \rightarrow \Sigma(M^{(6)} / M^{(5)}) \simeq \vee_i \mathbb{S}^7$  decomposes in  $\gamma : \vee_l \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \rightarrow \vee_i \mathbb{S}^7$  and  $\phi : \mathbb{S}^9 \rightarrow \vee_i \mathbb{S}^7$ . Up to homotopy, the map  $\phi$  is either 0 or  $\eta^2$  ( $\pi_2^s \cong \mathbb{Z}_2\{\eta^2\}$ ). If it is  $\eta^2 = \eta \circ \eta$ , then the map  $\phi^*$  is 0 due to the following.

$$\begin{array}{ccc}
[\mathbb{S}^7, \Sigma^7 \mathcal{F}_2] & \xrightarrow{(\eta^2)^*} & [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \\
\eta^* \searrow & & \nearrow \eta^* \\
& [\mathbb{S}^8, \Sigma^7 \mathcal{F}_2] = 0 &
\end{array}$$

Thus, we obtain  $\text{Im}(\Psi^*) = \text{Im}(\gamma^*)$ . This shows that in (3.11), the exact sequence in the second row induces a split exact sequence at  $[M^9/M^{(6)}, \Sigma^7 \mathcal{F}_2] / \text{Im}(\Psi^*)$ . Therefore, we get

$$[M^9, \Sigma^7 \mathcal{F}_2] \cong H^7(M; \mathbb{Z}_4) \oplus H^9(M; \mathbb{Z}_2). \quad (3.12)$$

As a result, we obtain the following in the spin case,

$$[M^9, PL/O] \cong H^7(M^9; \mathbb{Z}_{28}) \oplus H^8(M^9; \mathbb{Z}_2) \oplus H^9(M^9; \mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2).$$

2. Let  $M^9$  be a non-spin manifold. Using (3.8), we have  $[M^9, \Sigma^8 \mathcal{F}] \cong [M^9/M^{(6)}, \Sigma^8 \mathcal{F}]$ , and by (3.6) and (3.7),

$$[M^9/M^{(6)}, \Sigma^8 \mathcal{F}] \cong [C(\eta), \Sigma^8 \mathcal{F}] \oplus [M', \Sigma^8 \mathcal{F}] \text{ or } [C(\iota \circ \eta), \Sigma^8 \mathcal{F}] \oplus [M'', \Sigma^8 \mathcal{F}]. \quad (3.13)$$

Further,  $[\mathbb{S}^8, \Sigma^8 \mathcal{F}] \xrightarrow{\cong} [M(\mathbb{Z}_{2^r}, 7), \Sigma^8 \mathcal{F}] \cong \mathbb{Z}_2$ , and  $[M(\mathbb{Z}_{p^r}, 7), \Sigma^8 \mathcal{F}] = 0$  for all odd prime  $p$ . Hence

$$[M', \Sigma^8 \mathcal{F}] \cong H^8(M'; \mathbb{Z}_2) \text{ and } [M'', \Sigma^8 \mathcal{F}] \cong H^8(M''; \mathbb{Z}_2). \quad (3.14)$$

It remains to compute  $[C(\eta), \Sigma^8 \mathcal{F}]$  and  $[C(\iota \circ \eta), \Sigma^8 \mathcal{F}]$ .

The computation of  $[C(\eta), \Sigma^8 \mathcal{F}]$  follows easily from the following exact sequence

$$\dots [\mathbb{S}^8, \Sigma^8 \mathcal{F}] \xrightarrow{\cong} [\mathbb{S}^9, \Sigma^8 \mathcal{F}] \rightarrow [C(\eta), \Sigma^8 \mathcal{F}] \rightarrow [\mathbb{S}^7, \Sigma^8 \mathcal{F}] = 0.$$

Therefore, we get

$$[C(\eta), \Sigma^8 \mathcal{F}] = 0 \quad (3.15)$$

For the computation of  $[C(\iota \circ \eta), \Sigma^8 \mathcal{F}]$ , consider the following commutative diagram obtained from the cofiber sequence  $\mathbb{S}^8 \xrightarrow{\iota \circ \eta} M(\mathbb{Z}_2^r, 7) \rightarrow C(\iota \circ \eta)$ ,

$$\begin{array}{ccccccc}
 & & [\mathbb{S}^8, \Sigma^8 \mathcal{F}] & & & & \\
 & \nearrow \Sigma i^* & \downarrow \cong & & & & \\
 [\Sigma M(\mathbb{Z}_2^r, 7), \Sigma^8 \mathcal{F}] & \rightarrow & [\mathbb{S}^9, \Sigma^8 \mathcal{F}] & \rightarrow & [C(\iota \circ \eta), \Sigma^8 \mathcal{F}] & \rightarrow & [M(\mathbb{Z}_2^r, 7), \Sigma^8 \mathcal{F}] \xrightarrow{(\iota \circ \eta)^*} [\mathbb{S}^8, \Sigma^8 \mathcal{F}] \\
 & & & & & \uparrow d^* \cong & \nearrow \times 2 \\
 & & & & & [\mathbb{S}^8, \Sigma^8 \mathcal{F}] & 
 \end{array} \tag{3.16}$$

where the map  $\Sigma i^*$  is a part of the exact sequence obtained from the cofiber sequence  $\mathbb{S}^7 \xrightarrow{2^r} \mathbb{S}^7 \xrightarrow{i} M(\mathbb{Z}_2^r, 7)$ . Also, the map  $(\iota \circ \eta)^*$  is trivial which implies that  $[C(\iota \circ \eta), \Sigma^8 \mathcal{F}] \cong [M(\mathbb{Z}_2^r, 7), \Sigma^8 \mathcal{F}] \cong \mathbb{Z}_2$ . In conclusion, we get

$$[C(\eta), \Sigma^8 \mathcal{F}] \cong H^8(C(\eta); \mathbb{Z}_2) \text{ and } [C(\iota \circ \eta), \Sigma^8 \mathcal{F}] \cong H^8(C(\iota \circ \eta); \mathbb{Z}_2). \tag{3.17}$$

Therefore, combining (3.14) and (3.17) we obtain

$$[M, \Sigma^8 \mathcal{F}] \cong H^8(M; \mathbb{Z}_2). \tag{3.18}$$

Finally, let us compute  $[M^9, \Sigma^7 \mathcal{F}_2]$ , by taking into account two cases depending on the nature of the attaching map: whether the map  $Sq^2 \circ d_2 : H^6(M; \mathbb{Z}_4) \rightarrow H^9(M; \mathbb{Z}_2)$  is non-trivial or trivial.

In the case when  $Sq^2 \circ d_2$  is non-trivial, it is clear from the diagram (3.9) that

$$[M^9, \Sigma^7 \mathcal{F}_2] \cong [M^9, \Sigma^7 H\mathbb{Z}_4]. \tag{3.19}$$

Now, for the case  $Sq^2 \circ d_2 = 0$ , we need  $[C(\eta), \Sigma^7 \mathcal{F}_2]$  and  $[C(\iota \circ \eta), \Sigma^7 \mathcal{F}_2]$ . So let us first calculate these groups.

We have  $\eta : \mathbb{S}^8 \rightarrow \mathbb{S}^7$ ,  $\mathcal{F}_2 = \text{Fiber}(H\mathbb{Z}_4 \xrightarrow{Sq^2 \circ d_2} \Sigma^3 H\mathbb{Z}_2)$ , and the following commutative

square

$$\begin{array}{ccc}
 H\mathbb{Z}_2 & \xrightarrow{Sq^2 \circ Sq^1} & \Sigma^3 H\mathbb{Z}_2 \\
 \downarrow & & \downarrow = \\
 H\mathbb{Z}_4 & \xrightarrow{Sq^2 \circ d_2} & \Sigma^3 H\mathbb{Z}_2
 \end{array} \tag{3.20}$$

Let  $\mathcal{D} = \text{Fiber}(H\mathbb{Z}_2 \xrightarrow{Sq^2 \circ Sq^1} \Sigma^3 H\mathbb{Z}_2)$ . We have the following commutative diagram

$$\begin{array}{ccccc}
 \Sigma^7 H\mathbb{Z}_4 & \longrightarrow & \Sigma^7 \mathcal{D} & \longrightarrow & \Sigma^8 \mathcal{F} \\
 \downarrow = & & \downarrow & & \downarrow \\
 \Sigma^7 H\mathbb{Z}_4 & \longrightarrow & \Sigma^7 H\mathbb{Z}_2 & \xrightarrow{\Sigma^7 Sq^1} & \Sigma^8 H\mathbb{Z}_2 \\
 \downarrow & & \downarrow \Sigma^7(Sq^2 \circ Sq^1) & & \downarrow \Sigma^7 Sq^2 \\
 0 & \longrightarrow & \Sigma^{10} H\mathbb{Z}_2 & \xrightarrow{=} & \Sigma^{10} H\mathbb{Z}_2
 \end{array}$$

in which the rows and columns are cofiber sequences.

Note that  $[C(\eta), \Sigma^7 \mathcal{F}] = 0 = [C(\eta), \Sigma^8 \mathcal{F}]$  implies

$$[C(\eta), \Sigma^7 \mathcal{D}] \cong [C(\eta), \Sigma^7 H\mathbb{Z}_4] \cong \mathbb{Z}_4.$$

Now, using (3.20) we have the following commutative diagram of short exact sequences, wherein observe that if the bottom row splits, then so does the top row.

$$\begin{array}{ccccccc}
 0 & \longrightarrow & [C(\eta), \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [C(\eta), \Sigma^7 \mathcal{D}] & \longrightarrow & [C(\eta), \Sigma^7 H\mathbb{Z}_2] \longrightarrow 0 \\
 & & \downarrow = & & \downarrow & & \downarrow \\
 0 & \longrightarrow & [C(\eta), \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [C(\eta), \Sigma^7 \mathcal{F}_2] & \longrightarrow & [C(\eta), \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0
 \end{array}$$

Therefore,

$$[C(\eta), \Sigma^7 \mathcal{F}_2] \cong \mathbb{Z}_8. \tag{3.21}$$

To compute  $[C(\iota \circ \eta), \Sigma^7 \mathcal{F}_2]$ , consider the following commutative diagram

$$\begin{array}{ccccccc}
& & \mathbb{Z}_2 & & & & \\
& & \wr & & & & \\
0 & \longrightarrow & [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] & \longrightarrow & [C(\iota \circ \eta), \Sigma^7 \mathcal{F}_2] & \longrightarrow & [M(\mathbb{Z}_2^r, 7), \Sigma^7 \mathcal{F}_2] \longrightarrow 0 \\
& & \downarrow = & & \downarrow & & \downarrow \\
0 & \longrightarrow & [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] & \longrightarrow & [C(\eta), \Sigma^7 \mathcal{F}_2] & \longrightarrow & [\mathbb{S}^7, \Sigma^7 \mathcal{F}_2] \longrightarrow 0
\end{array} \tag{3.22}$$

where the rows are exact sequences induced from the cofiber sequences of  $C(\iota \circ \eta)$  and  $C(\eta)$ . To compute  $[M(\mathbb{Z}_2^r, 7), \Sigma^7 \mathcal{F}_2]$ , we use the fiber sequence (3.2) of  $\mathcal{F}_2$  that gives

$$[M(\mathbb{Z}_2^r, 7), \Sigma^7 \mathcal{F}_2] \cong H^7(M(\mathbb{Z}_2^r, 7); \mathbb{Z}_4),$$

and further, compute the cohomology using cofiber sequence  $\mathbb{S}^7 \xrightarrow{\times 2^r} \mathbb{S}^7 \rightarrow M(\mathbb{Z}_2^r, 7)$ , that gives

$$[M(\mathbb{Z}_2^r, 7), \Sigma^7 \mathcal{F}_2] \cong \begin{cases} \mathbb{Z}_4 & \text{if } r = 1, \\ \mathbb{Z}_2 & \text{if } r > 1. \end{cases} \tag{3.23}$$

A straightforward diagram chasing in (3.22) along with (3.21) shows the non-splitting of the short exact sequence at  $[C(\iota \circ \eta), \Sigma^7 \mathcal{F}_2]$ . Furthermore,

$$[C(\iota \circ \eta^2), \Sigma^7 \mathcal{F}_2] \cong \begin{cases} \mathbb{Z}_8 & \text{if } r = 1, \\ \mathbb{Z}_4 & \text{if } r > 1. \end{cases} \tag{3.24}$$

Now, consider the case when  $Sq^2 \circ d_2$  is trivial. This implies  $Sq^2 Sq^1 : H^6(M; \mathbb{Z}_2) \rightarrow H^9(M; \mathbb{Z}_2)$  is 0. Therefore

$$Sq^2 : \text{Coker}(Sq^1 : H^6(M; \mathbb{Z}_2) \rightarrow H^7(M; \mathbb{Z}_2)) \rightarrow H^9(M; \mathbb{Z}_2)$$



is well-defined and non-zero. Therefore, we have

$$\text{Coker}(Sq^1) = \text{Ker}(Sq^2) \oplus \mathbb{Z}_2.$$

Let  $q : H^*(M; \mathbb{Z}_4) \rightarrow H^*(M; \mathbb{Z}_2)$  be the map induced by the non-trivial morphism  $\mathbb{Z}_4 \rightarrow \mathbb{Z}_2$ .

Note that  $\text{Im}(H^7(\widehat{M}; \mathbb{Z}_4) \rightarrow H^7(M; \mathbb{Z}_4)) \subseteq \text{Ker}(Sq^2 \circ q)$ , where  $\widehat{M} = M'$  or  $M''$  as mentioned in (3.6) and (3.7). Then, using the exact sequence

$$0 \rightarrow \text{Coker}(Sq^1) \rightarrow H^7(M; \mathbb{Z}_4) \xrightarrow{q} \text{Ker}(Sq^1 : H^7(M; \mathbb{Z}_2) \rightarrow H^8(M; \mathbb{Z}_2)) \rightarrow 0,$$

observe that  $H^7(M; \mathbb{Z}_4) = K \oplus A$  where  $A = \mathbb{Z}_2$  or  $\mathbb{Z}_4$ , and  $K$  fits into the following possible non-trivial extension which is determined from the structure of  $H^7(M; \mathbb{Z}_4)$

$$0 \rightarrow \text{Ker}(Sq^2) \rightarrow K \rightarrow \text{Ker}(Sq^2 \oplus Sq^1) \rightarrow 0.$$

Additionally, note that  $K \subseteq H^7(M; \mathbb{Z}_4)$ , is in fact the image of  $H^7(\widehat{M}; \mathbb{Z}_4)$ . Thus, we have

$$[M, \Sigma^7 \mathcal{F}_2] \cong K \oplus \widetilde{A} \tag{3.25}$$

where  $\widetilde{A}$  is the non-trivial extension in

$$0 \rightarrow \mathbb{Z}_2 \rightarrow \widetilde{A} \rightarrow A \rightarrow 0.$$

This completes the proof. □

### 3.3 Smooth structures on 10-manifolds

We now consider at the analogue of Theorem 3.2.1 for closed, smooth manifold with  $H_1(M^{10}) = 0$ . To address this, we need information about the  $10^{th}$  Postnikov section  $\tau_{\leq 10} pl/o$  of  $pl/o$  computed in [21, Page 42]. In this context, observe that  $Sq^2 \circ Sq^2 \circ d_2 = 0$  leads to the construction

of a class  $\Phi : \mathcal{F}_2 \rightarrow \Sigma^4 H\mathbb{Z}_2$  using the following diagram

$$\begin{array}{ccc} \Sigma^{-1} H\mathbb{Z}_4 & \xrightarrow{Sq^2 \circ d_2} & \Sigma^2 H\mathbb{Z}_2 \longrightarrow \mathcal{F}_2 \\ & \downarrow Sq^2 & \searrow \Phi \\ & \Sigma^4 H\mathbb{Z}_2 & \end{array}$$

The operation  $\Phi$  defines a secondary cohomology operation from  $\text{Ker}(Sq^2 \circ d_2) (\subseteq H^i(M; \mathbb{Z}_4)) \rightarrow H^{i+4}(M; \mathbb{Z}_2)$ . Now, let

$$\mathcal{E} := \text{Fibre}(\mathcal{F}_2 \xrightarrow{\Phi} \Sigma^4 H\mathbb{Z}_2). \quad (3.26)$$

With this, the  $10^{th}$  Postnikov section of  $pl/o$  is given by

$$\tau_{\leq 10} pl/o \simeq \Sigma^8 \mathcal{F} \vee \Sigma^7 \mathcal{E} \vee \Sigma^7 H\mathbb{Z}_7 \vee \Sigma^9 H\mathbb{Z}_2 \vee \Sigma^{10} H\mathbb{Z}/3. \quad (3.27)$$

Let  $M^{10}$  be a closed, smooth manifold with  $H_1(M^{10}) = 0$ . Then there is a minimal cell structure [17, §4.C] on  $M^{10}/M^{(6)}$  of the form

$$M^{10}/M^{(6)} \simeq \left( \bigvee_l \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \right) \bigcup_f e^{10},$$

where the attaching map  $f$  lies in  $\pi_9 \left( \bigvee_l \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \right)$ . By the connectivity argument,

$$\pi_9 \left( \bigvee_l \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \right) \cong \oplus_l \pi_9(\mathbb{S}^7) \oplus_k \pi_9(\mathbb{S}^8) \oplus_{(p,r) \in J} \pi_9(M(\mathbb{Z}_{p^r}, 7)), \quad (3.28)$$

with  $\pi_9(\mathbb{S}^8) \cong \mathbb{Z}_2\{\eta\}$ ,  $\pi_9(\mathbb{S}^7) \cong \mathbb{Z}_2\{\eta^2\}$  and  $\pi_9(M(\mathbb{Z}_{p^r}, 7)) \cong \mathbb{Z}_2\{\iota \circ \eta^2\}$ .

If  $M^{10}$  is a spin manifold, then there exists a higher order cohomology operation  $\psi : H^6(M; \mathbb{Z}_4) \rightarrow H^{10}(M; \mathbb{Z}_2)$  corresponding to  $(Sq^2 \circ Sq^2) + (Sq^3 \circ Sq^1) = 0$  in order to detect the map  $\eta^2$  [41, Corollary 2, pg177]. Depending on either  $\psi$  is trivial or not, we have following possibilities:

1. If  $\psi$  is trivial, then

$$M^{10}/M^{(6)} \simeq M^{10}/M^{(6)} \simeq \vee_l \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \vee \mathbb{S}^{10}. \quad (3.29)$$

2. If  $\psi$  is non-trivial, then

$$M^{10}/M^{(6)} \simeq C(\eta^2) \vee M', \quad (3.30)$$

or

$$M^{10}/M^{(6)} \simeq C(\iota \circ \eta^2) \vee M'', \quad (3.31)$$

with  $M' \simeq \vee_{l-1} \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7)$  and  $M'' \simeq \vee_l \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J'} M(\mathbb{Z}_{p^r}, 7)$ .

The following theorem applies the splitting of the Postnikov section (3.27) to the case of 10-manifolds.

**Theorem 3.3.1.** *Let  $M^{10}$  be a closed smooth 10-manifold with  $H_1(M) = 0$ , and  $\Phi, \psi : H^6(M; \mathbb{Z}_4) \rightarrow H^{10}(M; \mathbb{Z}_2)$  be the secondary operations described in (3.26) and (3.29).*

1. *Let  $M^{10}$  be a spin manifold.*

(a) *If  $\Phi = 0$  then*

$$\begin{aligned} [M^{10}, PL/O] &\cong H^7(M^{10}; \mathbb{Z}_7) \oplus H^8(M^{10}; \mathbb{Z}_2) \oplus H^9(M^{10}; \mathbb{Z}_2) \oplus H^{10}(M^{10}; \mathbb{Z}_3) \\ &\oplus [M^{10}, \Sigma^7 \mathcal{E}]. \end{aligned}$$

*Furthermore, if the higher order cohomology operation  $\psi = 0$  then*

$$[M^{10}, \Sigma^7 \mathcal{E}] \cong H^{10}(M; \mathbb{Z}_2) \oplus H^7(M; \mathbb{Z}_4).$$

*On the other hand, if  $\psi \neq 0$ , then  $[M, \Sigma^7 \mathcal{E}] = K \oplus \tilde{A}$ , where  $K \subseteq H^7(M; \mathbb{Z}_4)$ , and  $\tilde{A}$  is the non-trivial extension satisfying the following sequence*

$$0 \rightarrow \mathbb{Z}_2 \rightarrow \tilde{A} \rightarrow A \rightarrow 0.$$

with  $A = \mathbb{Z}_4$  or  $\mathbb{Z}_2$ .

(b) If  $\Phi \neq 0$  then

$$[M^{10}, PL/O] \cong H^7(M^{10}; \mathbb{Z}_{28}) \oplus H^8(M^{10}; \mathbb{Z}_2) \oplus H^9(M^{10}; \mathbb{Z}_2) \oplus H^{10}(M^{10}; \mathbb{Z}_3).$$

2. If  $M^{10}$  is a non-spin manifold then

$$[M^{10}, PL/O] \cong H^7(M^{10}; \mathbb{Z}_7) \oplus H^{10}(M^{10}; \mathbb{Z}_3) \oplus \text{Ker}(Sq^2) \oplus \text{Ker}(Sq^2 \circ d_2),$$

where  $Sq^2 : H^8(M^{10}; \mathbb{Z}_2) \rightarrow H^{10}(M^{10}; \mathbb{Z}_2)$ , and  $Sq^2 \circ d_2 : H^7(M; \mathbb{Z}_4) \rightarrow H^{10}(M; \mathbb{Z}_2)$ .

*Proof.* Using the decomposition in (3.27), it is enough to compute  $[M^{10}, \Sigma^8 \mathcal{F}]$  and  $[M^{10}, \Sigma^7 \mathcal{E}]$ .

For  $[M^{10}, \Sigma^8 \mathcal{F}]$ , we have a long exact sequence from fibration (3.2)

$$\begin{array}{ccccccc} \cdots & \longrightarrow & [M, \Sigma^9 H\mathbb{Z}_2] & \longrightarrow & [M, \Sigma^8 \mathcal{F}] & \longrightarrow & [M, \Sigma^8 H\mathbb{Z}_2] \xrightarrow{Sq^2} [M, \Sigma^{10} H\mathbb{Z}_2] \\ & & \wr \parallel & & & & \\ & & 0 & & & & \end{array}$$

which implies

$$[M, \Sigma^8 \mathcal{F}] = \begin{cases} H^8(M; \mathbb{Z}_2) & \text{if } M \text{ is spin,} \\ \text{Ker}(Sq^2) & \text{if } M \text{ is not spin.} \end{cases} \quad (3.32)$$

For  $[M, \Sigma^7 \mathcal{E}]$ , we have the following exact sequence obtained from (3.26)

$$\begin{array}{ccccccccc} [M, \Sigma^8 H\mathbb{Z}_2] & & & & & & & & \\ \downarrow & \searrow Sq^2 & & & & & & & \\ [M, \Sigma^6 \mathcal{F}_2] & \xrightarrow{\Phi} & [M, \Sigma^{10} H\mathbb{Z}_2] & \longrightarrow & [M, \Sigma^7 \mathcal{E}] & \longrightarrow & [M, \Sigma^7 \mathcal{F}_2] & \xrightarrow{\Phi} & [M, \Sigma^{11} H\mathbb{Z}_2] \\ & & \wr \parallel & & & & \wr \parallel & & \\ & & \mathbb{Z}_2 & & & & 0 & & \end{array} \quad (3.33)$$

If either  $M$  is non-spin or the map  $\Phi \neq 0$ , it is evident that  $[M, \Sigma^7 \mathcal{E}] \cong [M, \Sigma^7 \mathcal{F}_2]$ . The latter

group can be computed using the long exact sequence

$$\begin{array}{c} \cdots [M, \Sigma^9 H\mathbb{Z}_2] \longrightarrow [M, \Sigma^7 \mathcal{F}_2] \longrightarrow [M, \Sigma^7 H\mathbb{Z}_4] \xrightarrow{Sq^2 \circ d_2} [M, \Sigma^{10} H\mathbb{Z}_2], \\ \parallel \\ 0 \end{array} \quad (3.34)$$

which together with (3.33) gives

$$[M, \Sigma^7 \mathcal{E}] \cong [M, \Sigma^7 \mathcal{F}_2] = \text{Ker}(Sq^2 \circ d_2). \quad (3.35)$$

Thus

$$[M, \Sigma^7 \mathcal{E}] = \begin{cases} H^7(M; \mathbb{Z}_4) & \text{if } M \text{ is spin and } \Phi \neq 0 \\ \text{Ker}(Sq^2 \circ d_2) & \text{if } M \text{ is non-spin.} \end{cases} \quad (3.36)$$

Now, assume that  $M$  is spin and  $\Phi = 0$ . In this case, from (3.33) and (3.34), we obtain the following short exact sequence

$$0 \rightarrow H^{10}(M; \mathbb{Z}_2) \rightarrow [M, \Sigma^7 \mathcal{E}] \rightarrow H^7(M; \mathbb{Z}_4) \rightarrow 0. \quad (3.37)$$

In order to compute  $[M, \Sigma^7 \mathcal{E}]$ , we first compute  $[C(\eta^2), \Sigma^7 \mathcal{E}]$  and  $[C(\iota \circ \eta^2), \Sigma^7 \mathcal{E}]$ . In this regard, consider the following commutative diagram whose rows and columns are cofiber sequences

$$\begin{array}{ccccc} \mathbb{S}^9 & \xrightarrow{\eta^2} & \mathbb{S}^7 & \longrightarrow & C(\eta^2) \\ \eta \downarrow & & \parallel & & \downarrow \\ \mathbb{S}^8 & \xrightarrow{\eta} & \mathbb{S}^7 & \longrightarrow & C(\eta) \\ \downarrow & & \downarrow & & \downarrow \\ \Sigma C(\eta) & \longrightarrow & * & \longrightarrow & \Sigma^2 C(\eta) \end{array}$$

This, together with (3.37) gives the following

$$\begin{array}{ccccccc}
& & [C(\eta), \Sigma^7 \mathcal{E}] & & & & \\
& & \downarrow & & & & \\
0 & \longrightarrow & \mathbb{Z}_2 & \longrightarrow & [C(\eta^2), \Sigma^7 \mathcal{E}] & \longrightarrow & \mathbb{Z}_4 \longrightarrow 0 \\
& & & & \downarrow & & \\
& & & & [\Sigma C(\eta), \Sigma^7 \mathcal{E}] & & 
\end{array}$$

For the non-spin case, we have  $[C(\eta), \Sigma^7 \mathcal{E}] \cong [C(\eta), \Sigma^7 \mathcal{F}_2] \cong \mathbb{Z}_8$ , and  $[\Sigma C(\eta), \Sigma^7 \mathcal{E}] = 0$  using (3.21). This implies that the short exact sequence for  $C(\eta^2)$  does not split, and we get

$$[C(\eta^2), \Sigma^7 \mathcal{E}] \cong \mathbb{Z}_8. \quad (3.38)$$

To compute  $[C(\iota \circ \eta^2), \Sigma^7 \mathcal{E}]$ , consider the following commutative diagram

$$\begin{array}{ccccccc}
& & \mathbb{Z}_2 & & & & \\
& & \wr & & & & \\
0 & \longrightarrow & [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] & \longrightarrow & [C(\iota \circ \eta^2), \Sigma^7 \mathcal{E}] & \longrightarrow & [M(\mathbb{Z}/2^r, 7), \Sigma^7 \mathcal{E}] \longrightarrow 0 \\
& & \downarrow = & & \downarrow & & \downarrow \\
0 & \longrightarrow & [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] & \longrightarrow & [C(\eta^2), \Sigma^7 \mathcal{E}] & \longrightarrow & [\mathbb{S}^7, \Sigma^7 \mathcal{E}] \longrightarrow 0 \\
& & & & & & \wr \\
& & & & & & \mathbb{Z}_4
\end{array} \quad (3.39)$$

which is obtained from the cofiber sequences of  $C(\eta^2)$  and  $C(\iota \circ \eta^2)$ . Since the short exact sequence at  $[C(\eta^2), \Sigma^7 \mathcal{E}]$  does not split, we conclude that the short exact sequence at  $[C(\iota \circ \eta^2), \Sigma^7 \mathcal{E}]$  does not split as well.

Thus, depending on  $r$  in  $M(\mathbb{Z}/2^r, 7)$ , (3.39) implies

$$[C(\iota \circ \eta^2), \Sigma^7 \mathcal{E}] \cong \begin{cases} \mathbb{Z}_8 & \text{if } r = 1, \\ \mathbb{Z}_4 & \text{if } r > 1. \end{cases} \quad (3.40)$$

Note that the 10-manifold spin case bifurcates into two sub-cases: whether the map  $\psi : H^6(M; \mathbb{Z}_4) \rightarrow H^{10}(M; \mathbb{Z}_2)$  (as described in (3.29)) is non-trivial or trivial.

Assume that  $\psi \neq 0$  for spin 10-manifold. This implies that the attaching map of the top cell  $\mathbb{S}^9 \rightarrow M/M^{(6)} \simeq (\mathbb{S}^7)^{\vee_l} \vee (\mathbb{S}^8)^{\vee_s} \vee_{p,r} M(\mathbb{Z}_{p^r}, 7)$  attaches by  $\eta^2$  onto some  $(\mathbb{S}^7)^{\vee_l}$  or some

$$M(\mathbb{Z}_{p^r}, 7).$$

Let  $K \subseteq H^7(M; \mathbb{Z}_4)$  be  $\text{Ker}\left(H^7(M; \mathbb{Z}_4) \rightarrow H^7(M; \mathbb{Z}_2) \xrightarrow{\Psi} H^{10}(M; \mathbb{Z}_2)\right)$ . Further, let  $H^7(M; \mathbb{Z}_4) = K \oplus A$  where  $A = \mathbb{Z}_2$  or  $\mathbb{Z}_4$  (similar to the 9-manifold scenario). Then,  $[M, \Sigma^7 \mathcal{E}] = K \oplus \tilde{A}$  where  $\tilde{A}$  is the non-trivial extension in  $0 \rightarrow \mathbb{Z}_2 \rightarrow \tilde{A} \rightarrow A \rightarrow 0$ .

On the other hand, assume  $\psi = 0$ . Analogous to the 9-manifold spin case, we get the following commutative diagram of short exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & [M^{10}, \Sigma^{10} H\mathbb{Z}_2] & \longrightarrow & [M^{10}, \Sigma^7 \mathcal{E}] & \longrightarrow & [M^{10}, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0 \\ & & \uparrow \cong & & \uparrow & & \uparrow \cong \\ 0 & \longrightarrow & [M^{10}/M^{(5)}, \Sigma^{10} H\mathbb{Z}_2] & \longrightarrow & [M^{10}/M^{(5)}, \Sigma^7 \mathcal{E}] & \longrightarrow & [M^{10}/M^{(5)}, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0. \end{array} \quad (3.41)$$

This implies that

$$[M^{10}, \Sigma^7 \mathcal{E}] \cong [M^{10}/M^{(5)}, \Sigma^7 \mathcal{E}].$$

For the computation of  $[M^{10}/M^{(5)}, \Sigma^7 \mathcal{E}]$ , consider the following

$$\begin{array}{ccccccc} 0 & \longrightarrow & [M^{10}/M^{(5)}, \Sigma^{10} H\mathbb{Z}_2] & \longrightarrow & [M^{10}/M^{(5)}, \Sigma^7 \mathcal{E}] & \longrightarrow & [M^{10}/M^{(5)}, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0 \\ & & \uparrow \cong & & \uparrow & & \uparrow \\ 0 & \longrightarrow & [M^{10}/M^{(6)}, \Sigma^{10} H\mathbb{Z}_2] & \longrightarrow & [M^{10}/M^{(6)}, \Sigma^7 \mathcal{E}] & \longrightarrow & [M^{10}/M^{(6)}, \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0 \\ & & \uparrow \phi^* & & \uparrow \Psi^* & & \uparrow \gamma^* \\ 0 & \longrightarrow & [\Sigma(M^{(6)}/M^{(5)}), \Sigma^{10} H\mathbb{Z}_2] & \longrightarrow & [\Sigma(M^{(6)}/M^{(5)}), \Sigma^7 \mathcal{E}] & \longrightarrow & [\Sigma(M^{(6)}/M^{(5)}), \Sigma^7 H\mathbb{Z}_4] \longrightarrow 0. \end{array} \quad (3.42)$$

Using (3.29), the map  $\Psi : M^{10}/M^{(6)} \rightarrow \Sigma(M^{(6)}/M^{(5)}) \simeq \vee_i \mathbb{S}^7$  decomposes in  $\gamma : \vee_l \mathbb{S}^7 \vee_k \mathbb{S}^8 \vee_{(p,r) \in J} M(\mathbb{Z}_{p^r}, 7) \rightarrow \vee_i \mathbb{S}^7$  and  $\phi : \mathbb{S}^{10} \rightarrow \vee_i \mathbb{S}^7$ . Moreover, note that the attaching map of the 10-cell onto the 6-cell is a multiple of  $v \in \pi_3^s$ . Thus, we need to compute  $v^* : [\mathbb{S}^7, \Sigma^7 \mathcal{E}] \rightarrow [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}]$ .

Using (3.27), we get the following commutative diagram (observe that it suffices to work 2-locally)

$$\begin{array}{ccc}
[\mathbb{S}^7, \Sigma^7 \mathcal{E}] & \xrightarrow{\nu^*} & [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] \\
\parallel & & \parallel \\
[\mathbb{S}^7, \tau_{\leq 10} pl/o] & \xrightarrow{\nu^*} & [\mathbb{S}^{10}, \tau_{\leq 10} pl/o] \\
\downarrow & & \downarrow \\
[\mathbb{S}^7, {}^{>6}\tau_{\leq 10} g/o] & \xrightarrow{\nu^*} & [\mathbb{S}^{10}, {}^{>6}\tau_{\leq 10} g/o]
\end{array}$$

Since the map  $[\mathbb{S}^{10}, \tau_{\leq 10} pl/o] \rightarrow [\mathbb{S}^{10}, {}^{>6}\tau_{\leq 10} g/o]$  is an isomorphism, and  $[\mathbb{S}^7, {}^{>6}\tau_{\leq 10} g/o] = 0$ , the map  $\nu^* = 0$ , and hence in (3.42) the induced map  $\phi^* = 0$ . Thus,  $\text{Im}(\Psi^*) = \text{Im}(\gamma^*)$ , which implies in the diagram (3.42) the sequence splits at  $[M^{10}/M^{(6)}, \Sigma^7 \mathcal{E}] / \text{Im}(\Psi^*)$  and consequently at  $[M^{10}/M^{(5)}, \Sigma^7 \mathcal{E}]$ . Therefore the exact sequence in (3.41) splits for  $M$ .

This completes the proof for all 10 dimensional manifold  $M^{10}$  with  $H_1(M^{10}) = 0$ .  $\square$



# Chapter 4

## Inertia Group of Manifolds of Dimensions Between 8 and 10

In this chapter we compute the concordance and homotopy inertia groups of smooth manifold  $M^n$  for  $8 \leq n \leq 10$ , and the inertia group of particular manifolds.

### 4.1 Concordance inertia group

Let  $M^n$  be a closed, oriented, smooth manifold. Consider a degree one map  $d : M^n \rightarrow \mathbb{S}^n$ , and the induced map  $d^* : [\mathbb{S}^n, PL/O] \rightarrow [M^n, PL/O]$ . Using the identification (2.5), we get  $[\mathbb{S}^n, PL/O] = \mathcal{C}_{PL}(\mathbb{S}^n)$  and  $[M^n, PL/O] = \mathcal{C}_{PL}(M^n)$ , and  $d^*$  maps  $[\Sigma^n]$  to  $[M^n \# \Sigma^n]$ . Then, observe that  $\ker(d^*)$  can be identified with  $I_c(M)$ . We utilize this identification to determine the concordance inertia group of 8, 9 and 10-dimensional manifolds.

**Theorem 4.1.1.** *Let  $M^8$  be a closed manifold. Then*

$$I_c(M^8) = 0.$$

*Proof.* The statement follows directly from the 8<sup>th</sup> Postnikov decomposition (3.1) of  $pl/o$ , wherein the map  $d^* : [\mathbb{S}^8, \tau_{\leq 8} pl/o] \rightarrow [M^8, \tau_{\leq 8} pl/o]$  induced by the collapse map  $d : M^8 \rightarrow M^8 \setminus \text{Int}(\mathbb{D}^8)$  is injective.  $\square$

**Theorem 4.1.2.** *Let  $M^9$  be a closed, oriented, smooth manifold with  $Sq^2 \circ d_2 : H^6(M; \mathbb{Z}_4) \rightarrow H^9(M; \mathbb{Z}_2)$ .*

1. *If  $M^9$  is spin, then  $I_c(M^9) = 0$ .*
2. *If  $M^9$  is non-spin and  $Sq^2 \circ d_2 = 0$ , then  $I_c(M^9) = \mathbb{Z}_2\{\eta \circ \varepsilon\}$ .*
3. *If  $M^9$  is non-spin and  $Sq^2 \circ d_2 \neq 0$ , then  $I_c(M^9) = \mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\}$ .*

*Proof.* Using the splitting (3.3), we have the following decomposition

$$[M^9, PL/O] \cong H^7(M^9; \mathbb{Z}_7) \oplus H^9(M^9; \mathbb{Z}_2) \oplus [M^9, \Sigma^8 \mathcal{F}] \oplus [M^9, \Sigma^7 \mathcal{F}_2]. \quad (4.1)$$

Since, the concordance inertia group of a manifold is isomorphic to the kernel of  $d^* : [\mathbb{S}^9, PL/O] \rightarrow [M^9, PL/O]$ , it is enough to check the  $\text{Ker}(d^*)$  in (4.1) componentwise. As  $M^9$  is oriented, the map  $d^* : H^9(\mathbb{S}^9; \mathbb{Z}_2) \rightarrow H^9(M^9; \mathbb{Z}_2)$  is injective on the top cohomology. Thus, we focus on the maps  $d^* : [\mathbb{S}^9, \Sigma^8 \mathcal{F}] \rightarrow [M^9, \Sigma^8 \mathcal{F}]$  and  $d^* : [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \rightarrow [M^9, \Sigma^7 \mathcal{F}_2]$ .

- (1) If  $M^9$  is a spin manifold, then by the proof of Theorem 3.2.1(1), we have

$$[M^9, \Sigma^8 \mathcal{F}] \cong H^8(M^9; \mathbb{Z}_2) \oplus H^9(M^9; \mathbb{Z}_2)$$

and

$$[M^9, \Sigma^7 \mathcal{F}_2] \cong H^7(M; \mathbb{Z}_4) \oplus H^9(M; \mathbb{Z}_2).$$

Thus, the induced maps  $d^* : [\mathbb{S}^9, \Sigma^8 \mathcal{F}] \rightarrow [M^9, \Sigma^8 \mathcal{F}]$  and  $d^* : [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \rightarrow [M^9, \Sigma^7 \mathcal{F}_2]$  are injective, where each of them has image isomorphic to  $H^9(M^9; \mathbb{Z}_2)$ . Therefore, the kernel of  $d^* : [\mathbb{S}^9, PL/O] \rightarrow [M^9, PL/O]$  is trivial for the spin manifold  $M^9$ .

(2) Let  $M^9$  be a non-spin manifold. From (3.6) and (3.7) we have

$$M^9 / M^{(6)} \simeq C(\eta) \vee M' \text{ or } C(\iota \circ \eta) \vee M''.$$

Let  $X = C(\eta)$  or  $C(\iota \circ \eta)$ . Since  $M'$  or  $M''$  is in the 8-skeleton of  $M^9 / M^{(6)}$ , the degree one map factors through  $X$ , and we have the following commutative diagram

$$\begin{array}{ccc} [\mathbb{S}^9, \Sigma^8 \mathcal{F}] & \xrightarrow{d^*} & [M^9, \Sigma^8 \mathcal{F}] \\ & \searrow d^* & \nearrow \\ & [X, \Sigma^8 \mathcal{F}] & \end{array}$$

It follows from (3.15) and (3.16) that the map  $[\mathbb{S}^9, \Sigma^8 \mathcal{F}] \cong \mathbb{Z}_2\{\eta \circ \varepsilon\} \rightarrow [X, \Sigma^8 \mathcal{F}]$  is trivial. Therefore, for the component  $\Sigma^8 \mathcal{F}$ , the induced map  $d^* : [\mathbb{S}^9, \Sigma^8 \mathcal{F}] \rightarrow [M^9, \Sigma^8 \mathcal{F}]$  is trivial as well.

For the map  $d^* : [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \rightarrow [M^9, \Sigma^7 \mathcal{F}_2]$ , consider the following commutative diagram

$$\begin{array}{ccc} [\mathbb{S}^9, \Sigma^9 H\mathbb{Z}_2] & \xrightarrow{\cong} & [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \cong \mathbb{Z}_2\{\mu\} \\ \cong \downarrow d^* & & \downarrow d^* \\ [M^9, \Sigma^6 H\mathbb{Z}_4] & \xrightarrow{Sq^2 d_2} & [M^9, \Sigma^9 H\mathbb{Z}_2] \longrightarrow [M^9, \Sigma^7 \mathcal{F}_2] \end{array} \quad (4.2)$$

whose rows are part of exact sequences obtained from the fibration sequence for  $\mathcal{F}_2 := \text{Fibre}(H\mathbb{Z}_4 \xrightarrow{Sq^2 \circ d_2} \Sigma^3 H\mathbb{Z}_2)$  as in (3.2). In the case  $Sq^2 \circ d_2 = 0$ , the map  $[M^9, \Sigma^9 H\mathbb{Z}_2] \rightarrow [M^9, \Sigma^7 \mathcal{F}_2]$  becomes injective, and so is the  $d^* : [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \rightarrow [M^9, \Sigma^7 \mathcal{F}_2]$  in (4.2).

Further, the map  $Sq^2 \circ d_2$  in (4.2) being non-trivial implies the map  $[M^9, \Sigma^9 H\mathbb{Z}_2] \rightarrow [M^9, \Sigma^7 \mathcal{F}_2]$  is trivial since  $[M^9, \Sigma^9 H\mathbb{Z}_2] \cong \mathbb{Z}_2$ . As a consequence, the map  $d^* : [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \rightarrow [M^9, \Sigma^7 \mathcal{F}_2]$  is trivial as well.

This completes the proof in all cases. □

Next, we discuss the 10-dimensional case.

**Theorem 4.1.3.** *Let  $M^{10}$  be a closed, simply-connected, smooth manifold, and let  $\Phi : H^6(M^{10}; \mathbb{Z}_4) \rightarrow H^{10}(M^{10}; \mathbb{Z}_2)$  be the secondary cohomology operation mentioned in (3.26).*

1. If  $M^{10}$  is a spin manifold and  $\Phi = 0$ , then

$$I_c(M^{10}) = 0.$$

2. If  $M^{10}$  is either a spin manifold with  $\Phi \neq 0$ , or a non-spin manifold, then

$$I_c(M^{10}) = \mathbb{Z}_2\{\eta \circ \mu\}.$$

*Proof.* The splitting (3.27) gives the following

$$\begin{aligned} [M^{10}, PL/O] &\cong [M^{10}, \Sigma^8 \mathcal{F}] \oplus [M^{10}, \Sigma^7 \mathcal{E}] \oplus H^7(M^{10}; \mathbb{Z}_7) \oplus H^9(M^{10}; \mathbb{Z}_2) \\ &\quad \oplus H^{10}(M^{10}; \mathbb{Z}/3). \end{aligned}$$

By using Theorem 3.2.1, we have  $[\mathbb{S}^{10}, PL/O] \cong [\mathbb{S}^{10}, \Sigma^{10} H\mathbb{Z}_3] \oplus [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}]$ . Analogous to Theorem 4.1.2, it is enough to check the componentwise kernel of  $d^* : [\mathbb{S}^{10}, \Sigma^{10} H\mathbb{Z}_3] \oplus [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] \rightarrow [M^{10}, \Sigma^{10} H\mathbb{Z}_3] \oplus [M^{10}, \Sigma^7 \mathcal{E}]$ .

Since  $M^{10}$  is simply-connected, the map  $d^* : [\mathbb{S}^{10}, \Sigma^{10} H\mathbb{Z}_3] \rightarrow [M^{10}, \Sigma^{10} H\mathbb{Z}_3]$  is injective. This implies that the generator  $\beta_1$  of  $[\mathbb{S}^{10}, \Sigma^{10} H\mathbb{Z}_3] = \mathbb{Z}_3\{\beta_1\}$  does not belong to  $I_c(M^{10})$ .

So it remains to compute the kernel of  $d^* : [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] = \mathbb{Z}_2\{\eta \circ \mu\} \rightarrow [M^{10}, \Sigma^7 \mathcal{E}]$ . For that, let us consider the following commutative diagram

$$\begin{array}{ccccc} [\mathbb{S}^{10}, \Sigma^8 H\mathbb{Z}_2] & & [\mathbb{S}^{10}, \Sigma^{10} H\mathbb{Z}_2] & \xrightarrow{\cong} & [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] \\ \downarrow & \searrow Sq^2 & \cong \downarrow d^* & & \downarrow d^* \\ [M^{10}, \Sigma^6 \mathcal{F}_2] & \xrightarrow{\Phi} & [M^{10}, \Sigma^{10} H\mathbb{Z}_2] & \longrightarrow & [M^{10}, \Sigma^7 \mathcal{E}] \end{array} \quad (4.3)$$

where the bottom row is a part of the exact sequence (3.33). From the diagram, it is clear that the map  $d^* : [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] \rightarrow [M^{10}, \Sigma^7 \mathcal{E}]$  is injective or trivial if and only if the map from  $[M^{10}, \Sigma^{10} H\mathbb{Z}_2] \rightarrow [M^{10}, \Sigma^7 \mathcal{E}]$  is injective or trivial, respectively. For the spin case, whenever  $\Phi = 0$ , the map  $[M^{10}, \Sigma^{10} H\mathbb{Z}_2] \rightarrow [M^{10}, \Sigma^7 \mathcal{E}]$  is injective, else it is trivial. On the other hand,  $\Phi$  is always non-trivial for the non-spin case. Hence the map  $[M^{10}, \Sigma^{10} H\mathbb{Z}_2] \rightarrow [M^{10}, \Sigma^7 \mathcal{E}]$  is trivial. This completes the proof in all cases.  $\square$

## 4.2 Homotopy inertia group

In surgery theory, there is a natural map  $f_{M^n} : \Theta_n \rightarrow S^{Diff}(M^n)$ , where  $S^{Diff}(M^n)$  is the homotopy smooth structure set of  $M^n$  [6, 54]. Then, the homotopy inertia group  $I_h(M^n)$  can be identified with  $\text{Ker}(f_{M^n})$ . If  $M^n$  is a closed, oriented, smooth manifold, then we have the following commutative square

$$\begin{array}{ccc} [\mathbb{S}^n, PL/O] & \xrightarrow{\psi_*} & [\mathbb{S}^n, G/O] \\ f_{M^n} \downarrow & & \downarrow d^* \\ S^{Diff}(M^n) & \xrightarrow{g'} & [M^n, G/O], \end{array} \quad (4.4)$$

where  $\psi : PL/O \rightarrow G/O$  is a natural fibration,  $d^*$  is induced from the degree one map, and  $g'$  is the part of surgery exact sequence. Note that, if  $n$  is even and  $M^n$  is simply-connected then the maps  $\psi_*$  and  $g'$  are injective.

Recall that,  $I_c(M^n) \subseteq I_h(M^n)$ . Hence it is enough to discuss those elements in  $\Theta_n \equiv [\mathbb{S}^n, PL/O]$  which are not in  $I_c(M^n)$ .

We observe the fact that follows from straightforward diagram chasing. Consider the following commutative diagram whose rows and columns are part of exact sequences of abelian groups, and  $\xi$  and  $\beta$  are surjective homomorphisms.

$$\begin{array}{ccccc} A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & C \\ \downarrow \xi & & \downarrow \phi & & \downarrow \psi \\ E & \xrightarrow{\gamma} & F & \xrightarrow{\eta} & G \\ & & \downarrow \theta & & \\ & & H & & \end{array} \quad (4.5)$$

**Lemma 4.2.1.** : *If  $y \notin \text{Im}(\phi)$  and  $\eta(y) = z \neq 0$  then  $z \notin \text{Im}(\psi)$ .*

*Proof.* We prove this by contradiction. Suppose  $z \in \text{Im}(\psi)$ . Then there exist elements  $b, c$  such that  $\beta(b) = c$  and  $\psi(c) = z$ . For  $\phi(b) = x$ , we have  $\eta(x) = z$  and  $\theta(x) = 0$ . Hence  $\eta(x - y) = 0$ , which also implies that there exists an element  $e$  such that  $\gamma(e) = x - y$ . As  $\xi$  is

surjective, commutativity of diagram allows us to affirm the existence of  $b'$  such that  $\phi(b') = x - y$ . Consequently,  $\theta(x - y) = \theta(x) - \theta(y) = 0$ , implying that  $\theta(y) = 0$ . However, this contradicts the fact that  $y \notin \text{Im}(\phi)$ . Therefore, the statement is established.  $\square$

Let  $M^{(6)}$  be the 6-skeleton of an  $n$ -manifold  $M$  for  $n \leq 10$ . Consider the following commutative diagram,

$$\begin{array}{ccc} [\Sigma M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \xlongequal{\quad} & [\Sigma M^{(6)}, K(\mathbb{Z}_{(2)}, 7)] \\ \downarrow j_* & & \downarrow j_* \\ [\Sigma M^{(6)}, \tau_{\leq 10} pl/o] & \xlongequal{\quad} & [\Sigma M^{(6)}, K(\mathbb{Z}_4, 7)] \end{array}$$

where the map  $j_*$  is induced from the fibration

$${}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl) \xrightarrow{j} \tau_{\leq 10} pl/o \xrightarrow{\psi} {}^{>6}\tau_{\leq 10} g/o. \quad (4.6)$$

Observe that, the map  $j_* : [\Sigma M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [\Sigma M^{(6)}, \tau_{\leq 10} pl/o]$  is surjective.

Further, consider the following commutative diagram

$$\begin{array}{ccccc} [\Sigma M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \rightarrow & [M/M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \xrightarrow{q^*} & [M, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \\ \downarrow j_* & & \downarrow j_* & & \downarrow j_* \\ [\Sigma M^{(6)}, \tau_{\leq 10} pl/o] & \longrightarrow & [M/M^{(6)}, \tau_{\leq 10} pl/o] & \xrightarrow{q^*} & [M, \tau_{\leq 10} pl/o] \\ & & \downarrow \psi_* & & \downarrow \psi_* \\ & & [M/M^{(6)}, {}^{>6}\tau_{\leq 10} g/o] & \xrightarrow{q^*} & [M, {}^{>6}\tau_{\leq 10} g/o] \end{array}$$

whose columns and rows are part of exact sequences of abelian groups obtained from the fibration in (4.6), and the cofibration  $M^{(6)} \hookrightarrow M \xrightarrow{q} M/M^{(6)}$ , respectively. Furthermore, the maps  $j_* : [\Sigma M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [\Sigma M^{(6)}, \tau_{\leq 10} pl/o]$  and

$q^* : [M/M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [M, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)]$  are surjective. Thus, the Lemma 4.2.1 holds for this commutative diagram, and hence we get the next proposition:

**Proposition 4.2.2.** *If  $y \notin \text{Im}(j_* : [M/M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [M/M^{(6)}, \tau_{\leq 10} pl/o])$  then  $q^*(y) \notin \text{Im}(j_* : [M, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [M, \tau_{\leq 10} pl/o])$  provided  $q^*(y) \neq 0$ .*

The map of our interest from (4.4), especially is  $d^* : [\mathbb{S}^n, {}^{>6}\tau_{\leq 10} g/o] \rightarrow [M, {}^{>6}\tau_{\leq 10} g/o]$ . For that, the way we are going to use Proposition 4.2.2 is as follows. Consider the following

diagram

$$\begin{array}{ccccc}
 [M/M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \xrightarrow{q^*} & [M, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & & \\
 \downarrow j_* & & \downarrow j_* & & \\
 [\mathbb{S}^n, \tau_{\leq 10} pl/o] & \xrightarrow{d^*} & [M/M^{(6)}, \tau_{\leq 10} pl/o] & \xrightarrow{q^*} & [M, \tau_{\leq 10} pl/o] \\
 \downarrow \psi_* & & \downarrow \psi_* & & \downarrow \psi_* \\
 [\mathbb{S}^n, {}^{>6}\tau_{\leq 10} g/o] & \xrightarrow{d^*} & [M/M^{(6)}, {}^{>6}\tau_{\leq 10} g/o] & \xrightarrow{q^*} & [M, {}^{>6}\tau_{\leq 10} g/o] \\
 & \searrow d^* & & & 
 \end{array} \tag{4.7}$$

whose columns are part of exact sequence using (4.6), the map  $d^*$  is induced degree one maps, and  $q^*$  is induced quotient map. For  $n = 8, 9$  and  $10$  the map  $\psi_* : [\mathbb{S}^n, \tau_{\leq 10} pl/o] \rightarrow [\mathbb{S}^n, {}^{>6}\tau_{\leq 10} g/o]$  is injective, surjective and isomorphism respectively. In most of the cases discussed below, we applied Proposition 4.2.2. For that, we use diagrams (4.2) and (4.7) together, and show that if  $0 \neq y \in d^*([\mathbb{S}^n, \tau_{\leq 10} pl/o]) \subseteq [M/M^{(6)}, \tau_{\leq 10} pl/o]$ , then  $y \notin \text{Im}(j_* : [M/M^{(6)}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [M/M^{(6)}, \tau_{\leq 10} pl/o])$ .

**Theorem 4.2.3.** *Let  $M^8$  be a closed, smooth manifold. Then*

$$I_h(M^8) = 0.$$

*Proof.* In (4.4), as  $[\mathbb{S}^8, PL/O] = \mathbb{Z}_2\{\varepsilon\}$  and the map  $\psi_* : [\mathbb{S}^8, PL/O] \rightarrow [\mathbb{S}^8, G/O]$  is injective, it is enough to prove that the image of  $\psi_*(\varepsilon)$  under  $d^* : [\mathbb{S}^8, G/O] \rightarrow [M^8, G/O]$  is non-zero. By Theorem 4.1.1, the map  $d^* : [\mathbb{S}^8, \tau_{\leq 8} pl/o] \rightarrow [M^8, \tau_{\leq 8} pl/o]$  is injective. Recall that,  ${}^{>6}\tau_{\leq 8}(\Sigma^{-1}g/pl)_{(2)} \simeq \Sigma^7 H\mathbb{Z}_{(2)}$  and  $\tau_{\leq 8}(pl/o) \simeq \Sigma^7 H\mathbb{Z}_4 \vee \Sigma^8 H\mathbb{Z}_2$ , such that the map  $\Sigma^7 H\mathbb{Z}_{(2)} \rightarrow \Sigma^8 H\mathbb{Z}_2$  is null homotopic. So we get that  $d^*([\mathbb{S}^8, \tau_{\leq 8} pl/o]) \cap \text{Im}(j_* : [M^8, {}^{>6}\tau_{\leq 8}(\Sigma^{-1}g/pl)] \rightarrow [M^8, \tau_{\leq 8} pl/o]) = \{0\}$ . Therefore, in (4.7),  $\psi_* : [M^8, \tau_{\leq 8} pl/o] \rightarrow [M^8, {}^{>6}\tau_{\leq 8} g/o]$  maps  $d^*([\mathbb{S}^8, \tau_{\leq 8} pl/o])$  injectively. Using this in (4.4), completes the proof.  $\square$

Above Theorem 4.2.3 along with the fact that every self-homotopy equivalence of  $\mathbb{RP}^8$  is homotopic to the identity (refer [4, Corollary 6]) gives the following result.

**Corollary 4.2.4.**  $I(\mathbb{RP}^8) = \{0\}$ .

In the 9-dimensional case, using (4.7) we first discuss the map  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 9} g/o] \rightarrow [M^9/M^{(6)}, >^6\tau_{\leq 9} g/o]$  in the next lemma.

**Lemma 4.2.5.** *Let  $M^9$  be a closed, oriented, smooth manifold with  $Sq^2 \circ d_2 : H^6(M^9; \mathbb{Z}_4) \rightarrow H^9(M^9; \mathbb{Z}_2)$ .*

1. *If  $M^9$  is spin then  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9/M^{(6)}, >^6\tau_{\leq 10} g/o]$  is injective.*
2. *Suppose  $M^9$  be non-spin and  $Sq^2 \circ d_2$  be trivial.*
  - (a) *If  $M^9$  satisfies (3.6) then  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9/M^{(6)}, >^6\tau_{\leq 10} g/o]$  is trivial.*
  - (b) *If  $M^9$  satisfies (3.7) then  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9/M^{(6)}, >^6\tau_{\leq 10} g/o]$  has  $\text{Ker}(d^*) = \mathbb{Z}_2\{\eta \circ \varepsilon\}$  and  $\text{Im}(d^*) = \mathbb{Z}_2\{\mu\}$ .*

*Proof.* Recall that  $[\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \cong [\mathbb{S}^9, \Sigma^8 \mathcal{F}] \oplus [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2]$ . Additionally,  $[\mathbb{S}^9, \Sigma^8 \mathcal{F}] = \mathbb{Z}_2\{\eta \circ \varepsilon\}$  and  $[\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] = \mathbb{Z}_2\{\mu\}$ .

1. If  $M^9$  is spin manifold then the  $M^9/M^{(6)}$  decomposition (3.5) directly gives the injectivity of the map  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9/M^{(6)}, >^6\tau_{\leq 10} g/o]$ .
2. Let  $M^9$  be a non-spin manifold and  $Sq^2 \circ d_2 = 0$ . Note that, from (3.6), (3.7) the degree map induces the following

$$\begin{array}{ccc}
 [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] & \xrightarrow{d^*} & [M^9/M^{(6)}, >^6\tau_{\leq 10} g/o] \\
 \searrow d^* & & \swarrow p^* \\
 & [X, >^6\tau_{\leq 10} g/o] &
 \end{array} \tag{4.8}$$

where  $X = C(\eta)$  or  $C(\iota \circ \eta)$  is the mapping cone of respective maps, and  $p : M^9/M^{(6)} \rightarrow X$  is the quotient map. Now, we will discuss the image of  $\{\mu\} \in [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2]$  under  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [X, >^6\tau_{\leq 10} g/o]$ .

- (a) If  $M^9$  satisfies (3.6) then  $X = C(\eta)$ . Consider the induced long exact sequence due to the cofiber sequence for  $\eta : \mathbb{S}^8 \rightarrow \mathbb{S}^7$

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & [\mathbb{S}^8, >^6\tau_{\leq 10} g/o] & \xrightarrow{\eta^*} & [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] & \longrightarrow & [C(\eta), >^6\tau_{\leq 10} g/o] \\
 & & & & & \searrow & \\
 & & & & & & [\mathbb{S}^7, >^6\tau_{\leq 10} g/o] \xrightarrow{\eta^*} [\mathbb{S}^8, >^6\tau_{\leq 10} g/o]
 \end{array}$$



Since  $[\mathbb{S}^7, >^6\tau_{\leq 10} g/o] = 0$  and  $\eta^* : [\mathbb{S}^8, >^6\tau_{\leq 10} g/o] \rightarrow [\mathbb{S}^9, >^6\tau_{\leq 10} g/o]$  is surjective, gives

$$[C(\eta), >^6\tau_{\leq 10} g/o] = 0$$

Therefore, in (4.8) the map  $d^* : [\mathbb{S}^9, \Sigma^7 \mathcal{F}_2] \rightarrow [M^9 / M^{(6)}, >^6\tau_{\leq 10} g/o]$  is trivial.

(b) If  $M^9$  satisfies (3.7) then  $X = C(\iota \circ \eta)$ . Now consider the induced long exact sequence from the cofiber sequence for  $\iota \circ \eta : \mathbb{S}^8 \rightarrow \mathbb{S}^7 \rightarrow M(\mathbb{Z}_{2^r}, 7)$

$$\begin{array}{ccccc}
 & & [\mathbb{S}^8, >^6\tau_{\leq 10} g/o] & & \\
 & \nearrow \text{dashed} & \downarrow \eta^* & & \\
 [\Sigma M(\mathbb{Z}_{2^r}, 7), >^6\tau_{\leq 10} g/o] & \longrightarrow & [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] & \xrightarrow{d^*} & [C(\iota \circ \eta), >^6\tau_{\leq 10} g/o] \\
 & \nwarrow \text{solid} & & & \\
 [M(\mathbb{Z}_{2^r}, 7), >^6\tau_{\leq 10} g/o] & \longrightarrow & [\mathbb{S}^8, >^6\tau_{\leq 10} g/o] & & \\
 & \searrow \text{dashed} & \uparrow \eta^* & & \\
 & & [\mathbb{S}^7, >^6\tau_{\leq 10} g/o] & & 
 \end{array}$$

where  $\text{Im}([\Sigma M(\mathbb{Z}_{2^r}, 7), >^6\tau_{\leq 10} g/o] \rightarrow [\mathbb{S}^8, >^6\tau_{\leq 10} g/o]) \cong \mathbb{Z}_2\{\varepsilon\} \subseteq \mathbb{Z}_2\{\bar{\nu}\} \oplus \mathbb{Z}_2\{\varepsilon\}$ .

This implies

$$\text{Ker}(d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [C(\iota \circ \eta), >^6\tau_{\leq 10} g/o]) = \mathbb{Z}_2\{\eta \circ \varepsilon\} \quad (4.9)$$

and

$$\text{Im}(d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [C(\iota \circ \eta), >^6\tau_{\leq 10} g/o]) = \mathbb{Z}_2\{\mu\}. \quad (4.10)$$

This completes the proof in all cases. □

**Theorem 4.2.6.** *Let  $M^9$  be a closed, oriented, smooth manifold with  $Sq^2 \circ d_2 : H^6(M^9; \mathbb{Z}_4) \rightarrow H^9(M^9; \mathbb{Z}_2)$ .*

1. *If  $M^9$  is spin then the map  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$  is injective.*
2. *Let  $M^9$  be non-spin and the map  $Sq^2 \circ d_2$  be trivial.*

- (a) If  $M^9$  satisfies (3.6) then  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$  maps  $\{\mu\}$  to 0.
- (b) If  $M^9$  satisfies (3.7) then  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$  maps  $\{\mu\}$  injectively.

*Proof.* 1. Let  $M^9$  be a spin manifold. Applying Proposition 4.2.2, using Lemma 4.2.5(1) and Theorem 4.1.2(1) together, we obtain the map  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$  is injective.

2. Let  $M^9$  be non-spin and the map  $Sq^2 \circ d_2$  be trivial.

- (a) If  $M^9$  satisfies (3.6), then from Lemma 4.2.5(2)(a) we know the map  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9 / M^{(6)}, >^6\tau_{\leq 10} g/o]$  is trivial. Since, the degree one map  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$  factors through  $[M^9 / M^{(6)}, >^6\tau_{\leq 10} g/o]$ , makes the statement clear in this case.
- (b) Now consider the case when  $M^9$  satisfies (3.7). From Lemma 4.2.5(2)(b) it follows that  $\mathbb{Z}_2\{\mu\}$  is in the  $\text{Im}\left(d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9 / M^{(6)}, >^6\tau_{\leq 10} g/o]\right)$ . As, under  $\psi_* : [\mathbb{S}^9, \tau_{\leq 10} pl/o] \rightarrow [\mathbb{S}^9, >^6\tau_{\leq 10} g/o]$  the group  $\mathbb{Z}_2\{\mu\}$  is mapped injectively, implies in (4.7),

$$d^*(\mu) \notin \text{Im}\left(j_* : [M^9 / M^{(6)}, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)] \rightarrow [M^9 / M^{(6)}, \tau_{\leq 10} pl/o]\right).$$

Also, it follows from Theorem 4.1.2(2)(a) that  $d^*(\mu)$  is non-zero in  $[M^9, \tau_{\leq 10} pl/o]$ . Finally using Proposition 4.2.2, we get that  $d^*(\mu)$  gets mapped non-trivially under  $\psi_* : [M^9, \tau_{\leq 10} pl/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$ . Therefore,  $\mu$  gets mapped non-trivially under  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$  as well. Hence the statement. □

It follows from [29] that, for dimensions  $8 \leq n \leq 10$ , the group  $\Theta_n$  fits into the following split short exact sequence

$$0 \rightarrow bP_{n+1} \rightarrow \Theta_n \rightarrow \pi_n^s / \text{Im}(J) \rightarrow 0,$$

where  $J : \pi_n(SO) \rightarrow \pi_n^s$  is the J-homomorphism, and  $\pi_n^s / \text{Im}(J) \subseteq \pi_n(F/O)$ . Note that,  $\pi_8^s / \text{Im}(J) = \mathbb{Z}_2\{\varepsilon\}$ ,  $\pi_9^s / \text{Im}(J) = \mathbb{Z}_2\{\mu\} \oplus \mathbb{Z}_2\{\eta \circ \varepsilon\}$ , and  $\pi_{10}^s / \text{Im}(J) = \mathbb{Z}_2\{\eta \circ \mu\} \oplus \mathbb{Z}_3\{\beta_1\}$ .

**Theorem 4.2.7.** *Let  $M^9$  be a closed, oriented, smooth manifold with the map  $Sq^2 \circ d_2 : H^6(M^9; \mathbb{Z}_4) \rightarrow H^9(M^9; \mathbb{Z}_2)$ .*

1. *If  $M^9$  is a spin manifold, then  $I_h(M^9) \cap \Theta_9 / bP_{10} = \{0\}$ .*
2. *If  $M^9$  is a spin simply-connected manifold, then  $I_h(M^9) = 0$ .*
3. *Suppose  $M^9$  is non-spin and the map  $Sq^2 \circ d_2 = 0$ .*
  - (a) *If  $M^9$  satisfies 3.6, then  $I_h(M^9) \supseteq \mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\}$ .*
  - (b) *If  $M^9$  satisfies 3.7, then  $I_h(M^9) \supseteq \mathbb{Z}_2\{\eta \circ \varepsilon\}$  and  $I_h(M^9) \not\supseteq \mathbb{Z}_2\{\mu\}$ .*
4. *If  $M^9$  is non-spin with  $Sq^2 \circ d_2 \neq 0$ , then  $I_h(M^9) \supseteq \mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\}$ .*

*Proof.* Recall that,  $I_h(M^n) = \text{Ker}(f_{M^n} : [\mathbb{S}^n, PL/O] \rightarrow S^{\text{Diff}}(M^n))$ , where  $f_{M^n}$  is part of the commutative square (4.4).

1. Let  $M^9$  be a spin manifold. Recollect that,  $\Theta_9 / bP_{10} = \mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\} \subseteq [\mathbb{S}^9, \tau_{\leq 10} pl/o]$ , and in (4.4) the map  $\psi_* : [\mathbb{S}^9, \tau_{\leq 10} pl/o] \rightarrow [\mathbb{S}^9, {}^{>6}\tau_{\leq 10} g/o]$  maps  $\mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\}$  injectively. Furthermore, by Theorem 4.2.6(1), the map  $d^* : [\mathbb{S}^9, {}^{>6}\tau_{\leq 10} g/o] \rightarrow [M^9, {}^{>6}\tau_{\leq 10} g/o]$  is injective, thereby making the statement clear in this case.
2. If  $M^9$  is simply-connected spin manifold then the result [8, Proposition II.2] says that  $I_h(M^9) \subseteq \text{Coker}(J_9) = \mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\}$ . Hence, from statement (1) we get  $I_h(M^9) = 0$ .
3. Let  $M^9$  be a non-spin manifold and the map  $Sq^2 \circ d_2 = 0$ .
  - (a) Suppose  $M^9$  satisfies 3.6. Consider the following commutative diagram

$$\begin{array}{ccccc}
 & & [\mathbb{S}^9, PL/O] & & \\
 & \nearrow \mathbf{d} & \downarrow \psi_* & & \\
 [\Sigma M_0^9, G/O] & \xrightarrow{(\Sigma h)^*} & [\mathbb{S}^9, G/O] & \xrightarrow{d^*} & [M^9, G/O]
 \end{array}$$

where  $M_0^9 = M^9 \setminus \text{int}(\mathbb{D}^9)$ , and  $\mathbf{d} : [\Sigma M_0^9, G/O] \rightarrow [\mathbb{S}^9, PL/O]$  is the map given in [8, Proposition 3.1], such that  $I_h(M^9) = \text{Im}(\mathbf{d})$ . The bottom row is obtained from cofiber sequence  $M^9 \xrightarrow{d} \mathbb{S}^9 \xrightarrow{\Sigma h} \Sigma M_0^9$  such that,  $h$  is the top cell attaching map of  $M^9$  and  $d$  is the degree one map.

Using Theorem 4.1.2(2)(a) and Theorem 4.2.6(1)(a), together gives  $d^* : [\mathbb{S}^9, G/O] \rightarrow [M^9, G/O]$  is trivial. Hence, in the above diagram,  $(\Sigma h)^* : [\Sigma M_0^9, G/O] \rightarrow [\mathbb{S}^9, G/O]$  is surjective. As  $\mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\} \subseteq [\mathbb{S}^9, PL/O]$  makes the statement clear in this case.

- (b) If  $M^9$  satisfies 3.7, then by Theorem 4.1.2(2)(a) we know that  $\mathbb{Z}_2\{\eta \circ \mu\} \subseteq I_h(M^9)$ . Now, in (4.4) recall that,  $\psi_*(\mu)$  is non-zero in  $[\mathbb{S}^9, >^6\tau_{\leq 10} g/o]$ . And also Theorem 4.2.6(2)(b) shows that  $\mathbb{Z}_2\{\mu\}$  is mapped injectively under the map  $d^* : [\mathbb{S}^9, >^6\tau_{\leq 10} g/o] \rightarrow [M^9, >^6\tau_{\leq 10} g/o]$ . Therefore,  $f_{M^9}(\mu)$  is also non-zero, which complete the proof for this case.

4. The statement for the case when  $M^9$  is non-spin and the map  $Sq^2 \circ d_2 \neq 0$  is clear from Theorem 4.1.2(2)(b) since  $\mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus \mathbb{Z}_2\{\mu\} = I_c(M^9) \subseteq I_h(M^9)$ .

□

Recall that,  $L^9(m)$  is a closed, oriented, smooth 9-manifold, and its inertia group is discussed below.

**Theorem 4.2.8.** *Let  $m$  be a positive integer and  $n$  be a non-negative integer.*

1. *If  $m = 2n + 1$ , then  $I(L^9(m)) = 0$ .*
2. *If  $m = 4n + 2$ , then  $I(L^9(m)) = \mathbb{Z}_2\{\eta \circ \varepsilon\}$ .*
3. *If  $m = 4n$ , then  $I(L^9(m)) = \mathbb{Z}_2\{\eta \circ \varepsilon\} \oplus bP_{10}$ .*

*Proof.* Note that, if  $m$  is odd then  $L^9(m)$  is a spin manifold; otherwise, it is non-spin. Observe that, if  $L^9(m)$  is non-spin, then  $Sq^2 \circ d_2 = 0$ , and  $L^9(m)$  satisfies (3.7). Additionally,  $bP_{10} \subseteq I_h(L^9(m))$  if and only if  $4|m$  by [31, Theorem 4.2]. Combining this together with the fact that; an orientation-preserving self-homotopy equivalence of a lens space is homotopic to the identity, and using Theorem 4.2.7, completes the proof. □

Particularly when  $m = 2$ ,  $L^{2n+1}(m)$  is nothing but the usual real projective space  $\mathbb{RP}^{2n+1}$ . The following remark is part of Theorem 4.2.7 and serves as a special case of  $m = 2$  in Theorem 4.2.8.

**Remark 4.2.9.** *There exists an unique exotic sphere  $\Sigma \in \Theta_9$  such that  $\mathbb{RP}^9 \# \Sigma$  is diffeomorphic to  $\mathbb{RP}^9$ .*

Now, in 10-dimensional case for the homotopy inertia group we just need to check the case when  $M^{10}$  is spin and  $\Phi = 0$ . For the same, recall the homotopy decompositions of  $M^{10}/M^{(6)}$  (3.29), (3.30) and (3.31). Since  $\pi_{10}(pl/o) \cong \pi_{10}(g/o) \cong \mathbb{Z}_2\{\eta \circ \mu\} \oplus \mathbb{Z}_3\{\beta_1\}$ , it suffices to work locally at prime 2 and 3. Let us first discuss the 3-localized case.

**Theorem 4.2.10.** *Let  $M^{10}$  be a closed, oriented, smooth manifold. Then the induced degree one map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(3)} \rightarrow [M^{10}, >^6\tau_{\leq 10} g/o]_{(3)}$  is injective.*

*Proof.* Recall that  $g/o_{(3)} \simeq \text{cok}(J)_{(3)} \times bso_{(3)}$ . Since  $[\mathbb{S}^{10}, bso]_{(3)} = 0$ ; we get  $[\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(3)} = [\mathbb{S}^{10}, \text{cok}(J)]_{(3)}$ . Note that

$$\pi_i(\text{cok}(J))_{(3)} \begin{cases} 0 & \text{if } i \leq 9 \\ \mathbb{Z}_3 & \text{if } i = 10 \end{cases}$$

This implies  $\tau_{\leq 10} \text{cok}(J)_{(3)} = K(\mathbb{Z}_3, 10)$ . Therefore the statement is true because  $d^* : H^{10}(\mathbb{S}^{10}; \mathbb{Z}_3) \rightarrow H^{10}(M^{10}; \mathbb{Z}_3)$  is injective.  $\square$

Now let us work 2-locally.

**Theorem 4.2.11.** *Let  $M^{10}$  be a closed, smooth 10-manifold with  $H_1(M) = 0$ . Then the induced degree one map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [M^{10}, >^6\tau_{\leq 10} g/o]_{(2)}$  is injective if  $M^{10}$  satisfy (3.29) or (3.31), and is trivial if  $M^{10}$  satisfy (3.30).*

*Proof.* Consider the case when  $M^{10}$  satisfies (3.29). Since  $[\mathbb{S}^{10}, >^6\tau_{\leq 10}(\Sigma^{-1}g/pl)]_{(2)} = 0$ ,  $d^*([\mathbb{S}^{10}, \tau_{\leq 10} pl/o]_{(2)}) \cap j_*([M^{10}/M^{(6)}, >^6\tau_{\leq 10}(\Sigma^{-1}g/pl)]_{(2)}) = \{0\}$ . From Theorem 4.1.3, we know that  $d^*(\eta \circ \mu)$  is non-zero in  $[M^{10}, \tau_{\leq 10} pl/o]$ . Therefore, by Proposition 4.2.2 the map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [M^{10}, >^6\tau_{\leq 10} g/o]_{(2)}$  is injective.

For the remaining cases, the homotopy decompositions of  $M^{10}/M^{(6)}$  in (3.30) or (3.31), gives the following commutative diagram

$$\begin{array}{ccc} [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o] & \xrightarrow{d^*} & [M^{10}/M^{(6)}, >^6\tau_{\leq 10} g/o] \\ & \searrow d^* & \swarrow p^* \\ & [X, >^6\tau_{\leq 10} g/o] & \end{array}$$

where  $X = C(\eta^2)$  or  $C(\iota \circ \eta^2)$ , and  $p : M^{10}/M^{(6)} \rightarrow X$  is the quotient map. According to our interest for homotopy inertia group, as  $p^*$  is injective, so first we will check the image of  $\{\eta \circ \mu\} \in [\mathbb{S}^{10}, \Sigma^7 \mathcal{C}]$  under  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o] \rightarrow [X, >^6\tau_{\leq 10} g/o]$ .

Now, suppose  $M^{10}$  satisfies (3.30). Consider the cofiber sequence for  $\eta^2 : \mathbb{S}^9 \rightarrow \mathbb{S}^7$

$$\begin{array}{ccccc} [\mathbb{S}^8, >^6\tau_{\leq 10} g/o]_{(2)} & \xrightarrow{(\eta^2)^*} & [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} & \longrightarrow & [C(\eta^2), >^6\tau_{\leq 10} g/o]_{(2)} \\ & & & \searrow & \\ & & & & [\mathbb{S}^7, >^6\tau_{\leq 10} g/o]_{(2)} \xrightarrow{(\eta^2)^*} [\mathbb{S}^9, >^6\tau_{\leq 10} g/o]_{(2)} \end{array}$$

Here, the group  $[\mathbb{S}^7, >^6\tau_{\leq 10} g/o]_{(2)} = 0$ . Further, we have

$$\begin{array}{ccccc} [\mathbb{S}^8, >^6\tau_{\leq 10} g/o]_{(2)} & \cong & [\mathbb{S}^8, \text{cok}(J)]_{(2)} & \oplus & [\mathbb{S}^8, \text{bso}]_{(2)} \\ \downarrow (\eta^2)^* & & \downarrow (\eta^2)^* & & \downarrow (\eta^2)^* \\ [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} & \cong & [\mathbb{S}^{10}, \text{cok}(J)]_{(2)} & \oplus & [\mathbb{S}^{10}, \text{bso}]_{(2)} \end{array}$$

where  $[\mathbb{S}^{10}, \text{cok}(J)]_{(2)} = 0$  and  $(\eta^2)^* : [\mathbb{S}^8, \text{bso}]_{(2)} \rightarrow [\mathbb{S}^{10}, \text{bso}]_{(2)}$  is non-zero [1, Proposition 7.1]. Therefore, the map  $(\eta^2)^* : [\mathbb{S}^8, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)}$  is surjective, proving that in the exact sequence 4.2

$$[C(\eta^2), >^6\tau_{\leq 10} g/o]_{(2)} = 0. \quad (4.11)$$

This gives in (4.2) the map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [M^{10}/M^{(6)}, >^6\tau_{\leq 10} g/o]_{(2)}$  is trivial. As a result, in (4.7) the map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [M^{10}, >^6\tau_{\leq 10} g/o]_{(2)}$  is trivial.

For the case when  $M^{10}$  satisfies (3.31), consider the exact sequence

$$\begin{array}{ccc}
[\Sigma M(\mathbb{Z}_{2^r}, 7), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \rightarrow & [\mathbb{S}^{10}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \xrightarrow{d^*} [C(\iota \circ \eta^2), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \\
& \nwarrow & \nearrow \\
[M(\mathbb{Z}_{2^r}, 7), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \xrightarrow{\quad} & [\mathbb{S}^9, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \\
& \searrow & \nearrow \\
& [\mathbb{S}^7, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \xrightarrow{(\eta^2)^*} [\mathbb{S}^9, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)]
\end{array}$$

where  $[\mathbb{S}^{10}, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)]_{(2)} = 0$ , and the map

$(\eta^2)^* : [\mathbb{S}^7, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [\mathbb{S}^9, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)]$  is trivial. Thus,

$$[C(\iota \circ \eta^2), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \cong [M(\mathbb{Z}_{2^r}, 7), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)]$$

The cofiber sequence for  $\mathbb{S}^7 \xrightarrow{2^r} \mathbb{S}^7 \rightarrow M(\mathbb{Z}_{2^r}, 7)$  induces the following

$$\begin{array}{ccc}
[\mathbb{S}^8, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \xrightarrow{\times 2^r} & [\mathbb{S}^8, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \rightarrow [M(\mathbb{Z}_{2^r}, 7), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \\
& \nwarrow & \nearrow \\
[\mathbb{S}^7, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] & \xrightarrow{\times 2^r} & [\mathbb{S}^7, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)]
\end{array}$$

Note that  $[\mathbb{S}^8, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] = 0$  and  $[\mathbb{S}^7, {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] \cong \mathbb{Z}$  together implies  $[M(\mathbb{Z}_{2^r}, 7), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] = 0$ , which gives,

$$[C(\iota \circ \eta^2), {}^{>6}\tau_{\leq 10}(\Sigma^{-1}g/pl)] = 0 \quad (4.12)$$

Therefore, it follows from (4.6) that the map

$$\psi_* : [C(\iota \circ \eta^2), \tau_{\leq 10} pl/o] \rightarrow [C(\iota \circ \eta^2), {}^{>6}\tau_{\leq 10} g/o] \quad (4.13)$$

is injective. Consequently, in (4.7),  $d^*(\eta \circ \mu) \notin j_*([M^{10}/M^{(6)}, {}^{>6}\tau_{\leq 10}\Sigma^{-1}(g/pl)])$ . Since, according to Theorem 4.1.3(1)(a), the element  $d^*(\eta \circ \mu)$  is non-zero in  $[M^{10}, \tau_{\leq 10} pl/o]$ , and by Proposition 4.2.2, the element  $d^*(\eta \circ \mu)$  gets mapped non-trivially under  $\psi_* : [M^{10}, \tau_{\leq 10} pl/o] \rightarrow [M^{10}, {}^{>6}\tau_{\leq 10} g/o]$ . Therefore, the map  $d^* : [\mathbb{S}^{10}, {}^{>6}\tau_{\leq 10} g/o] \rightarrow [M^{10}, {}^{>6}\tau_{\leq 10} g/o]$  maps  $\eta \circ \mu$  non-trivially, completing the proof. □

**Theorem 4.2.12.** *Let  $M^{10}$  be a closed, simply-connected, smooth 10-manifold.*

1. If  $M^{10}$  is a spin manifold and  $\Phi = 0$  then

$$I_h(M^{10}) = \begin{cases} \{0\} & \text{if } M^{10} \text{ satisfies (3.29) or (3.31),} \\ \mathbb{Z}_2\{\eta \circ \mu\} & \text{if } M^{10} \text{ satisfies (3.30).} \end{cases}$$

2. If  $M^{10}$  is a spin manifold and  $\Phi \neq 0$  or is a non-spin manifold, then  $I_h(M^{10}) = \mathbb{Z}_2\{\eta \circ \mu\}$ .

*Proof.* The proof follows from (4.4) using for the simply-connected 10-manifolds together with Theorem 4.2.10 and Theorem 4.2.11.  $\square$

**Remark 4.2.13.** Theorem 4.2.10 gives more general result: Let  $M^{10}$  be a closed, oriented, smooth manifold. Then  $I_h(M^{10}) \cap (\Theta_{10})_{(3)} = \{0\}$ .

### 4.3 Inertia group of $\mathbb{RP}^{10}$

Now, we compute the homotopy inertia group of a non simply-connected 10-manifold by considering the proof techniques used for Theorem 4.2.12 with minimal adjustments.

**Theorem 4.3.1.** 1.  $I_c(\mathbb{RP}^{10}) = \mathbb{Z}_3\{\beta_1\}$

2.  $I_h(\mathbb{RP}^{10}) = \mathbb{Z}_3\{\beta_1\} \oplus \mathbb{Z}_2\{\eta \circ \mu\}$

*Proof.* Since  $[\mathbb{S}^{10}, \tau_{\leq 10} pl/o] = [\mathbb{S}^{10}, >^6 \tau_{\leq 10} g/o] = \mathbb{Z}_3\{\beta_1\} \oplus \mathbb{Z}_2\{\eta \circ \mu\}$ , we work locally at prime 2 and 3.

As  $\mathbb{RP}_{(3)}^{10} \simeq *$ , hence  $d^* : [\mathbb{S}^{10}, \tau_{\leq 10} pl/o]_{(3)} \rightarrow [\mathbb{RP}^{10}, \tau_{\leq 10} pl/o]_{(3)}$  is trivial. Thus,

$$\mathbb{Z}_3\{\beta_1\} \subseteq I_c(\mathbb{RP}^{10}) \subseteq I_h(\mathbb{RP}^{10}).$$

1. Now we work 2-locally, so it is enough to study the map  $d^* : [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] \rightarrow [\mathbb{RP}^{10}, \Sigma^7 \mathcal{E}]$ .

In  $H^*(\mathbb{RP}^{10}, \mathbb{Z}_2)$  the formula  $Sq^2(x^8) = 0$  implies that the attaching map of the 10-cell in  $\mathbb{RP}^{10}/\mathbb{RP}^7$  is  $\mathbb{S}^9 \xrightarrow{(2,0)} \mathbb{S}^9 \vee \mathbb{S}^8$ , thus

$$\mathbb{RP}^{10}/\mathbb{RP}^7 \simeq \mathbb{S}^8 \vee M(\mathbb{Z}_2, 9).$$



This cofiber sequence induces the following

$$\begin{array}{ccccc}
 & & \mathbb{Z}_2 & & \\
 & & \Downarrow & & \\
 \dots [\mathbb{S}^{10} \vee \mathbb{S}^9, \Sigma^7 \mathcal{E}] & \xrightarrow{\times(2,0)} & [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] & \xrightarrow{d^*} & [\mathbb{RP}^{10}/\mathbb{RP}^7, \Sigma^7 \mathcal{E}] \\
 & & \nwarrow & & \\
 & & [\mathbb{RP}^{10}, \Sigma^7 \mathcal{E}] & \longrightarrow & [\mathbb{S}^9 \vee \mathbb{S}^8, \Sigma^7 \mathcal{E}] \longrightarrow [\mathbb{S}^9, \Sigma^7 \mathcal{E}]
 \end{array}$$

giving that the map  $d^* : [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] \rightarrow [\mathbb{RP}^{10}/\mathbb{RP}^7, \Sigma^7 \mathcal{E}]$  is injective.

As  $\Sigma^7 \mathcal{E}$  is 6-connected, the cofiber sequence  $\mathbb{RP}^5 \hookrightarrow \mathbb{RP}^{10} \rightarrow \mathbb{RP}^{10}/\mathbb{RP}^5$  gives

$$[\mathbb{RP}^{10}, \Sigma^7 \mathcal{E}] \cong [\mathbb{RP}^{10}/\mathbb{RP}^5, \Sigma^7 \mathcal{E}].$$

So, now we are going to prove that the induced map  $[\mathbb{RP}^{10}/\mathbb{RP}^7, \Sigma^7 \mathcal{E}] \rightarrow [\mathbb{RP}^{10}/\mathbb{RP}^5, \Sigma^7 \mathcal{E}]$ , from the cofiber sequence  $\mathbb{RP}^7/\mathbb{RP}^5 \rightarrow \mathbb{RP}^{10}/\mathbb{RP}^5 \rightarrow \mathbb{RP}^{10}/\mathbb{RP}^7$ , is injective. This concludes that our desired map  $d^* : [\mathbb{S}^{10}, \Sigma^7 \mathcal{E}] \rightarrow [\mathbb{RP}^{10}, \Sigma^7 \mathcal{E}]$  is injective.

Now let us understand the following commutative diagram localized at prime 2

$$\begin{array}{ccccc}
 [\mathbb{S}^7, >^6 \tau_{\leq 10} \Sigma^{-1}(g/pl)] & \xrightarrow{(\eta^2)^*} & [\mathbb{S}^9, >^6 \tau_{\leq 10} \Sigma^{-1}(g/pl)] & & \\
 \parallel & & \uparrow i^* & & \\
 [\Sigma \mathbb{RP}^7/\mathbb{RP}^5, >^6 \tau_{\leq 10} \Sigma^{-1}(g/pl)] & \rightarrow & [\mathbb{RP}^{10}/\mathbb{RP}^7, >^6 \tau_{\leq 10} \Sigma^{-1}(g/pl)] & & \\
 \downarrow j_* & & \downarrow j_* & & \\
 [\Sigma \mathbb{RP}^7/\mathbb{RP}^5, \Sigma^7 \mathcal{E}] & \longrightarrow & [\mathbb{RP}^{10}/\mathbb{RP}^7, \Sigma^7 \mathcal{E}] & \longrightarrow & [\mathbb{RP}^{10}/\mathbb{RP}^5, \Sigma^7 \mathcal{E}]
 \end{array}$$

The rows of this diagram are part of exact sequences induced from cofibrations. In this context, the map  $j_*$  is induced by the composition of maps

$$j : >^6 \tau_{\leq 10} \Sigma^{-1}(g/pl) \rightarrow \tau_{\leq 10} pl/o,$$

followed by the projection map onto the  $\tau_{\leq 10} pl/o$  component  $\Sigma^7 \mathcal{E}$ .

The surjectivity of the map

$$j_* : [\Sigma(\mathbb{RP}^7/\mathbb{RP}^5), >^6 \tau_{\leq 10} \Sigma^{-1}(g/pl)] \rightarrow [\Sigma \mathbb{RP}^7/\mathbb{RP}^5, \Sigma^7 \mathcal{E}],$$

as well as the equality of groups  $[\mathbb{S}^7, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)]$  and  $[\Sigma\mathbb{RP}^7/\mathbb{RP}^5, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)]$ , follow from the fact that  $\mathbb{RP}^7/\mathbb{RP}^5 \simeq \mathbb{S}^6 \vee \mathbb{S}^7$  and the homotopy decompositions of  $>^6\tau_{\leq 10}\Sigma^{-1}(g/pl)$  and  $\tau_{\leq 10} pl/o$ .

The map

$$\iota^* : [\mathbb{RP}^{10}/\mathbb{RP}^7, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)] \rightarrow [\mathbb{S}^9, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)]$$

is induced from the inclusion map  $\iota : \mathbb{S}^9 \rightarrow M(\mathbb{Z}_2, 9)$ . Additionally,  $\iota^*$  is an isomorphism which can be easily checked using the cofiber sequence  $\mathbb{S}^9 \xrightarrow{\times 2} \mathbb{S}^9 \xrightarrow{\iota} M(\mathbb{Z}_2, 9) \vee \mathbb{S}^8 \simeq \mathbb{RP}^{10}/\mathbb{RP}^7$ .

Since the map

$$[\mathbb{S}^7, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)] \rightarrow [\mathbb{S}^9, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)]$$

induced by  $\eta^2 : \mathbb{S}^9 \rightarrow \mathbb{S}^7$  is the zero map, the top commutative square shows that the map

$$[\Sigma\mathbb{RP}^7/\mathbb{RP}^5, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)] \rightarrow [\mathbb{RP}^{10}/\mathbb{RP}^7, >^6\tau_{\leq 10}\Sigma^{-1}(g/pl)]$$

is trivial. Therefore, in the bottom commutative diagram the map  $[\Sigma\mathbb{RP}^7/\mathbb{RP}^5, \Sigma^7\mathcal{E}] \rightarrow [\mathbb{RP}^{10}/\mathbb{RP}^7, \Sigma^7\mathcal{E}]$  is a trivial map. Since it is a part of exact sequence, concludes the map

$$[\mathbb{RP}^{10}/\mathbb{RP}^7, \Sigma^7\mathcal{E}] \rightarrow [\mathbb{RP}^{10}/\mathbb{RP}^5, \Sigma^7\mathcal{E}]$$

is injective. This complete the proof of (1).

2. For  $I_h(\mathbb{RP}^{10})$ , as we have  $\mathbb{Z}/3\{\beta_1\} \subseteq I_c(\mathbb{RP}^{10}) \subseteq I_h(\mathbb{RP}^{10})$ . Therefore, it is enough to work 2-locally. Consider the diagram (4.4), in which we note that  $\psi_* : [\mathbb{S}^{10}, PL/O] \rightarrow [\mathbb{S}^{10}, G/O]$  and  $g' : S^{\text{Diff}}(M^n) \rightarrow [M^n, G/O]$  are injective. So it is enough to show that the map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [\mathbb{RP}^{10}, >^6\tau_{\leq 10} g/o]_{(2)}$  is trivial. Note that, this map  $d^*$

factors through  $[\mathbb{RP}^{10}/\mathbb{RP}^6, >^6\tau_{\leq 10} g/o]_{(2)}$ , as shown in the diagram below,

$$\begin{array}{ccc}
 [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} & \xrightarrow{d^*} & [\mathbb{RP}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \\
 & \searrow d^* & \swarrow q^* \\
 & [\mathbb{RP}^{10}/\mathbb{RP}^6, >^6\tau_{\leq 10} g/o]_{(2)} & 
 \end{array} \quad (4.14)$$

where  $q : \mathbb{RP}^{10} \rightarrow \mathbb{RP}^{10}/\mathbb{RP}^6$  is the quotient map. From this, it is sufficient to prove that the map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [\mathbb{RP}^{10}/\mathbb{RP}^6, >^6\tau_{\leq 10} g/o]_{(2)}$  is trivial.

As  $\mathbb{RP}^{10}/\mathbb{RP}^7 \simeq \mathbb{S}^8 \vee M(\mathbb{Z}/2, 9)$  and  $\Sigma\mathbb{RP}^7/\mathbb{RP}^6 \simeq \mathbb{S}^8$ , consider the following commutative diagram

$$\begin{array}{ccccc}
 [\Sigma\mathbb{RP}^7/\mathbb{RP}^6, >^6\tau_{\leq 10} g/o]_{(2)} & \xrightarrow{f^*} & [\mathbb{RP}^{10}/\mathbb{RP}^7, >^6\tau_{\leq 10} g/o]_{(2)} & \xrightarrow{q^*} & [\mathbb{RP}^{10}/\mathbb{RP}^6, >^6\tau_{\leq 10} g/o]_{(2)} \\
 \parallel & & \parallel & & \uparrow d^* \\
 [\mathbb{S}^8, >^6\tau_{\leq 10} g/o]_{(2)} & \xrightarrow{f^*} & [\mathbb{S}^8 \vee M(\mathbb{Z}/2, 9), >^6\tau_{\leq 10} g/o]_{(2)} & \xleftarrow{d^*} & [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)}
 \end{array} \quad (4.15)$$

whose top row is a part of exact sequence induced from the cofiber sequence  $\mathbb{RP}^{10}/\mathbb{RP}^6 \xrightarrow{q} \mathbb{RP}^{10}/\mathbb{RP}^7 \xrightarrow{f} \Sigma\mathbb{RP}^7/\mathbb{RP}^6$ . Furthermore, the injectivity of the map  $d^* : [\mathbb{S}^{10}, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [\mathbb{S}^8 \vee M(\mathbb{Z}/2, 9), >^6\tau_{\leq 10} g/o]_{(2)}$  can be verified using a long exact sequence induced from a cofibration sequence  $\mathbb{S}^9 \xrightarrow{\times 2} \mathbb{S}^9 \rightarrow M(\mathbb{Z}/2, 9)$ .

Observe that, the connecting map  $f : \mathbb{RP}^{10}/\mathbb{RP}^7 \rightarrow \Sigma\mathbb{RP}^7/\mathbb{RP}^6$  is homotopic to the map  $(2, \phi) : \mathbb{S}^8 \vee M(\mathbb{Z}/2, 9) \rightarrow \mathbb{S}^8$ . Here, the map  $\phi$  fits in the following homotopy commutative triangle

$$\begin{array}{ccc}
 M(\mathbb{Z}/2, 9) & \xrightarrow{\phi} & \mathbb{S}^8 \\
 \uparrow \iota & \nearrow \eta & \\
 \mathbb{S}^9 & & 
 \end{array}$$

According to [12, Lemma 2.3], there is a non-split short exact sequence

$$\begin{array}{ccccccc}
 0 & \longrightarrow & [\mathbb{S}^{10}, \mathbb{S}^8]_{(2)} & \xrightarrow{d^*} & [M(\mathbb{Z}/2, 9), \mathbb{S}^8]_{(2)} & \longrightarrow & [\mathbb{S}^9, \mathbb{S}^8]_{(2)} \longrightarrow 0 \\
 & & \wr \parallel & & & & \wr \parallel \\
 & & \mathbb{Z}/2 = \{\eta^2\} & & & & \mathbb{Z}/2 = \{\eta\}
 \end{array} \quad (4.16)$$

which makes  $\phi$  the generator of  $[M(\mathbb{Z}/2, 9), \mathbb{S}^8]_{(2)} = \mathbb{Z}/4$ . In (4.16), observe that  $d^*(\eta^2) = 2\phi$ . This implies that the induced map  $(2\phi)^* : [\mathbb{S}^8, >^6\tau_{\leq 10} g/o]_{(2)} \rightarrow [M(\mathbb{Z}/2, 9), >^6\tau_{\leq 10} g/o]_{(2)}$

fits into the following commutative diagram

$$\begin{array}{ccc}
 [\mathbb{S}^8, >^6 \tau_{\leq 10} g/o]_{(2)} & & \\
 (\eta^2)^* \downarrow & \searrow (2\phi)^* & \\
 [\mathbb{S}^{10}, >^6 \tau_{\leq 10} g/o]_{(2)} & \xrightarrow{d^*} & [M(\mathbb{Z}/2, 9), >^6 \tau_{\leq 10} g/o]_{(2)}.
 \end{array}$$

Since  $(\eta^2)^*$  is surjective implies

$$\begin{aligned}
 \text{Im}\left(d^*: [\mathbb{S}^{10}, >^6 \tau_{\leq 10} g/o]_{(2)} \rightarrow [M(\mathbb{Z}/2, 9), >^6 \tau_{\leq 10} g/o]_{(2)}\right) &\subseteq \text{Im}((2\phi)^*) \\
 &\subseteq \text{Im}(\phi^*) \subseteq \text{Im}((2, \phi)^*) = \text{Im}(f^*).
 \end{aligned}$$

Now, it follows from the top row exact sequence in (4.15) that

$$d^*: [\mathbb{S}^{10}, >^6 \tau_{\leq 10} g/o]_{(2)} \rightarrow [\mathbb{RP}^{10}/\mathbb{RP}^6, >^6 \tau_{\leq 10} g/o]_{(2)} \text{ is trivial, which completes the proof.}$$

□

The following corollary is an immediate consequence of Theorem 4.3.1, and the fact that  $I_h(\mathbb{RP}^{10}) \subseteq I(\mathbb{RP}^{10}) \subseteq \Theta_{10}$ .

**Corollary 4.3.2.**  $I(\mathbb{RP}^{10}) = \Theta_{10}$ .

## **Part II**



# Chapter 5

## On the Connected Sum of Manifolds

In this chapter, our interest is to compute certain structure sets and cohomotopy groups of the connected sum of manifolds. One of the primary motives for studying the connected sum of manifolds comes from geometric topology, wherein it plays a vital role in the classification of manifolds up to T-stabilization. It is well studied in [20, 26, 32, 33].

In the category of manifolds, the classification problem involves studying different structure sets. Interestingly, many of these sets like stable cohomotopy groups, real or complex  $K$ -groups, smooth normal invariant sets, etc., exhibit a bijective correspondence with  $[M, Y]$ , for a manifold  $M$  and an infinite loop space  $Y$  [30, 34, 43, 51]. It is worth noting that the set  $[M, Y]$  becomes more accessible from the viewpoint of algebraic topology as it fits into an exact sequence of abelian groups. In this chapter, spaces of our interest are the connected sum of closed, oriented, smooth  $n$ -manifolds  $M_i$ , denoted by  $\#^{i \in I} M_i$ , for  $i \in I = \{1, 2, \dots, k\}$ .

Let  $M_i$  be a closed, oriented, smooth  $n$ -manifold for  $i \in I = \{1, 2, \dots, k\}$ . To obtain the connected sum of  $M_i$ 's first delete a smaller open  $n$ -dimensional disk from the top cell of each  $M_i$  and denote the obtained manifold with boundary by  $\overset{o}{M}_i$ . Additionally, delete  $k$  number of smaller pairwise disjoint  $n$ -dimensional open disks  $\bigsqcup_{i \in I} \overset{o}{\mathbb{D}}_i^n$  from the standard sphere  $\mathbb{S}^n$ . Now, glue the boundary  $\mathbb{S}_i^{n-1}$  of the  $\overset{o}{M}_i$  homeomorphically onto the boundary  $\mathbb{S}_i^{n-1}$  in  $\mathbb{S}^n$  for each

$i \in I$ , in such a way that the resulting space is a closed, oriented, smooth  $n$ -manifold (pictorial presentation up to homotopy is given in Figure 5.1). Let us denote the connected sum of  $M_i$ 's by  $\#^{i \in I} M_i$ .

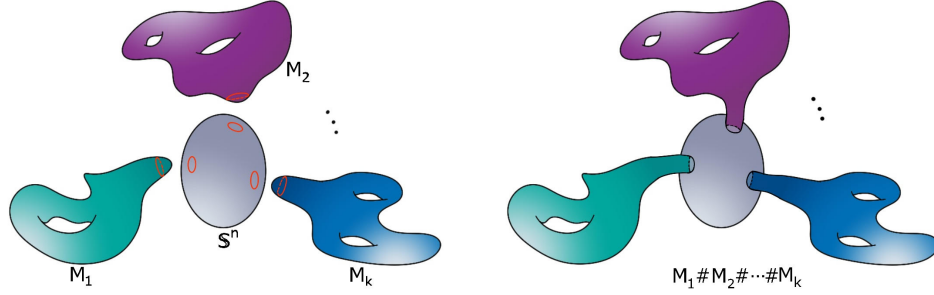


Figure 5.1: Connected sum of  $k$ -many  $n$ -manifolds

Now, for the  $\#^{i \in I} M_i$  and for any homotopy associative  $H$ -space  $Y$ , it is natural to ask the question, if we know  $[M_i, Y]$  for  $i \in I$ , then using this can we compute  $[\#^{i \in I} M_i, Y]$ ? Formally we ask the following questions:

**Question 5.0.1.** *Can we express  $[\#^{i \in I} M_i, Y]$  in terms of  $[M_i, Y]$  for  $i \in I$ ?*

**Question 5.0.2.** *How is the group  $[\#^{i \in I} M_i, Y]$  related with each group  $[M_i, Y]$ ? Under what conditions can one establish injective or surjective relations among these groups?*

To address Question 5.0.1, we present a Theorem 5.1.1 which gives a short exact sequence for computing the group  $[\#^{i \in I} M_i, Y]$  in terms of values associated with  $[M_i, Y]$ . Additionally, in consideration of Question 5.0.2, we establish conditions on the collapse map  $\psi : M_1 \# M_2 \rightarrow M_1$  to clarify the relation between the groups  $[M_1, Y]$  and  $[M_1 \# M_2, Y]$  in Theorem 5.1.3. Furthermore, it is demonstrated that the concordance inertia group of a connected sum of manifolds is the sum of the individual manifold concordance inertia groups.

In geometric topology, a typical way to stabilize a manifold is to study the homotopy decomposition of the connected sum of a manifold with projective spaces. In particular, the study of the connected sum of projective spaces is essential. To illustrate the applications of the theorems discussed in Sections 5.1 and 5.2, we present computations for the connected sum of projective spaces in Sections 5.3 and 5.4 particularly for  $Y = PL/O$  and  $SF$ . Recall that,  $PL/O$  is a homotopy fiber of the natural map  $BO \rightarrow BPL$ . Here,  $BO$  is the classifying spaces for stable real vector bundles, and  $BPL$  is the classifying space for the group  $PL = \bigcup_n PL_n$  where  $PL_n$  is the group of self  $PL$ -homeomorphisms of  $\mathbb{R}^n$  (Chapter 2 or Lemma 10.6 of [54]). The space



$SF$  is the limit of the spaces  $SF_n$  over  $n$ , where  $SF_n$  consists of base point preserving degree one maps from  $\mathbb{S}^{n-1}$  to itself (Chapter 2 or Sections 2 and 3 of [37]).

## 5.1 Maps from the connected sum of manifolds to infinite loop spaces

Let  $M_1, M_2, \dots, M_k$  be closed, oriented, smooth  $n$ -dimensional manifolds with homotopy decomposition  $M_i \simeq M_i^{(n-1)} \cup_{h_i} \mathbb{D}^n$ , where  $h_i : \mathbb{S}_i^{n-1} \rightarrow M_i^{(n-1)}$  is the attaching map for the top cell with  $i \in I = \{1, 2, \dots, k\}$ . Then we have

$$\#^{i \in I} M_i \simeq \left( \bigvee_{i \in I} M_i^{(n-1)} \right) \bigcup_{\xi} \mathbb{D}^n,$$

where  $\xi = (\bigvee_{i \in I} h_i) \circ p$  is the attaching map for the top cell with  $p : \mathbb{S}^{n-1} \rightarrow \bigvee_{i \in I} \mathbb{S}_i^{n-1}$  as the pinch map. Thus we have the cofiber sequence  $\mathbb{S}^{n-1} \xrightarrow{\xi} \bigvee_{i \in I} M_i^{(n-1)} \xrightarrow{\iota} \#^{i \in I} M_i$ . From this sequence, we obtain the following long exact sequence by applying the contravariant functor  $[-, Y]$  for an infinite loop space  $Y$

$$\begin{array}{ccccccc} \dots & \longrightarrow & \bigoplus_{i \in I} [\Sigma^{S+1} M_i^{(n-1)}, Y] & \xrightarrow{(\Sigma^{S+1} \xi)^*} & [\mathbb{S}^{n+S}, Y] & \xrightarrow{(\Sigma^S d)^*} & [\Sigma^S (\#^{i \in I} M_i), Y] \\ & & & & \searrow (\Sigma^S \iota)^* & & \\ & & \bigoplus_{i \in I} [\Sigma^S M_i^{(n-1)}, Y] & \longrightarrow & \dots & \xrightarrow{(\Sigma^S \xi)^*} & [\mathbb{S}^n, Y] \\ & & & & \searrow d^* & & \\ & & \#^{i \in I} M_i, Y & \xrightarrow{\iota^*} & \bigoplus_{i \in I} [M_i^{(n-1)}, Y] & \xrightarrow{\xi^*} & [\mathbb{S}^{n-1}, Y], \end{array} \quad (5.1)$$

where  $d^*$  is induced by the degree one map  $d : \#^{i \in I} M_i \rightarrow \mathbb{S}^n$  and  $\Sigma^S$  denotes the positive  $s$ -fold suspension.

We use the property of the pinch map in sequence (5.1) to derive the following theorem:

**Theorem 5.1.1.** *Let  $Y$  be an infinite loop space. Then there exists a short exact sequence*

$$0 \longrightarrow \operatorname{Coker}\left((\Sigma^{S+1}\xi)^*\right) \xrightarrow{\widetilde{(\Sigma^S d)^*}} [\Sigma^S(\#^{i \in I} M_i), Y] \xrightarrow{(\Sigma^S \iota)^*} \operatorname{Ker}\left((\Sigma^S \xi)^*\right) \longrightarrow 0,$$

where the map  $\widetilde{(\Sigma^S d)^*}$  is induced by the degree one map  $\Sigma^S d : \Sigma^S(\#^{i \in I} M_i) \rightarrow \mathbb{S}^{n+s}$ ,

$$\operatorname{Coker}\left((\Sigma^{S+1}\xi)^*\right) = \pi_{n+s}(Y) \Big/ \sum_{i \in I} \operatorname{Im}\left((\Sigma^{S+1} h_i)^*\right)$$

and

$$\operatorname{Ker}\left((\Sigma^S \xi)^*\right) = \operatorname{Ker}\left(\sum_{i \in I} (\Sigma^S h_i)^*\right).$$

*Proof.* Using the long exact sequence (5.1), we obtain the following short exact sequence

$$0 \longrightarrow \pi_{n+s}(Y) \Big/ \operatorname{Ker}\left((\Sigma^S d)^*\right) \xrightarrow{\widetilde{(\Sigma^S d)^*}} [\Sigma^S(\#^{i \in I} M_i), Y] \xrightarrow{(\Sigma^S \iota)^*} \operatorname{Im}\left((\Sigma^S \iota)^*\right) \longrightarrow 0,$$

where  $\widetilde{(\Sigma^S d)^*}$  is the induced map by the degree one map  $(\Sigma^S d) : \Sigma^S(\#^{i \in I} M_i) \rightarrow \mathbb{S}^{n+s}$ . Using exactness of the sequence (5.1), we have

$$\operatorname{Im}\left((\Sigma^S \iota)^*\right) = \operatorname{Ker}\left((\Sigma^S \xi)^*\right)$$

and

$$\operatorname{Ker}\left((\Sigma^S d)^*\right) = \operatorname{Im}\left((\Sigma^{S+1} \xi)^*\right).$$

This offers the existence of the required short exact sequence.

Now we express the  $\operatorname{Im}\left((\Sigma^{S+1} \xi)^*\right)$  in terms of  $\operatorname{Im}\left((\Sigma^{S+1} h_i)^*\right)$ , and the  $\operatorname{Ker}\left((\Sigma^S \xi)^*\right)$  in terms of  $(\Sigma^S h_i)^*$ . Note that  $\xi = (\vee_{i \in I} h_i) \circ p$ , where  $p$  is the pinch map. For any  $m \in \mathbb{N}$ , the

pinch map  $p : \mathbb{S}^m \rightarrow \bigvee_{i \in I} \mathbb{S}_i^m$  induces the map  $p^* : \bigoplus_{i \in I} [\mathbb{S}_i^m, Y] \rightarrow [\mathbb{S}^m, Y]$  with the property that it maps any element  $(x_1, x_2, \dots, x_k)$  to  $\sum_{i \in I} x_i$ . This implies that for any positive  $s$ -fold suspension of  $\xi$ , we have  $(\Sigma^s \xi)^* = \sum_{i \in I} (\Sigma^s h_i)^*$  which leads to the following:

$$\mathrm{Im}\left((\Sigma^{s+1} \xi)^*\right) = \sum_{i \in I} \mathrm{Im}\left((\Sigma^{s+1} h_i)^*\right) \quad (5.2)$$

and

$$\mathrm{Ker}\left((\Sigma^s \xi)^*\right) = \mathrm{Ker}\left(\sum_{i \in I} (\Sigma^s h_i)^*\right). \quad (5.3)$$

This completes the proof.  $\square$

If we consider a  $k$ -fold connected sum of a manifold  $M$ ,  $\#^k M$ , in Theorem 5.1.1, then we have the following version:

**Theorem 5.1.2.** *Let  $M$  be a closed, oriented, smooth  $n$ -manifold. Then the*

$$\mathrm{Coker}\left((\Sigma^{s+1} \xi)^*\right) = \pi_{n+s}(Y) / \mathrm{Im}\left((\Sigma^{s+1} h)^*\right)$$

and

$$\mathrm{Ker}\left((\Sigma^s \xi)^*\right) \cong \left(\bigoplus_{k=1} [\Sigma^s M^{(n-1)}, Y]\right) \oplus \mathrm{Ker}\left((\Sigma^s h)^*\right),$$

where  $\xi = (\bigvee_k h) \circ p$  is the attaching map for  $\#^k M$  with  $h : \mathbb{S}^{n-1} \rightarrow M^{(n-1)}$  as the attaching map for  $M$  and  $p$  is the pinch map.

*Proof.* We have  $\xi = (\bigvee_k h) \circ p$ , hence as a particular case of (5.2), we obtain

$$\mathrm{Im}\left((\Sigma^{s+1} \xi)^*\right) = \mathrm{Im}\left((\Sigma^{s+1} h)^*\right).$$

Moreover, we have

$$\mathrm{Ker}((\Sigma^S \xi)^*) \cong (\oplus_{k-1} [\Sigma^S M^{(n-1)}, Y]) \oplus \mathrm{Ker}((\Sigma^S h)^*).$$

This holds since any element  $(x_1, x_2, \dots, x_k)$  of  $\oplus_k [\Sigma^S M^{(n-1)}, Y]$  lies in  $\mathrm{Ker}((\Sigma^S \xi)^*)$  if and only if  $\sum_{i \in I} (\Sigma^S h)^*(x_i) = (\Sigma^S h)^*(\sum_{i \in I} x_i)$  is 0. Therefore the sum  $\sum_{i \in I} x_i = x$  for some  $x \in \mathrm{Ker}((\Sigma^S \xi)^*)$  which implies  $x_k = x - (\sum_{i=1}^{k-1} x_i)$ . Hence the map

$$\begin{aligned} (\oplus_{k-1} [\Sigma^S M^{(n-1)}, Y]) \oplus \mathrm{Ker}(\Sigma^S h^*) &\longrightarrow \mathrm{Ker}((\Sigma^S \xi)^*) \\ (x_1, x_2, \dots, x_{k-1}, x) &\mapsto (x_1, x_2, \dots, x_{k-1}, x - (\sum_{i=1}^{k-1} x_i)) \end{aligned}$$

is indeed an isomorphism, and this completes the proof.  $\square$

We conclude this section by exhibiting a relation between the groups  $[M_1, Y]$  and  $[M_1 \# M_2, Y]$ . In particular, we present conditions for the injectivity and surjectivity of the induced collapse map from  $[M_1, Y]$  to  $[M_1 \# M_2, Y]$ .

**Theorem 5.1.3.** *Let  $M_1$  and  $M_2$  be closed, oriented, smooth  $n$ -manifolds with  $M_i = M_i^{(n-1)} \cup_{h_i} \mathbb{D}^n$ , where  $h_i : \mathbb{S}^{n-1} \rightarrow M_i^{(n-1)}$  is the attaching map for  $i = 1, 2$ . Let  $\psi : M_1 \# M_2 \rightarrow M_1$  be the collapse map and  $Y$  be an infinite loop space. Then for the attaching map  $\xi := (h_1 \vee h_2) \circ p$  of  $M_1 \# M_2$ , where  $p$  is the pinch map, the following holds:*

- (i) *The map  $\psi^* : [M_1, Y] \rightarrow [M_1 \# M_2, Y]$  is injective if and only if  $\mathrm{Im}((\Sigma \xi)^*) = \mathrm{Im}((\Sigma h_1)^*)$ .*
- (ii) *The map  $\psi^* : [M_1, Y] \rightarrow [M_1 \# M_2, Y]$  is surjective if and only if  $\mathrm{Ker}(\xi^*) \cong \mathrm{Ker}(h_1^*)$ .*

*Proof.* Consider the following homotopy commutative diagram

$$\begin{array}{ccccccc} \mathbb{S}^{n-1} & \xrightarrow{\xi} & M_1^{(n-1)} \vee M_2^{(n-1)} & \xhookrightarrow{\iota} & M_1 \# M_2 & \xrightarrow{d} & \dots \\ \downarrow g & & \downarrow \phi & & \downarrow \psi & & \\ \mathbb{S}^{n-1} & \xrightarrow{h_1} & M_1^{(n-1)} & \xhookrightarrow{\iota_1} & M_1 & \xrightarrow{d} & \dots \end{array}$$

where  $\xi = (h_1 \vee h_2) \circ p$  for the pinch map  $p$ , horizontal sequences are cofiber sequences associated to maps  $\xi$  and  $h_1$ , and maps  $g$ ,  $\phi$  and  $\psi$  are collapse maps. Now, applying the contravariant

functor  $[-, Y]$  onto this diagram, we obtain the following commutative diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & \pi_n(Y) / \text{Im}((\Sigma h_1)^*) & \xrightarrow{\widetilde{d}^*} & [M_1, Y] & \xrightarrow{\iota_1^*} & \text{Ker}(h_1^*) \longrightarrow 0 \\
& & \downarrow \widetilde{(\Sigma g)^*} & & \downarrow \psi^* & & \downarrow \widetilde{\phi}^* \\
0 & \longrightarrow & \pi_n(Y) / \text{Im}((\Sigma \xi)^*) & \xrightarrow{\widetilde{d}^*} & [M_1 \# M_2, Y] & \xrightarrow{\iota^*} & \text{Ker}(\xi^*) \longrightarrow 0
\end{array} \tag{5.4}$$

where horizontal short exact sequences are obtained using Theorem 5.1.1. Observe that, the induced map  $\widetilde{(\Sigma g)^*} : \pi_n(Y) / \text{Im}((\Sigma h_1)^*) \rightarrow \pi_n(Y) / \text{Im}((\Sigma \xi)^*)$ , by  $(\Sigma g)^*$ , is surjective; and the induced map  $\widetilde{\phi}^* : \text{Ker}(h_1^*) \rightarrow \text{Ker}(\xi^*)$ , by  $\phi^*$ , is injective.

- (i) Let  $\psi^* : [M_1, Y] \rightarrow [M_1 \# M_2, Y]$  be injective. This implies  $\widetilde{(\Sigma g)^*}$  is an isomorphism. Note that we always have  $\text{Im}((\Sigma h_1)^*) \subseteq \text{Im}((\Sigma \xi)^*)$ . Therefore, we obtain  $\text{Im}((\Sigma h_1)^*) = \text{Im}((\Sigma \xi)^*)$ .

On the other hand,  $\text{Im}((\Sigma h_1)^*) = \text{Im}((\Sigma \xi)^*)$  implies the map  $\widetilde{(\Sigma g)^*}$  is injective. Using the fact that map  $\widetilde{\phi}^*$  is injective in (5.4), by five lemma we obtain the result.

- (ii) Suppose the map  $\psi^* : [M_1, Y] \rightarrow [M_1 \# M_2, Y]$  is surjective. Therefore, in (5.4), the map  $\widetilde{\phi}^*$  turns out to be surjective, and hence an isomorphism.

Conversely, suppose the map  $\widetilde{\phi}^*$  be an isomorphism. Then, using the fact that  $\widetilde{(\Sigma g)^*}$  is surjective, by five lemma we conclude that the map  $\psi^*$  is surjective, which completes the proof.

□

Recall that, the kernel of the induced degree one map  $d^* : [\mathbb{S}^n, PL/O] \rightarrow [M, PL/O]$  can be identified with the concordance inertia group of  $M$ ,  $I_c(M)$ . In Theorem 5.1.3 (i), if we consider  $Y = PL/O$ , then the equivalent statement we have is as follows:

**Corollary 5.1.4.** *The map  $\psi^* : \mathcal{C}(M_1) \rightarrow \mathcal{C}(M_1 \# M_2)$  is injective if and only if  $I_c(M_1 \# M_2) = I_c(M_1)$ .*

## 5.2 Sum formula and the concordance structure set of projective spaces

Recall that, in case of standard  $n$ -sphere, the set  $\mathcal{C}(\mathbb{S}^n)$  coincides with the group  $\Theta_n$  of  $h$ -cobordism classes of homotopy  $n$ -spheres, for all  $n \geq 5$  [29]. As mentioned in Chapter 2, the subgroups of  $\Theta_n$  play an important role in the classification problem. In [57], Wilkens demonstrated that  $I(M\#N)$  is not equal to  $I(M) \oplus I(N)$  in general. In contrast to this finding, Section 5.2 demonstrates that this equality holds for the concordance inertia group. In particular, we see that, the concordance inertia group of individual manifolds holds the complete information needed to determine the concordance inertia group of their connected sum.

In this section, we begin by establishing a crucial result that, the concordance inertia groups of individual manifolds hold the complete information needed to determine the concordance inertia group of their connected sum.

Let  $M_i$  be a closed, oriented, smooth  $n$ -manifold with the attaching map  $h_i : \mathbb{S}^{n-1} \rightarrow M_i^{(n-1)}$ , for  $i \in I$ . Recall that, if  $d : M_i \rightarrow \mathbb{S}^n$  is a degree one map, then the kernel of the induced map  $d^* : [\mathbb{S}^n, PL/O] \rightarrow [M_i, PL/O]$  can be identified with the concordance inertia group of  $M_i$ ,  $I_c(M_i)$ . The induced long exact sequence associated with the map  $h_i$ ,

$$\cdots \longrightarrow [\Sigma M_i^{n-1}, PL/O] \xrightarrow{(\Sigma h_i)^*} [M_i, PL/O] \xrightarrow{d^*} [\mathbb{S}^n, PL/O]$$

gives

$$I_c(M_i) = \text{Ker}(d^*) = \text{Im}((\Sigma h_i)^*).$$

Now, let us consider  $Y = PL/O$  in the sequence (5.1). Then we have  $\text{Ker}(d^*) = \text{Im}((\Sigma \xi)^*)$ . Combining this with the equation (5.2), we have the following theorem.

**Theorem 5.2.1.** *Let  $M_1, M_2, \dots, M_k$  be closed, oriented, smooth  $n$ -dimensional manifolds. Then*

$$I_c(\#^{i \in I} M_i) = \sum_{i \in I} I_c(M_i) .$$

In particular, if  $M_i = M$ , for all  $i = 1, 2, \dots, k$ , then

$$I_c(\#^k M) = I_c(M) .$$

**Observation 5.2.2.** *The group  $I_c(\#^{i \in I} M_i)$  is trivial if and only if each  $I_c(M_i)$  is trivial for  $i \in I$ .*

Let  $A$  be  $(n-1)$ -connected closed, oriented, smooth  $2n$ -manifold for  $n \geq 2$ . According to Wall's theorem,  $A$  can be represented as the minimal cell decomposition  $A = (\vee_r \mathbb{S}^n) \cup_h \mathbb{D}^{2n}$ , where  $h : \mathbb{S}^{2n-1} \rightarrow \vee_r \mathbb{S}^n$  is the attaching map. Within the range of  $4 \leq n \leq 6$ , by utilizing the cofiber sequence derived from the attaching map  $h$  and the fact that  $\Theta_m = 0$  for  $m = 3, 4, 5, 6, 12$ , it is apparent that  $I_c(A) = 0$ . Thus, because of Theorem 5.2.1, for any closed, oriented, smooth  $2n$ -dimensional manifold  $M$  with  $4 \leq n \leq 6$  we get  $I_c(M \# A) = I_c(M)$ , and Corollary 5.1.4 says that the map  $\psi^* : \mathcal{C}(M) \rightarrow \mathcal{C}(M \# A)$  is injective. In these cases, we also get that the map  $\psi^*$  is surjective. This can be easily verified with the help of the cofiber sequence  $\vee_r \mathbb{S}^n \xrightarrow{\iota} M \# A \xrightarrow{q} M$ , where  $\iota$  is the inclusion of  $(2n-1)$ -skeleton of  $A$ . As an outcome, we get the following result.

**Corollary 5.2.3.** *Let  $M$  and  $A$  be closed, oriented, smooth  $2n$ -dimensional manifolds. In addition, if  $A$  is a  $(n-1)$ -connected then, for  $4 \leq n \leq 6$ ,*

$$\mathcal{C}(M \# A) \cong \mathcal{C}(M) .$$

For example, if  $M^{2n}$  is a closed, oriented, smooth manifold then we have the following results for any  $k \in \mathbb{N}$ :

- (i)  $\mathcal{C}(M^8 \# (\#^k \mathbb{H}\mathbb{P}^2)) \cong \mathcal{C}(M^8)$ .
- (ii)  $\mathcal{C}(M^{10} \# (\#^k (\mathbb{S}^5 \times \mathbb{S}^5))) \cong \mathcal{C}(M^{10})$ .
- (iii)  $\mathcal{C}(M^{12} \# (\#^k (\mathbb{S}^6 \times \mathbb{S}^6))) \cong \mathcal{C}(M^{12})$ .

Recall that, the homotopy inertia group  $I_h(M^n)$  consists of  $\Sigma^n \in \Theta_n$  such that there exists a diffeomorphism  $M^n \# \Sigma^n \rightarrow M^n$  that is homotopic to the canonical homeomorphism  $h_{\Sigma^n} : M^n \# \Sigma^n \rightarrow M^n$ . The sum formula is also valid for the homotopy inertia group under certain conditions on manifolds, stated below.

**Proposition 5.2.4.** *Let  $M_1, M_2, \dots, M_k$  be closed, simply-connected, smooth  $n$ -dimensional manifolds such that for each  $M_i$  odd degree cohomologies with integral and  $\mathbb{Z}_2$ -coefficients vanish. Then*

$$I_h(\#^{i \in I} M_i) = \sum_{i \in I} I_h(M_i) .$$

*Proof.* By [24, Corollary 3.2], if  $M_i$  is a closed, simply-connected, smooth manifold such that odd degree cohomologies with integral and  $\mathbb{Z}_2$ -coefficients vanish, then  $I_h(M_i) = I_c(M_i)$ . Thus, assuming conditions from the statement we obtain  $\sum_{i \in I} I_h(M_i) = \sum_{i \in I} I_c(M_i)$  and  $I_h(\#^{i \in I} M_i) = I_c(\#^{i \in I} M_i)$ . Combining this with Theorem 5.2.1 completes the proof.  $\square$

**Remark 5.2.5.** *Let  $k, n \in \mathbb{N}$ .*

1. *If  $n \leq 8$  then  $I_h(\#^k \mathbb{CP}^n) = 0$ , since  $I_c(\mathbb{CP}^n) = 0$  by [27].*
2. *If  $n \leq 4$  then  $I_h(\#^k \mathbb{HP}^n) = 0$ , since  $I_c(\mathbb{HP}^n) = 0$  by [2].*
3. *By [23]  $I_c(\mathbb{OP}^2) = 0$  hence  $I_h(\#^k \mathbb{OP}^2) = 0$ .*

As applications of the results obtained thus far, discussed in Sections 5.1 and 5.2, we conduct computations in the subsequent sections. Specifically, we focus on the  $k$ -fold connected sum of projective spaces, as well as cases where  $Y = PL/O$  and  $SF$ .

### 5.3 Concordance structure set of $k$ -fold connected sum of projective spaces

Recall that,  $\mathbb{CP}^n = \mathbb{CP}^{n-1} \cup_{h_C} \mathbb{D}^{2n}$ , where  $h_C : \mathbb{S}^{2n-1} \rightarrow \mathbb{CP}^{n-1}$  is the Hopf map. The groups  $\mathcal{C}(\mathbb{CP}^n)$  are computed for  $n = 3, 4$  in [22] and for  $5 \leq n \leq 8$  in [25]. Using Theorems 5.1.1 and 5.1.2, we now compute the concordance structure set of  $k$ -fold connected sum of complex projective spaces.

**Corollary 5.3.1.** (i) *There is an isomorphism*

$$d^* : \Theta_8 \rightarrow \mathcal{C}(\#^k \mathbb{CP}^4),$$



where  $\Theta_8 \cong \mathbb{Z}_2$ .

(ii) *There is a short exact sequence*

$$0 \rightarrow \Theta_{10} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{CP}^5) \xrightarrow{\iota^*} \oplus_k \mathcal{C}(\mathbb{CP}^4) \rightarrow 0,$$

where  $\Theta_{10} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$  and  $\mathcal{C}(\mathbb{CP}^4) \cong \mathbb{Z}_2$ .

(iii) *There is an isomorphism*

$$\iota^* : \mathcal{C}(\#^k \mathbb{CP}^6) \rightarrow \oplus_{k-1} \mathcal{C}(\mathbb{CP}^5) \oplus \text{Ker}(h_C^*),$$

where  $\mathcal{C}(\mathbb{CP}^5) \cong \mathbb{Z}_2^2 \oplus \mathbb{Z}_3$  and  $\text{Ker}(h_C^* : \mathcal{C}(\mathbb{CP}^5) \rightarrow \Theta_{11}) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$ .

(iv) *There is a short exact sequence*

$$0 \rightarrow \Theta_{14} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{CP}^7) \xrightarrow{\iota^*} \oplus_{k-1} \mathcal{C}(\mathbb{CP}^6) \oplus \text{Ker}(h_C^*) \rightarrow 0,$$

where  $\Theta_{14} \cong \mathbb{Z}_2$ ,  $\mathcal{C}(\mathbb{CP}^6) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$  and  $\text{Ker}(h_C^* : \mathcal{C}(\mathbb{CP}^6) \rightarrow \Theta_{13}) \cong \mathbb{Z}_2$ .

(v) *There is a short exact sequence*

$$0 \rightarrow \Theta_{16} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{CP}^8) \xrightarrow{\iota^*} \oplus_{k-1} \mathcal{C}(\mathbb{CP}^7) \oplus \text{Ker}(h_C^*) \rightarrow 0,$$

where  $\Theta_{16} \cong \mathbb{Z}_2$ ,  $\mathcal{C}(\mathbb{CP}^7) \cong \mathbb{Z}_2^2$  and  $\text{Ker}(h_C^* : \mathcal{C}(\mathbb{CP}^7) \rightarrow \Theta_{15}) \cong \mathbb{Z}_2$ .

*Proof.* Using Remark 5.2.5 (i), for  $n \leq 8$ , we conclude that the group  $I_c(\#^k \mathbb{CP}^n)$  is trivial which implies that the induced map  $d^* : \Theta_{2n} \rightarrow \mathcal{C}(\#^k \mathbb{CP}^n)$  is injective. Therefore, for  $\#^k \mathbb{CP}^n$  the short exact sequence derived from Theorems 5.1.1 and 5.1.2 for  $Y = PL/O$  is as follows:

$$0 \rightarrow \Theta_{2n} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{CP}^n) \xrightarrow{\iota^*} \oplus_{k-1} \mathcal{C}(\mathbb{CP}^{n-1}) \oplus \text{Ker}(h_C^*) \rightarrow 0, \quad (5.5)$$

where  $h_C : \mathbb{S}^{2n-1} \rightarrow \mathbb{CP}^{n-1}$  is the Hopf map. Then we have the following:

- (i) Consider  $n = 4$  in (5.5). It has been proved that  $\mathcal{C}(\mathbb{CP}^3)$  is trivial in [22, Theorem 2.3 (i)]. This implies that  $d^* : \Theta_8 \rightarrow \mathcal{C}(\#^k \mathbb{CP}^4)$  is an isomorphism. In particular, we get that

$$\mathcal{C}(\#^k \mathbb{CP}^4) = \{[\#^k \mathbb{CP}^4], [\#^k \mathbb{CP}^4 \# \Sigma^8]\},$$

where  $\Sigma^8$  is an exotic 8-sphere. This conveys that the  $\#^k \mathbb{CP}^4$  has exactly two distinct smooth structures up to concordance, and at most two distinct smooth structures up to diffeomorphism.

- (ii) It is evident from (i) that  $\mathcal{C}(\mathbb{CP}^4) \cong \mathbb{Z}_2$  which is also computed in [22, Theorem 2.3 (ii)]. Note that the group  $\Theta_{10} = \mathbb{Z}_2 \oplus \mathbb{Z}_3$ . Moreover, by [42, Proposition 2.6] the composition  $\mathbb{S}^9 \xrightarrow{h_C} \mathbb{CP}^4 \xrightarrow{d} \mathbb{S}^8$  is null-homotopic. This implies that the induced map  $h_C^* : \mathcal{C}(\mathbb{CP}^4) \rightarrow \Theta_9$  is trivial. Therefore, by substituting the corresponding groups for  $n = 5$  in the sequence (5.5), we get the required short exact sequence.
- (iii) Let  $n = 6$  in the sequence (5.5). As the group  $\Theta_{12}$  is trivial, we infer that the induced map  $\iota^*$  is a monomorphism, and hence an isomorphism. Further, in [25, Theorem 2.7 (ii)], it is proved that  $\mathcal{C}(\mathbb{CP}^5) \cong \mathbb{Z}_2^2 \oplus \mathbb{Z}_3$ , and the image of the induced map  $h_C^* : \mathcal{C}(\mathbb{CP}^5) \rightarrow \Theta_{11}$  is isomorphic to  $\mathbb{Z}_2$ . Therefore, by the isomorphism theorem of groups, we conclude that  $\text{Ker}(h_C^*)$  is isomorphic to  $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ . Thus we obtain the required isomorphism.
- (iv) Consider the sequence (5.5) for  $n = 7$ . In [25, Theorem 2.7(iii)], it is proved that  $\mathcal{C}(\mathbb{CP}^6) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$ , and the induced map  $h_C^* : \mathcal{C}(\mathbb{CP}^6) \rightarrow \Theta_{13}$  is an epimorphism, where  $\Theta_{13} = \mathbb{Z}_3$ . This implies that  $\text{Ker}(h_C^*) \cong \mathbb{Z}_2$ .
- (v) It has been proved in [25, Theorem 2.9] that the group  $\mathcal{C}(\mathbb{CP}^7) \cong \mathbb{Z}_2^2$ , and the image of the induced map  $h_C^* : \mathcal{C}(\mathbb{CP}^7) \rightarrow \Theta_{15}$  is isomorphic to  $\mathbb{Z}_2$ . Since the group  $\Theta_{16} = \mathbb{Z}_2$ , we obtain the required short exact sequence from the sequence (5.5) by putting  $n = 8$ .

□

From Corollary 5.3.1 (i) and (iii) we have  $\mathcal{C}(\#^k \mathbb{CP}^4) \cong \mathbb{Z}_2$  and  $\mathcal{C}(\#^k \mathbb{CP}^6) \cong \mathbb{Z}_2^{2k-1} \oplus \mathbb{Z}_3^k$ , respectively. Further, we explicitly compute the concordance structure sets of  $\#^k \mathbb{CP}^5$  and  $\#^k \mathbb{CP}^7$ . The proofs exhibit that the sequences in Corollary 5.3.1 (ii) and (iv) split.

**Theorem 5.3.2.** *Let  $k \in \mathbb{N}$ . Then*

$$(i) \mathcal{C}(\#^k \mathbb{CP}^5) \cong \mathbb{Z}_2^{k+1} \oplus \mathbb{Z}_3$$

$$(ii) \mathcal{C}(\#^k \mathbb{CP}^7) \cong \mathbb{Z}_2^{k+1} \oplus \mathbb{Z}_3^{k-1}$$

*Proof.* The localization at a prime ideal is an exact functor on the category of abelian groups. We use this property extensively to compute required groups by deducing splittings of the below sequences from Corollary 5.3.1 (ii) and (iv),

$$0 \rightarrow \Theta_{10} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{CP}^5) \xrightarrow{\iota^*} \oplus_k \mathcal{C}(\mathbb{CP}^4) \rightarrow 0$$

and

$$0 \rightarrow \Theta_{14} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{CP}^7) \xrightarrow{\iota^*} \oplus_{k-1} \mathcal{C}(\mathbb{CP}^6) \oplus \text{Ker}(h^*) \rightarrow 0,$$

respectively. The primes occurring in the primary decomposition of groups  $\mathcal{C}(\#^k \mathbb{CP}^5)$  and  $\mathcal{C}(\#^k \mathbb{CP}^7)$  are (2) and (3). Note that the localization of these short exact sequences at (3), results in  $\mathcal{C}(\#^k \mathbb{CP}^5)_{(3)} \cong \mathbb{Z}_3$  and  $\mathcal{C}(\#^k \mathbb{CP}^7)_{(3)} \cong \mathbb{Z}_3^{k-1}$  respectively. Thus the splittings of the above sequences are equivalent to the non-existence of order 4 element in middle groups.

(i) Let  $\alpha : \vee_k \mathbb{CP}^3 \rightarrow \#^k \mathbb{CP}^5$  be the inclusion map. Since  $\alpha$  is inclusion of the 6-skeleton, we have

$$\#^k \mathbb{CP}^5 / \vee_k \mathbb{CP}^3 \simeq \left( \vee_k \mathbb{CP}^4 / \vee_k \mathbb{CP}^3 \right) \bigcup_{\beta \circ \xi} \mathbb{D}^{10},$$

where  $\xi : \mathbb{S}^9 \rightarrow \vee_k \mathbb{CP}^4$  is the attaching map to obtain  $\#^k \mathbb{CP}^5$  and  $\beta : \vee_k \mathbb{CP}^4 \rightarrow \vee_k \mathbb{CP}^4 / \vee_k \mathbb{CP}^3$  is the quotient map. Furthermore, the space  $\vee_k \mathbb{CP}^4 / \vee_k \mathbb{CP}^3$  is homotopy equivalent to  $\vee_k (\mathbb{CP}^4 / \mathbb{CP}^3) \simeq \vee_k \mathbb{S}^8$ . Hence we have  $\beta \simeq \vee_k d$ , where  $d : \mathbb{CP}^4 \rightarrow \mathbb{S}^8$  is the degree one map. In [42, Proposition 2.6], it is showed that the composition  $d \circ h_C : \mathbb{S}^9 \rightarrow \mathbb{CP}^4 \rightarrow \mathbb{S}^8$  is nullhomotopic, where  $h_C$  is the Hopf map. Therefore, in the following homotopy com-

mutative diagram,

$$\begin{array}{ccccc}
 & & \vee_k \mathbb{S}^9 & & \\
 & \nearrow p & & \searrow \vee_k h_C & \\
 \mathbb{S}^9 & \xrightarrow{\xi} & \vee_k \mathbb{CP}^4 & \xrightarrow{\beta \simeq \vee_k d} & \vee_k \mathbb{S}^8
 \end{array}$$

we have  $\beta \circ \xi \simeq (\vee_k(d \circ h_C)) \circ p$  to be nullhomotopic. This implies that  $\#^k \mathbb{CP}^5 / \vee_k \mathbb{CP}^3 \simeq \vee_k \mathbb{S}^8 \vee \mathbb{S}^{10}$ .

Now consider the cofiber sequence  $\vee_k \mathbb{CP}^3 \xrightarrow{\alpha} \#^k \mathbb{CP}^5 \xrightarrow{q} \vee_k \mathbb{S}^8 \vee \mathbb{S}^{10}$  and apply the functor  $[-, PL/O]$  on it. Then the induced map  $q^* : \oplus_k \Theta_8 \oplus \Theta_{10} \rightarrow \mathcal{C}(\#^k \mathbb{CP}^5)$  is surjective because the group  $\mathcal{C}(\mathbb{CP}^3)$  is trivial. Finally, the groups  $\Theta_{8(2)} \cong \mathbb{Z}_2$  and  $\Theta_{10(2)} \cong \mathbb{Z}_2$  implies that  $\mathcal{C}(\#^k \mathbb{CP}^5)$  has no order 4 element.

- (ii) Consider the cofiber sequence  $\sqcup_{k-1} \mathbb{S}^{13} \xrightarrow{\gamma} \#^k \mathbb{CP}^7 \xrightarrow{\rho} \vee_k \mathbb{CP}^7$ , where the map  $\gamma$  is the inclusion of  $\sqcup_{k-1} \mathbb{S}^{n-1}$  on gluing parts during the connected sum operation in  $\#^k \mathbb{CP}^7$  such that the cofiber of  $\gamma$  is  $\vee_k \mathbb{CP}^7$ . In the long exact sequence obtained by functor  $[-, PL/O]$ , the induced map  $\rho_{(2)}^* : \oplus_k \mathbb{CP}_{(2)}^7 \rightarrow \mathcal{C}(\#^k \mathbb{CP}^7)_{(2)}$  is surjective since  $\Theta_{13(2)}$  is trivial. Therefore the group  $\mathcal{C}(\mathbb{CP}^7)_{(2)} \cong \mathbb{Z}_2^2$  implies that  $\mathcal{C}(\#^k \mathbb{CP}^7)$  has no order 4 element.

This completes the proof. □

Now we show analogous results hold in the case of  $k$ -fold connected sum of quaternionic projective spaces. Recall that  $\mathbb{HP}^n = \mathbb{HP}^{n-1} \cup_{h_Q} \mathbb{D}^{4n}$ , where the attaching map  $h_Q : \mathbb{S}^{4n-1} \rightarrow \mathbb{HP}^{n-1}$  is the Hopf map. Computations of groups  $\mathcal{C}(\mathbb{HP}^n)$  for  $n \leq 4$  are done in [3, Theorem 3.1]. Here, by virtue of Theorems 5.1.1 and 5.1.2 in case of  $\mathcal{C}(\#^k \mathbb{HP}^n)$  we obtain the following:

**Corollary 5.3.3.** (i) *There is an isomorphism*

$$d^* : \Theta_8 \rightarrow \mathcal{C}(\#^k \mathbb{HP}^2),$$

where  $\Theta_8 \cong \mathbb{Z}_2$ .

(ii) *There is an isomorphism*

$$\iota^* : \mathcal{C}(\#^k \mathbb{HP}^3) \rightarrow \oplus_k \mathcal{C}(\mathbb{HP}^2),$$

where  $\mathcal{C}(\mathbb{HP}^2) \cong \mathbb{Z}_2$ .

(iii) *There is a short exact sequence*

$$0 \rightarrow \Theta_{16} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{HP}^4) \xrightarrow{\iota^*} \oplus_{k-1} \mathcal{C}(\mathbb{HP}^3) \xrightarrow{0} 0,$$

where  $\Theta_{16} \cong \mathbb{Z}_2$  and  $\mathcal{C}(\mathbb{HP}^3) \cong \mathbb{Z}_2$ .

*Proof.* It follows from Proposition 5.2.4 and Remark 5.2.5 that the induced map  $d^* : \Theta_{4n} \rightarrow \mathcal{C}(\#^k \mathbb{HP}^n)$  is injective for  $n \leq 4$ . Therefore, using Theorems 5.1.1 and 5.1.2 for  $\#^k \mathbb{HP}^n$  and  $Y = PL/O$  we obtain the following short exact sequence

$$0 \rightarrow \Theta_{4n} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{HP}^n) \xrightarrow{\iota^*} \oplus_{k-1} \mathcal{C}(\mathbb{HP}^{n-1}) \oplus \text{Ker}(h_Q^*) \rightarrow 0 \quad (5.6)$$

where  $h_Q : \mathbb{S}^{4n-1} \rightarrow \mathbb{HP}^{n-1}$  is the Hopf map. Then we have the following:

- (i) The group  $\mathcal{C}(\mathbb{HP}^1)$  is trivial because  $\mathbb{HP}^1$  is homeomorphic to  $\mathbb{S}^4$  and  $\mathcal{C}(\mathbb{S}^4)$  is trivial. Thus for  $n = 2$ , the induced map  $d^*$  in the sequence (5.6) is an isomorphism.
- (ii) Let  $n = 3$  in the sequence (5.6). The group  $\Theta_{12}$  is trivial, which results in an isomorphism of the map  $\iota^*$ . Next, it follows from (i) that the group  $\mathcal{C}(\mathbb{HP}^2) \cong \mathbb{Z}_2$ , and by [3, Theorem 3.1 (i)] the induced map  $h_Q^* : \mathcal{C}(\mathbb{HP}^2) \rightarrow \Theta_{11}$  is trivial. This completes the proof of (ii).
- (iii) It is evident from (ii) that the group  $\mathcal{C}(\mathbb{HP}^3) \cong \mathbb{Z}_2$ . In [3, Theorem 3.1 (ii)], it is proved that the induced map  $h_Q^* : \mathcal{C}(\mathbb{HP}^3) \rightarrow \Theta_{15}$  is a monomorphism. Since the group  $\Theta_{16} = \mathbb{Z}_2$ , we obtain the required short exact sequence from the sequence (5.6) by substituting  $n = 4$ .

□

The existence of a short exact sequence in the corollary below is similar to Corollary 5.3.3 for the connected sum of octonionic projective space  $\mathbb{OP}^2$ .

**Corollary 5.3.4.** *There is a short exact sequence*

$$0 \rightarrow \Theta_{16} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{O}P^2) \xrightarrow{\iota^*} \oplus_k \Theta_8 \rightarrow 0,$$

where  $\Theta_{16} \cong \mathbb{Z}_2$  and  $\Theta_8 \cong \mathbb{Z}_2$ .

*Proof.* Recall that  $\mathbb{O}P^2 = \mathbb{O}P^1 \cup_{h_O} \mathbb{D}^{16}$ , where the attaching map  $h_O : \mathbb{S}^{15} \rightarrow \mathbb{O}P^1$  is the Hopf map. In [23, Theorem 2.3 (iii)], it is proved that the concordance inertia group of  $\mathbb{O}P^2$  is trivial. By Theorem 5.2.1, we conclude that for all  $k \in \mathbb{N}$ , the concordance inertia group of  $\#^k \mathbb{O}P^2$  is trivial. Hence the induced map  $d^* : \Theta_{16} \rightarrow \mathcal{C}(\#^k \mathbb{O}P^2)$  is injective. Hence using Theorems 5.1.1 and 5.1.2 for  $\mathcal{C}(\#^k \mathbb{O}P^2)$ , we obtain the following short exact sequence:

$$0 \rightarrow \Theta_{16} \xrightarrow{d^*} \mathcal{C}(\#^k \mathbb{O}P^2) \xrightarrow{\iota^*} \oplus_{k-1} \mathcal{C}(\mathbb{O}P^1) \oplus \text{Ker}(h_O^*) \rightarrow 0$$

Moreover, by [24, Theorem 4.5], the induced map  $h_O^* : \Theta_8 \rightarrow \Theta_{15}$ , maps the non-trivial element of  $\Theta_8$  to a null homotopic map  $f : \mathbb{S}^{15} \rightarrow \mathbb{S}^8 \rightarrow PL/O$ . As a result, we obtain the required short exact sequence.  $\square$

Similarly using [3, Theorem 3.1 (i)] and [25, Theorem 2.7], we have the concluding result.

**Corollary 5.3.5.**  $\mathcal{C}((\#^k \mathbb{C}P^6) \# (\#^l \mathbb{H}P^3)) \cong \mathbb{Z}_2^{2k+l-1} \oplus \mathbb{Z}_3^k$

## 5.4 Stable cohomotopy groups of $k$ -fold connected sum of complex projective spaces

In this section, we calculate the reduced 0<sup>th</sup> stable cohomotopy groups  $\pi_s^0(\#^k \mathbb{C}P^n)$  for  $3 \leq n \leq 8$  and  $k \geq 2$ . These computations rely on pre-existing results and computations that are presented below. In [7], Brumfiel focused on the computations of 0<sup>th</sup> stable cohomotopy groups of individual copies of the complex projective spaces in lower dimensions; the groups are as follows:

**Theorem 5.4.1** ([7]).

1.  $\pi_s^0(\mathbb{CP}^3) \cong \mathbb{Z}_2$
2.  $\pi_s^0(\mathbb{CP}^4) \cong \mathbb{Z}_2^2$
3.  $\pi_s^0(\mathbb{CP}^5) \cong \mathbb{Z}_2^2 \oplus \mathbb{Z}_3$
4.  $\pi_s^0(\mathbb{CP}^6) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$

Furthermore, R. Kasilingam [25] computed the following groups:

**Theorem 5.4.2** ([25]).

1.  $\pi_s^0(\mathbb{CP}^7) \cong \mathbb{Z}_2^3$
2.  $\pi_s^0(\mathbb{CP}^8) \cong \mathbb{Z}_2^3$

Our computations mainly utilize the following short exact sequence.

**Theorem 5.4.3** (Chapter 5, Theorems 5.1.1 and 5.1.2). *Let  $M$  be a closed, oriented, smooth  $n$ -manifold. Then there exists a short exact sequence*

$$0 \longrightarrow \pi_n(Y) / \text{Im}((\Sigma h)^*) \xrightarrow{\tilde{d}^*} [\#^{i \in I} M, Y] \xrightarrow{\iota^*} \left( \bigoplus_{k=1} [M^{(n-1)}, Y] \right) \oplus \text{Ker}(h^*) \longrightarrow 0, \quad (5.7)$$

where  $\xi = (\vee_k h) \circ p$  is the attaching map for  $\#^k M$  with  $h : \mathbb{S}^{n-1} \rightarrow M^{(n-1)}$  as the attaching map for  $M$  and  $p$  is the pinch map.

In (5.7), consider  $M_i = \mathbb{CP}^n$  for all  $i$ , and substituting  $Y$  to be the infinite loop space  $SF$ . Recall that,  $SF$  is the limit of the spaces  $SF_n$  over  $n$ , where  $SF_n$  consist of base point preserving degree one maps from  $\mathbb{S}^{n-1}$  to itself. Therefore we have the following short exact sequence,

$$0 \longrightarrow \pi_{2n}^s / \text{Im}((\Sigma h)^*) \xrightarrow{\tilde{d}^*} \pi_s^0(\#^k \mathbb{CP}^n) \xrightarrow{\iota^*} \bigoplus_{k=1} \pi_s^0(\mathbb{CP}^{n-1}) \oplus \text{Ker}(h^*) \longrightarrow 0. \quad (5.8)$$

The next theorem uses the computations done for the groups  $\text{Im}((\Sigma h)^*)$  and  $\text{Ker}(h^*)$  in [7] and [25, Proposition 2.8, Theorem 2.9] for  $3 \leq n \leq 8$ .

**Theorem 5.4.4.** (i) *There is an isomorphism*

$$d^* : \pi_6^s \rightarrow \pi_s^0(\#^k \mathbb{CP}^3),$$

where  $\pi_6^s = \mathbb{Z}_2$ .

(ii) *There is a short exact sequence*

$$0 \rightarrow \pi_8^s / \mathbb{Z}_2 \xrightarrow{\tilde{d}^*} \pi_s^0(\#^k \mathbb{CP}^4) \xrightarrow{\iota^*} \bigoplus_k \pi_s^0(\mathbb{CP}^3) \rightarrow 0,$$

where  $\pi_8^s = \mathbb{Z}_2^2$  and  $\pi_s^0(\mathbb{CP}^3) \cong \mathbb{Z}_2$ .

(iii) *There is a short exact sequence*

$$0 \rightarrow \pi_{10}^s \xrightarrow{d^*} \pi_s^0(\#^k \mathbb{CP}^5) \xrightarrow{\iota^*} \bigoplus_{k-1} \pi_s^0(\mathbb{CP}^4) \oplus \text{Ker}(h^*) \rightarrow 0,$$

where  $\pi_{10}^s = \mathbb{Z}_2 \oplus \mathbb{Z}_3$ ,  $\pi_s^0(\mathbb{CP}^4) \cong \mathbb{Z}_2^2$  and  $\text{Ker}(h^* : \pi_s^0(\mathbb{CP}^4) \rightarrow \pi_9^s) \cong \mathbb{Z}_2$ .

(iv) *There is an isomorphism*

$$\iota^* : \pi_s^0(\#^k \mathbb{CP}^6) \rightarrow \bigoplus_{k-1} \pi_s^0(\mathbb{CP}^5) \oplus \text{Ker}(h^*),$$

where  $\pi_s^0(\mathbb{CP}^5) \cong \mathbb{Z}_2^2 \oplus \mathbb{Z}_3$  and  $\text{Ker}(h^* : \pi_s^0(\mathbb{CP}^5) \rightarrow \pi_{11}^s) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$ .

(v) *There is a short exact sequence*

$$0 \rightarrow \pi_{14}^s \xrightarrow{d^*} \pi_s^0(\#^k \mathbb{CP}^7) \xrightarrow{\iota^*} \bigoplus_{k-1} \pi_s^0(\mathbb{CP}^6) \oplus \text{Ker}(h^*) \rightarrow 0,$$

where  $\pi_{14}^s = \mathbb{Z}_2^2$ ,  $\pi_s^0(\mathbb{CP}^6) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$  and  $\text{Ker}(h^* : \pi_s^0(\mathbb{CP}^6) \rightarrow \pi_{13}^s) \cong \mathbb{Z}_2$ .

(vi) *There is a short exact sequence*

$$0 \rightarrow \pi_{16}^s / \mathbb{Z}_2 \xrightarrow{\tilde{d}^*} \pi_s^0(\#^k \mathbb{CP}^8) \xrightarrow{\iota^*} \bigoplus_{k-1} \pi_s^0(\mathbb{CP}^7) \oplus \text{Ker}(h^*) \rightarrow 0,$$

where  $\pi_{16}^s = \mathbb{Z}_2^2$ ,  $\pi_s^0(\mathbb{CP}^7) \cong \mathbb{Z}_2^3$  and  $\text{Ker}(h^* : \pi_s^0(\mathbb{CP}^7) \rightarrow \pi_{15}^s) \cong \mathbb{Z}_2^2$ .

From Theorem 5.4.4, we obtain that  $\pi_s^0(\#^k \mathbb{CP}^3) \cong \mathbb{Z}_2$  and  $\pi_s^0(\#^k \mathbb{CP}^6) \cong \mathbb{Z}_2^{2k-1} \oplus \mathbb{Z}_3^k$ . Now we compute the groups  $\pi_s^0(\#^k \mathbb{CP}^4)$  and  $\pi_s^0(\#^k \mathbb{CP}^7)$  by proving that the aforementioned short



exact sequences (ii) and (v) split. The group  $\pi_s^0(\#^k \mathbb{CP}^5)$  is computed in the last chapter of the thesis, as the proof requires additional results discussed therein.

We begin by constructing a cofiber sequence: For  $i \in I = \{1, 2, \dots, k\}$ , let  $M_i$  be a closed, oriented, smooth  $n$ -manifold with the attaching map  $h_i : \mathbb{S}_i^{n-1} \rightarrow M_i^{(n-1)}$ . Consider, for  $I' = \{1, 2, \dots, k-1\}$ , the map  $\mu : \sqcup_{i \in I'} \mathbb{S}_i^{n-1} \hookrightarrow \#^{i \in I} M_i$  that maps each  $\mathbb{S}_i^{n-1}$  on the gluing part of  $M_i$  while taking the connected sum. Observe that the homotopy cofiber of the map  $\mu$  is  $\vee_{i \in I} M_i$ .

Now, consider the map  $\lambda$  which is the composition map  $j \circ \vee_{i \in I} p_i \circ \sqcup_{i \in I'} \lambda_i$  as shown in the following homotopy commutative diagram,

$$\begin{array}{ccc}
 \sqcup_{i \in I'} \mathbb{S}_i^{n-1} & \xrightarrow{\lambda} & \#^{i \in I} M_i \\
 \sqcup_{i \in I'} \lambda_i \downarrow & & \uparrow j \\
 \vee_{i \in I} \widehat{M}_i & \xrightarrow[\vee_{i \in I} p_i]{} & \vee_{i \in I} M_i^{(n-1)},
 \end{array} \tag{5.9}$$

where  $\widehat{M}_i$  is obtained by deleting a smaller open disc from the top cell of  $M_i$  while taking the connected sum, and  $\lambda_i : \mathbb{S}_i^{n-1} \hookrightarrow \widehat{M}_i$  is the inclusion on that boundary part. Note that, the important reason of constructing  $\lambda$  in such a way is that for all  $i \in I$ , the map  $p_i$  is a homotopy equivalence, and  $p_i \circ \lambda_i$  is homotopic to the attaching map  $h_i$ .

Consider the following diagram obtained from cofiber sequences in associated with maps  $\mu$  and  $\lambda$

$$\begin{array}{ccccccc}
 \sqcup_{i \in I'} \mathbb{S}_i^{n-1} & \xrightarrow{\mu} & \#^{i \in I} M_i & \xrightarrow{\rho} & \vee_{i \in I} M_i & \longrightarrow & \dots \\
 \downarrow \text{Id} & & \downarrow \text{Id} & & \downarrow \beta & & \\
 \sqcup_{i \in I'} \mathbb{S}_i^{n-1} & \xrightarrow{\lambda} & \#^{i \in I} M_i & \xrightarrow{q} & C_\lambda & \longrightarrow & \dots
 \end{array} \tag{5.10}$$

in which the leftmost square commutes up to homotopy. This results in the existence of the map  $\beta : \vee_{i \in I} M_i \rightarrow C_\lambda$  such that the second square commutes up to homotopy, and consequently, all squares commute up to homotopy. Applying the contravariant functor  $[-, Y]$  on the above sequences yields  $[C_\lambda, Y] \cong \oplus_{i \in I} [M_i, Y]$ , for an homotopy associative  $H$ -group  $Y$ .

In the forthcoming proof of the theorem, the cofiber sequence resulting from the constructed map  $\lambda$  plays a vital role.

**Theorem 5.4.5.** *Let  $k \in \mathbb{N}$ . Then*

- (i)  $\pi_s^0(\#^k \mathbb{CP}^4) \cong \mathbb{Z}_2^{k+1}$
- (ii)  $\pi_s^0(\#^k \mathbb{CP}^7) \cong \mathbb{Z}_2^{k+2} \oplus \mathbb{Z}_3^{k-1}$

*Proof.* (i) Consider the short exact sequence from Theorem 5.4.4 (ii),

$$0 \longrightarrow \pi_8^s / \mathbb{Z}_2 \xrightarrow{\tilde{d}^*} \pi_s^0(\#^k \mathbb{CP}^4) \xrightarrow{l^*} \bigoplus_k \pi_s^0(\mathbb{CP}^3) \longrightarrow 0,$$

The splitting of this sequence is equivalent to the non-existence of order 4 element in the group  $\pi_s^0(\#^k \mathbb{CP}^4)$ . Consider the map  $\lambda : \sqcup_{k-1} \mathbb{S}^7 \rightarrow \#^k \mathbb{CP}^4$  from (5.9) and the cofiber sequence associated to  $\lambda$  from (5.10). We know that the map  $\lambda$  factors through  $(\vee_{k-1} h)$  up to homotopy, where  $h : \mathbb{S}^7 \rightarrow \mathbb{CP}^3$  is the Hopf map. In [7], it is proved that the induced map  $h^* : \pi_s^0(\mathbb{CP}^3) \rightarrow \pi_7^s$  is trivial, and consequently  $\lambda^*$  as well. Thus, in the long exact sequence obtained from  $\lambda$  by applying functor  $[-, SF]$ , the induced map  $q^* : \bigoplus_k \pi_s^0(\mathbb{CP}^4) \rightarrow \pi_s^0(\#^k \mathbb{CP}^4)$  is surjective. Finally, the group  $\pi_s^0(\mathbb{CP}^4) \cong \mathbb{Z}_2^2$  concludes that the  $\pi_s^0(\#^k \mathbb{CP}^4)$  has no order 4 element.

(ii) Now consider the short exact sequence from Theorem 5.4.4 (v)

$$0 \longrightarrow \pi_{14}^s \xrightarrow{d^*} \pi_s^0(\#^k \mathbb{CP}^7) \xrightarrow{l^*} \bigoplus_{k-1} \pi_s^0(\mathbb{CP}^6) \oplus \text{Ker}(h^*) \longrightarrow 0,$$

The primes occurring in the primary decomposition of the group  $\pi_s^0(\#^k \mathbb{CP}^7)$  are 2 and 3. Note that the localization of the short exact sequence at 3, results in  $\pi_s^0(\#^k \mathbb{CP}^7)_{(3)} \cong \mathbb{Z}_3^{k-1}$ . Thus, the sequence splits if the group  $\pi_s^0(\#^k \mathbb{CP}^7)$  has no order 4 element. Consider the map  $\lambda : \sqcup_{k-1} \mathbb{S}^{13} \rightarrow \#^k \mathbb{CP}^7$  from (5.9) and the cofiber sequence associated to  $\lambda$  from (5.10). The induced map  $\lambda^* : \pi_s^0(\#^k \mathbb{CP}^7)_{(2)} \rightarrow \pi_{13(2)}^0$ , in the long exact sequence of cohomotopy groups, is trivial as the group  $\pi_{13(2)}^s = 0$ . This implies the map  $q^* : \bigoplus_k \pi_s^0(\mathbb{CP}^7)_{(2)} \rightarrow \pi_s^0(\#^k \mathbb{CP}^7)_{(2)}$  is surjective. Since the group  $\pi_s^0(\mathbb{CP}^7) \cong \mathbb{Z}_2^3$  implies  $\pi_s^0(\#^k \mathbb{CP}^7)$  has no order 4 element.

This completes the proof for both cases.





## Chapter 6

# Topological K-groups of $k$ -fold Connected Sum of Complex Projective Spaces

In 1950, Alexander Grothendieck introduced Topological K-theory, and subsequently developed by Michael Atiyah and Friedrich Hirzebruch. The approach primarily focuses on the study of vector bundles over topological spaces, providing a powerful tool for their classification and understanding. K-theory is one of the first examples of an extraordinary cohomology theory. Beyond its applications in the classification of bundles, there are associated cohomology classes known as Stiefel-Whitney classes, which measure orientability and spin-structure of vector bundles. Moreover, Stiefel-Whitney classes generate all characteristic classes, which are cohomology invariants for vector bundles when considering cohomology with  $\mathbb{Z}_2$ -coefficients. Furthermore, these cohomologies with  $\mathbb{Z}$ -coefficients are connected to Pontryagin and Euler classes.

Let us fix the notations  $\tilde{K}^s(X)$  for the reduced complex  $s^{\text{th}}$  K-group of a space  $X$ , and  $\tilde{KO}^s(X)$  for the reduced real  $s^{\text{th}}$  K-group, where  $s \in \mathbb{Z}$ .

In this chapter, we compute all the reduced real and complex K-groups of  $\#^k \mathbb{CP}^n$ , for all  $k, n \geq 2$ , along with the generators. At  $k = 1$ , these groups are studied in [16] for all  $n \in \mathbb{N}$ .

In Chapter 7, we use these groups for the computation of normal invariant set of  $\#^k \mathbb{CP}^n$ , which plays a crucial role in the manifold classification problem.

Note that, the group  $\widetilde{KO}^0(\#^k \mathbb{CP}^n)$  coincides with the homotopy group  $[\#^k \mathbb{CP}^n, BSO]$ , and  $\widetilde{K}^0(\#^k \mathbb{CP}^n)$  coincides with  $[\#^k \mathbb{CP}^n, BSU]$ . Recall that, the spaces  $BSO$  and  $BSU$  are classifying spaces for the stable vector bundle over  $\mathbb{R}$  and  $\mathbb{C}$  respectively. More generally, for a space  $X$ , we have following identifications

$$\widetilde{K}^{-s}(X) = [\Sigma^s X, BSU],$$

$$\widetilde{KO}^{-s}(X) = [\Sigma^s X, BSO]$$

for  $s \in \mathbb{N}$ . The Bott periodicity theorem says that  $\widetilde{K}^s(X)$  are 2-periodic and  $\widetilde{KO}^s(X)$  are 8-periodic, i.e.,

$$\widetilde{K}^s(X) \cong \widetilde{K}^{s+2}(X),$$

$$\widetilde{KO}^s(X) \cong \widetilde{KO}^{s+8}(X)$$

for  $s \in \mathbb{Z}$ . Therefore, in case of complex vector bundles, it is enough to calculate  $\widetilde{K}^s(X)$  for any two consecutive integers  $s \in \mathbb{Z}$ , and for real vector bundles, it is enough to calculate  $\widetilde{KO}^s(X)$  for any eight consecutive integers  $s \in \mathbb{Z}$ .

For the forthcoming computations, we often use the following short exact sequence, obtained from (5.7) by considering  $M_i = \mathbb{CP}^n$  for all  $i \in I$ ,

$$0 \longrightarrow \pi_{2n+s}(Y) / \text{Im}(\Sigma^{s+1}h)^* \xrightarrow{\widetilde{d}^*} [\Sigma^s \#^k \mathbb{CP}^n, Y] \xrightarrow{\iota^*} \bigoplus_{k=1} [\Sigma^s \mathbb{CP}^{n-1}, Y] \oplus \text{Ker}(\Sigma^s h)^* \longrightarrow 0, \quad (6.1)$$

where  $d : \Sigma^s \#^k \mathbb{CP}^n \rightarrow \mathbb{S}^{2n+s}$  is the degree one map,  $\iota : \vee_k \Sigma^s \mathbb{CP}^{n-1} \rightarrow \Sigma^s \#^k \mathbb{CP}^n$  be an inclusion map, and  $h : \mathbb{S}^{2n-1} \rightarrow \mathbb{CP}^{n-1}$  be the Hopf map.

## 6.1 Complex K-groups

Let us consider the following short exact sequence which is obtained by substituting  $Y = BSU$  in the sequence (6.1), along with the identification  $[\Sigma^s X, BSU] = \tilde{K}^{-s}(X)$  for  $s \in \mathbb{N}$ ,

$$0 \longrightarrow \tilde{K}^{-s}(\mathbb{S}^{2n}) / \text{Im}(\Sigma^{s+1}h)^* \xrightarrow{\tilde{d}^*} \tilde{K}^{-s}(\#^k \mathbb{CP}^n) \xrightarrow{\iota^*} \bigoplus_{k-1} \tilde{K}^{-s}(\mathbb{CP}^{n-1}) \oplus \text{Ker}(\Sigma^s h)^* \longrightarrow 0. \quad (6.2)$$

Due to Bott periodicity, it is enough to compute the groups  $\tilde{K}^0(\#^k \mathbb{CP}^n)$  and  $\tilde{K}^{-1}(\#^k \mathbb{CP}^n)$  for all  $k, n \geq 2$ .

Let  $\omega$  be a generator of the free abelian group  $\tilde{K}^0(\mathbb{S}^{2n})$ . Let  $H$  be the canonical complex line bundle over  $\mathbb{CP}^n$ , and let  $\eta := H - 1$ . Then the set  $\{\eta^j \mid j = 1, 2, \dots, n\}$  forms an integral basis for the free abelian group  $\tilde{K}^0(\mathbb{CP}^n)$  for all  $n \in \mathbb{N}$  [16].

Let us consider the case where  $s = 0$  in the sequence (6.2). In this situation, we have  $\tilde{K}^{-1}(\mathbb{CP}^{n-1}) = 0$ , which implies that the map  $\Sigma h^* : \tilde{K}^{-1}(\mathbb{CP}^{n-1}) \rightarrow \tilde{K}^0(\mathbb{S}^{2n})$  becomes trivial. Furthermore, the group  $\tilde{K}^0(\mathbb{S}^{2n-1})$  is also zero, leading to  $\text{Ker}(h)^* = \tilde{K}^0(\mathbb{CP}^{n-1})$ . Since both  $\tilde{K}^0(\mathbb{S}^{2n})$  and  $\tilde{K}^0(\mathbb{CP}^{n-1})$  are free abelian groups, we obtain a split short exact sequence, which allows us to make the following statement.

**Theorem 6.1.1.** *The  $\tilde{K}^0(\#^k \mathbb{CP}^n)$  is a free module with a basis*

$$\{d^*(\omega)\} \cup \{\eta_i^j \mid i = 1, 2, \dots, k, j = 1, 2, \dots, n-1\}$$

for  $k, n \in \mathbb{N}$ .

**Note:** In this section, we are using the same notation  $\eta_i^j$  instead of  $\gamma^*(\eta_i^j)$  if  $\gamma^*$  denotes the left inverse of  $\iota^*$  associated to the split exact sequence.

Next, let us consider  $s = 1$  in the sequence (6.2). Then since  $\tilde{K}^{-1}(\mathbb{S}^{2n})$  and  $\tilde{K}^{-1}(\mathbb{CP}^{n-1})$  are both trivial, for all  $n \in \mathbb{N}$ , we get the following result.

**Theorem 6.1.2.** *The group  $\tilde{K}^{-1}(\#^k \mathbb{CP}^n)$  is trivial for all  $k, n \in \mathbb{N}$ .*

With these computations, we have determined the complex K-groups of  $\#^k \mathbb{CP}^n$  for all

values of  $k, n \geq 2$ .

## 6.2 Real K-groups

Now let us proceed with the computations of the real  $K$ -groups of  $\#^k \mathbb{CP}^n$ . Due to Bott periodicity it is enough to compute  $\widetilde{KO}^m(\#^k \mathbb{CP}^n)$  within the range  $-7 \leq m \leq 0$ . To accomplish this, we primarily use the exact sequence introduced in the proof of the following lemma.

**Lemma 6.2.1.** *Let  $M_i$  be an  $n$ -manifold with  $(n-1)$ -skeleton  $M_i^{(n-1)}$  for  $i \in I = \{1, 2, \dots, k\}$ , and the space  $Y$  be a homotopy-commutative  $H$ -group. If the groups  $[M_k, Y]$ ,  $[M_i^{(n-1)}, Y]$  are trivial for all  $i \in I' = \{1, 2, \dots, k-1\}$  then the group  $[\#^{i \in I} M_i, Y]$  is also trivial.*

*Proof.* The proof follows directly by considering the following induced long exact sequence

$$\cdots \longrightarrow [\bigvee_{i \in I'} \Sigma M_i^{(n-1)}, Y] \xrightarrow{c^*} [M_k, Y] \xrightarrow{q^*} [\#^{i \in I} M_i, Y] \xrightarrow{\iota^*} [\bigvee_{i \in I'} M_i^{(n-1)}, Y] \cdots \quad (6.3)$$

from the cofiber sequence  $\bigvee_{i \in I'} M_i^{(n-1)} \xrightarrow{\iota} \#^{i \in I} M_i \xrightarrow{q} M_k \xrightarrow{c} \bigvee_{i \in I'} \Sigma M_i^{(n-1)}$ , where  $\iota$  is an inclusion map,  $q$  is the quotient map, and  $c$  is the connecting map in the cofiber sequence.  $\square$

In the subsequent theorem, we introduce the variables  $\eta$ ,  $\alpha$ ,  $\beta$ , and  $\gamma$  as replacements for  $\mu_0$ ,  $\mu_1$ ,  $\mu_2$ , and  $\mu_3$  respectively, from [16, Theorem 2]. This substitution is intended to minimize confusion caused by excessive subscripts. Similarly, the symbols  $\sigma$  and  $\tau$  retain the same notation as in [16, Theorem 2]. Thus,  $\eta_i$ ,  $\sigma_i$ , and  $\tau_i$  respectively correspond to  $\eta$ ,  $\sigma$ , and  $\tau$  for all  $i \in I'$ .

Also, to simplify the notation, we will use  $f^*$  instead of  $(\Sigma^s f)^*$ , as the latter can be understood to be the maps  $\iota$ ,  $q$ , or  $c$  from the sequence (6.3).

**Theorem 6.2.2.** *Let  $k, n \geq 2$ , and let  $I' = \{1, 2, \dots, k-1\}$ . Then*

(1) (a)  $\widetilde{KO}^0(\#^k \mathbb{CP}^{4n})$  is a free module with a basis

$$\{q^*(\eta_k^j), q^*(\eta_k^{2n}), \eta_i^j \mid i \in I', j = 1, 2, \dots, 2n-1\}.$$



(b)  $\widetilde{KO}^0(\#^k \mathbb{CP}^{4n+1})$  is a free module with a basis

$$\{q^*(\eta_k^j), q^*(\eta_k^{2n+1}), \eta_i^j \mid 2q^*(\eta_k^{2n+1}) = 0, i \in I', j = 1, 2, \dots, 2n\}.$$

(c)  $\widetilde{KO}^0(\#^k \mathbb{CP}^{4n+2})$  is a free module with a basis

$$\{q^*(\eta_k^j), \eta_i^j \mid 2\eta_i^{2n+1} = 0, i \in I', j = 1, 2, \dots, 2n+1\}.$$

(d)  $\widetilde{KO}^0(\#^k \mathbb{CP}^{4n+3})$  is a free module with a basis

$$\{q^*(\eta_k^j), \eta_i^j \mid i \in I', j = 1, 2, \dots, 2n+1\}.$$

$$(2) \quad \widetilde{KO}^{-1}(\#^k \mathbb{CP}^n) = 0$$

(3) (a)  $\widetilde{KO}^{-2}(\#^k \mathbb{CP}^{4n})$  is a free module with a basis

$$\{q^*(\alpha\eta_k^j), q^*(\alpha\eta_k^{2n-1}), \alpha\eta_i^j, \sigma_i \mid 2\sigma_i = \alpha\eta_i^{2n-1}, i \in I', j = 0, 1, \dots, 2n-2\}.$$

(b)  $\widetilde{KO}^{-2}(\#^k \mathbb{CP}^{4n+1})$  is a free module with a basis

$$\{q^*(\alpha\eta_k^j), q^*(\alpha\eta_k^{2n}), \alpha\eta_i^j \mid i \in I', j = 0, 1, \dots, 2n-1\}.$$

(c)  $\widetilde{KO}^{-2}(\#^k \mathbb{CP}^{4n+2})$  is a free module with a basis

$$\{q^*(\alpha\eta_k^j), \alpha\eta_i^j \mid i \in I', j = 0, 1, \dots, 2n\}.$$

(d)  $\widetilde{KO}^{-2}(\#^k \mathbb{CP}^{4n+3})$  is a free module with a basis

$$\{q^*(\alpha\eta_k^j), q^*(\sigma_k), \alpha\eta_i^j \mid 2\sigma_k = \alpha\eta_k^{2n+1}, i \in I', j = 0, 1, \dots, 2n\}.$$

$$(4) \quad \widetilde{KO}^{-3}(\#^k \mathbb{CP}^n) = \begin{cases} \mathbb{Z}_2^{k-1} & ; n = 4m \\ \mathbb{Z}_2 & ; n = 4m + 3 \\ 0 & ; n = 4m + 1 \text{ or } n = 4m + 2. \end{cases}$$

(5) (a)  $\widetilde{KO}^{-4}(\#^k \mathbb{CP}^{4n})$  is a free module with a basis

$$\{q^*(\beta \eta_k^j), \beta \eta_i^j \mid 2\beta \eta_i^{2n-1} = 0, i \in I', j = 0, 1, \dots, 2n-1\}.$$

(b)  $\widetilde{KO}^{-4}(\#^k \mathbb{CP}^{4n+1})$  is a free module with a basis

$$\{q^*(\beta \eta_k^j), \beta \eta_i^j \mid i \in I', j = 0, 1, \dots, 2n-1\}.$$

(c)  $\widetilde{KO}^{-4}(\#^k \mathbb{CP}^{4n+2})$  is a free module with a basis

$$\{q^*(\beta \eta_k^j), q^*(\beta \eta_k^{2n}), \beta \eta_i^j \mid i \in I', j = 0, 1, \dots, 2n-1\}.$$

(d)  $\widetilde{KO}^{-4}(\#^k \mathbb{CP}^{4n+3})$  is a free module with a basis

$$\{q^*(\beta \eta_k^j), q^*(\beta \eta_k^{2n+1}), \beta \eta_i^j \mid 2\beta \eta_k^{2n+1} = 0, i \in I', j = 0, 1, \dots, 2n\}.$$

$$(6) \quad \widetilde{KO}^{-5}(\#^k \mathbb{CP}^n) = 0.$$

(7) (a)  $\widetilde{KO}^{-6}(\#^k \mathbb{CP}^{4n})$  is a free module with a basis

$$\{q^*(\gamma \eta_k^j), \gamma \eta_i^j \mid i \in I' \text{ and } j = 0, 1, \dots, 2n-1\}.$$

(b)  $\widetilde{KO}^{-6}(\#^k \mathbb{CP}^{4n+1})$  is a free module with a basis

$$\{q^*(\gamma \eta_k^j), q^*(\tau_k), \gamma \eta_i^j \mid 2\tau_k = \gamma \eta_k^{2n}, i \in I' \text{ and } j = 0, 1, \dots, 2n-1\}.$$

(c)  $\widetilde{KO}^{-6}(\#^k \mathbb{CP}^{4n+2})$  is a free module with a basis

$$\{q^*(\gamma\eta_k^j), q^*(\gamma\eta_k^{2n}), \gamma\eta_i^j, \tau_i \mid 2\tau_i = \gamma\eta_i^{2n}, i \in I' \text{ and } j = 0, 1, \dots, 2n-1\}.$$

(d)  $\widetilde{KO}^{-6}(\#^k \mathbb{CP}^{4n+3})$  is a free module with a basis

$$\{q^*(\gamma\eta_k^j), q^*(\gamma\eta_k^{2n+1}), \gamma\eta_i^j \mid i \in I' \text{ and } j = 0, 1, \dots, 2n\}.$$

$$(8) \quad \widetilde{KO}^{-7}(\#^k \mathbb{CP}^n) = \begin{cases} \mathbb{Z}_2^{k-1} & ; n = 4m + 2 \\ \mathbb{Z}_2 & ; n = 4m + 1 \\ 0 & ; n = 4m \text{ or } n = 4m + 3. \end{cases}$$

*Proof.* The proof primarily relies on a specific instance of the sequence (6.3), which can be stated as follows:

$$\dots \oplus_{k-1} \widetilde{KO}^{-s-1}(\mathbb{CP}^{n-1}) \xrightarrow{c^*} \widetilde{KO}^{-s}(\mathbb{CP}^n) \xrightarrow{q^*} \widetilde{KO}^{-s}(\#^k \mathbb{CP}^n) \xrightarrow{i^*} \oplus_{k-1} \widetilde{KO}^{-s}(\mathbb{CP}^{n-1}) \dots \quad (6.4)$$

We will establish that this sequence exhibits a splitting at  $\widetilde{KO}^{-s}(\#^k \mathbb{CP}^n)$  for  $k, n \geq 2$  and  $0 \leq s \leq 7$ . As a result, we obtain a basis for the group  $\widetilde{KO}^{-s}(\#^k \mathbb{CP}^n)$  in terms of the generators of  $\widetilde{KO}^{-s}(\mathbb{CP}^n)$  and  $\widetilde{KO}^{-s}(\mathbb{CP}^{n-1})$ .

The proof of (1) will be presented following the computation of  $\widetilde{KO}^{-7}(\#^k \mathbb{CP}^n)$ .

The proof of (2) follows immediately by considering  $s = 1$  in the sequence (6.4) using the fact that  $\widetilde{KO}^{-1}(\mathbb{CP}^n) = 0$  for all  $n \in \mathbb{N}$ .

To establish (3), we consider  $s = 2$  in the sequence (6.4). Remarkably, the map  $c^*$  is always trivial in this case. This can be attributed to the fact that  $\widetilde{KO}^{-3}(\mathbb{CP}^{n-1})$  is either 0 or  $\mathbb{Z}_2$ , while  $\widetilde{KO}^{-2}(\mathbb{CP}^n)$  is a free group for any  $n \in \mathbb{N}$ . Moreover, the group  $\widetilde{KO}^{-1}(\mathbb{CP}^n)$  is trivial, leading to the conclusion that the short exact sequence splits.

Now, to prove (4), let us examine the case of  $s = 3$  in the sequence (6.4). The computation of  $\widetilde{KO}^{-3}(\#^k \mathbb{CP}^n)$  becomes straightforward when utilizing this sequence for  $n = 4m, 4m + 1$ , and  $4m + 2$ . For  $n = 4m + 3$ , the sequence offers two options: either 0 or  $\mathbb{Z}_2$ . To eliminate one of the choices, we can consider the sequence (6.1) for  $Y = BSO$  and  $s = 3$ . By employing the

identification  $[\Sigma^s X, BSO] = \widetilde{KO}^{-s}(X)$ , we obtain the following result:

$$0 \longrightarrow \widetilde{KO}^{-s}(\mathbb{S}^{2n}) / \text{Im}(\Sigma^{s+1}h)^* \xrightarrow{\widetilde{d}^*} \widetilde{KO}^{-s}(\#^k \mathbb{CP}^n) \xrightarrow{\iota^*} \bigoplus_{k-1} \widetilde{KO}^{-s}(\mathbb{CP}^{n-1}) \oplus \text{Ker}(\Sigma^s h)^* \longrightarrow 0. \quad (6.5)$$

By considering  $s = 3$  and  $n = 4m + 3$ , and substituting the known value  $\widetilde{KO}^{-3}(\mathbb{CP}^{4m+2}) = 0$  into this sequence, we observe that  $\widetilde{d}^*$  becomes an isomorphism. Furthermore, when we examine this sequence for  $k = 1$ , we find that  $\text{Im}(\Sigma^4 h)^*$  is trivial. This completes the proof for this particular case.

The proofs of (5)(b), (5)(c), and (5)(d) follow directly from applying the sequence (6.4) with  $s = 4$ . Regarding (5)(a), if we utilize the sequence (6.4) for  $\widetilde{KO}^{-4}(\#^k \mathbb{CP}^{4m})$ , we can establish that the torsion part of  $\widetilde{KO}^{-4}(\#^k \mathbb{CP}^{4m})$  is  $\mathbb{Z}_2^t$ , where the maximum possible value of  $t$  in the exact sequence is  $k - 1$ . Finally, using the sequence (6.5) for  $s = 4$  and  $n = 4m$  we establish that  $t$  is precisely equal to  $k - 1$ .

The proof of (6) follows a similar approach to that of (2), as  $\widetilde{KO}^{-5}(\mathbb{CP}^n)$  is zero for all  $n \in \mathbb{N}$ .

Likewise, the proof of (7) follows an analogous approach to that of (3).

To prove (8), consider  $s = 7$  in 6.4. The cases  $n = 4m, 4m + 2$ , and  $4m + 3$  follow immediately from the sequence. Further, the computation of the remaining case  $n = 4m + 1$  follows analogously the way we computed  $\widetilde{KO}^{-3}(\#^k \mathbb{CP}^{4m+3})$  using sequence 6.5.

The proof of (1)(a),(b) and (d) follows immediately using sequence (6.4) for  $s = 0$ . If we consider  $n = 4m + 2$  in 6.4 we get  $\widetilde{KO}^0(\#^k \mathbb{CP}^{4m+2}) = \mathbb{Z}^{2kt+1} \oplus \mathbb{Z}_2^k$  or  $\mathbb{Z}^{2kt+1} \oplus \mathbb{Z}_2^{k-1}$ . Finally, to eliminate one possibility, use the sequence (6.5) for  $s = 0$  and  $n = 4m + 2$ . This completes the proof for all cases.

□

## Chapter 7

# Tangential Smoothings of $k$ -fold Connected Sum of Complex Projective Spaces

In surgery theory, one of the key objectives is to determine the diffeomorphism classes of tangential homotopic manifolds (see [5, 11, 18, 49]). This involves the study of various structure sets, many of them we have already computed for  $\#^k \mathbb{CP}^n$  in Chapter 5. Here, for a manifold  $M$ , recall that the set  $\mathcal{N}(M)$  is the collection normal cobordism classes of degree one normal maps on  $M$ . In [51], Sullivan established a significant bijection between  $\mathcal{N}(M)$  and the group  $[M, F/O]$ , where  $F/O$  denotes the homotopy fiber of the natural map  $BSO \rightarrow BSF$ . Recall from Section 2.4 that,  $BSO$  and  $BSF$  are the classifying spaces for stable real vector bundles and stable spherical fibrations, respectively, (the reader can also refer to [54, Lemma 10.6], and Sections 2 and 3, [37]). Brumfiel's result (Page 401, [7]), is focused on the splitting of  $[\mathbb{CP}^n, F/O]$ . Note that we have computed the normal invariant set of  $\#^k \mathbb{CP}^n$  in Theorem 7.1.3. Our Theorem 7.1.3 extends the Brumfiel's result to  $k$ -fold connected sum of complex projective spaces. Moreover, it is observed that the torsion part of this group is isomorphic to  $\pi_s^0(\#^k \mathbb{CP}^n)$  for all  $k, n \in \mathbb{N}$ , and the free part of  $[\#^k \mathbb{CP}^n, F/O]$  corresponds to the free part of the group  $\widetilde{KO}(\#^k \mathbb{CP}^n)$ .

Finally in Section 7.2, we compute the smooth and  $PL$  tangential structure set of  $\#^k \mathbb{CP}^n$ . For that we require more results which we discuss now.

## 7.1 Smooth normal invariant set

Let  $\mathcal{N}(M)$  denote the set of smooth normal invariants of  $M$ . Recall that  $\mathcal{N}(M)$  occurs in the smooth surgery exact sequence, and the group  $[M, F/O]$  are in bijection [51]. Consequently, studying the group  $[M, F/O]$  becomes crucial for gaining a deeper understanding of surgery theory. For  $\#^k \mathbb{CP}^n$ , we can examine the following long exact sequence obtained from the fiber sequence (2.3), to know  $\mathcal{N}(\#^k \mathbb{CP}^n)$ ;

$$\begin{array}{ccccccc} \cdots & \longrightarrow & [\#^k \mathbb{CP}^n, SO] & \xrightarrow{a_*} & [\#^k \mathbb{CP}^n, SF] & \xrightarrow{b_*} & [\#^k \mathbb{CP}^n, F/O] \\ & & & & \searrow c_* & & \\ & & [\#^k \mathbb{CP}^n, BSO] & \xrightarrow{f_*} & [\#^k \mathbb{CP}^n, BSF] & \longrightarrow & \cdots \end{array} \quad (7.1)$$

In this sequence, observe that, for all  $k, n \in \mathbb{N}$  the group  $[\#^k \mathbb{CP}^n, SO]$  coincides with the  $\widetilde{KO}^{-1}(\#^k \mathbb{CP}^n)$ , and which is trivial by Theorem 6.2.2 (2). Therefore, in (7.1), the map  $b_*$  is injective. To understand  $c_*$ , let us first shift our attention to compute the kernel of the map  $f_* : \widetilde{KO}^0(\#^k \mathbb{CP}^n) \rightarrow [\#^k \mathbb{CP}^n, BSF]$ . Note that the case  $k = 1$  is already established in the [7, pg 401]. The result says that the kernel of the map  $f_* : \widetilde{KO}^0(\mathbb{CP}^n) \rightarrow [\mathbb{CP}^n, BSF]$  is torsion-free for all  $n \in \mathbb{N}$ , where  $f : BSO \rightarrow BSF$  represents the canonical fibration. In the subsequent discussion, we aim to prove a similar result for the  $k$ -fold connected sum of  $\mathbb{CP}^n$ .

**Theorem 7.1.1.** *The kernel of  $f_* : \widetilde{KO}^0(\#^k \mathbb{CP}^n) \rightarrow [\#^k \mathbb{CP}^n, BSF]$  is torsion free for all  $k, n \geq 2$ .*

*Proof.* For the cases when  $n = 4m$  or  $4m + 3$ , the result is straightforward since the groups  $\widetilde{KO}^0(\#^k \mathbb{CP}^n)$  are free.

In the remaining cases, we will demonstrate that the torsion part of  $\widetilde{KO}^0(\#^k \mathbb{CP}^n)$  maps injectively under  $f_*$ . To do so, we consider the following commutative diagram with the map  $J_*$

obtained from the  $J$ -homomorphism;

$$\begin{array}{ccccccc}
0 & \longrightarrow & \widetilde{KO}^0(\mathbb{S}^{2n}) & \xrightarrow{d^*} & \widetilde{KO}^0(\#^k \mathbb{CP}^n) & \xrightarrow{\iota^*} & \oplus_k \widetilde{KO}^0(\mathbb{CP}^{n-1}) \\
\downarrow & & J_* \downarrow & & f_* \downarrow & & \downarrow \oplus_k f_* \\
\oplus_k [\Sigma \mathbb{CP}^{n-1}, BSF] & \xrightarrow{(\Sigma \xi)^*} & [\mathbb{S}^{2n}, BSF] & \xrightarrow{d^*} & [\#^k \mathbb{CP}^n, BSF] & \xrightarrow{\iota^*} & \oplus_k [\mathbb{CP}^{n-1}, BSF],
\end{array} \tag{7.2}$$

the horizontal rows are part of long exact sequences obtained by the cofiber sequence (5.1) for  $\#^{i \in I} M_i = \#^k \mathbb{CP}^n$ ,  $Y = BSO$  and  $BSF$  respectively, and the map  $\xi = (\vee_k h_C) \circ p$  such that  $h$  is the Hopf map and  $p$  is the pinch map.

For the case when  $n = 4m + 1$ , we have  $\widetilde{KO}^0(\#^k \mathbb{CP}^{4m+1}) \cong \mathbb{Z}^{2km} \oplus \mathbb{Z}_2$  according to Theorem 6.2.2 (b). Therefore, in order to demonstrate that  $\text{Ker}(f_*)$  is torsion-free, it suffices to show that  $\text{Im}(f_*(\mathbb{Z}_2))$  is non-zero.

We observe that  $\widetilde{KO}^0(\mathbb{S}^{8m+2}) \cong \mathbb{Z}_2$ , and the injectivity of the map  $J_*$  implies that  $\text{Im}(J_*) \cong \mathbb{Z}_2$  [47]. Thus, our objective is to show that the composition  $d^* \circ J_*$  is non-zero, which is equivalent to proving that  $\text{Im}(J_*) \not\subseteq \text{Ker}(d^*) = \text{Im}((\Sigma \xi)^*)$ . Note that, as indicated in the proof of Theorem 5.1.2, we have  $\text{Im}((\Sigma \xi)^*) = \text{Im}((\Sigma h)^*)$ . Hence, the problem reduces to verifying that  $\text{Im}(J_*) \not\subseteq \text{Im}((\Sigma h)^*)$ .

To prove this consider the following commutative diagram for an individual copy of  $\mathbb{CP}^n$

$$\begin{array}{ccccccc}
\cdots 0 & \longrightarrow & \widetilde{KO}^0(\mathbb{S}^{8m+2}) = \mathbb{Z}_2 & \xrightarrow{d^*} & \widetilde{KO}^0(\mathbb{CP}^{4m+1}) & \cdots \\
\downarrow & & J_* \downarrow & & f_* \downarrow & \\
\cdots [\Sigma \mathbb{CP}^{4m}, BSF] & \xrightarrow{(\Sigma h)^*} & [\mathbb{S}^{8m+2}, BSF] & \xrightarrow{d^*} & [\mathbb{CP}^{4m+1}, BSF] & \cdots,
\end{array}$$

where  $\widetilde{KO}^0(\mathbb{CP}^{4m+1}) \cong \mathbb{Z}^{2m} \oplus \mathbb{Z}_2$ . According to [7], the kernel of  $f_* : \widetilde{KO}^0(\mathbb{CP}^{4m+1}) \rightarrow [\mathbb{CP}^{4m+1}, BSF]$  is torsion-free. Therefore,  $f_*(\text{Im}(d^*))$  is non-zero since  $d^*$  is injective. As a result,  $d^*(\text{Im}(J_*) = \mathbb{Z}_2)$  is also non-zero. Consequently, we can conclude that  $\text{Im}(J_*) \not\subseteq \text{Ker}(d^*) = \text{Im}((\Sigma h)^*)$ , which completes the proof in this case.

Lastly, consider the case  $n = 4m + 2$  in (7.2). Since  $\widetilde{KO}^0(\mathbb{S}^{8m+4}) = \mathbb{Z}$  and by Theorem 6.2.2 (c) the  $\widetilde{KO}^0(\#^k \mathbb{CP}^{4m+2}) = \mathbb{Z}^{2km+1} \oplus \mathbb{Z}_2^{k-1}$ , we can observe that the map  $\iota^*|_{\mathbb{Z}_2^{k-1}}$  is a monomorphism for the  $\mathbb{Z}_2^{k-1}$  component. Additionally, since the kernel of  $f_* : \widetilde{KO}^0(\mathbb{CP}^{4m+1}) \rightarrow$

$[\mathbb{CP}^{4m+1}, BSF]$  is a free group [47], the map  $f_*|_{\mathbb{Z}_2}$  is also a monomorphism. This implies that the composition  $(\oplus_k f_*) \circ \iota^*|_{\mathbb{Z}_2^{k-1}}$  is a monomorphism. Consequently, we can conclude that  $f_*|_{\mathbb{Z}_2^{k-1}} : \widetilde{KO}^0(\#^k \mathbb{CP}^{4m+2}) \rightarrow [\#^k \mathbb{CP}^{4m+2}, BSF]$  is a monomorphism, completing the proof.  $\square$

Note that, from [7], it is known for all  $n \in \mathbb{N}$ , the groups  $[\mathbb{CP}^n, SF]$  and  $[\mathbb{CP}^n, BSF]$  are finite abelian groups. Thus, using the long exact sequence (6.3) for  $Y = SF$  and  $BSF$  respectively, we conclude that the groups  $[\#^k \mathbb{CP}^n, SF]$  and  $[\#^k \mathbb{CP}^n, BSF]$  are also finite abelian for all  $k, n \in \mathbb{N}$ . Recall that, in the Theorem 7.1.1, we established that the kernel of the induced map  $f_*$  is a free abelian group. Using the fact that the group  $[\#^k \mathbb{CP}^n, BSF]$  is finite abelian, we can conclude the following remark:

**Remark 7.1.2.** *The kernel of  $f_* : [\#^k \mathbb{CP}^n, BSO] \rightarrow [\#^k \mathbb{CP}^n, BSF]$  is isomorphic to the free part of  $[\#^k \mathbb{CP}^n, BSO]$ , for all  $k, n \in \mathbb{N}$ .*

Now, we have enough information to presents a result for the group  $[\#^k \mathbb{CP}^n, F/O]$  with  $k, n \geq 2$ , which is  $k$ -fold connected sum analogue to the Brumfiles result from [8, page 401]. Note that, corresponding to the exact sequence (7.1), we have the following short exact sequence

$$0 \rightarrow [\#^k \mathbb{CP}^n, SF] \xrightarrow{b_*} [\#^k \mathbb{CP}^n, F/O] \xrightarrow{c_*} \text{Ker}(f_*) \rightarrow 0, \quad (7.3)$$

where  $[\#^k \mathbb{CP}^n, SF]$  is a finite abelian group and  $\text{Ker}(f_*)$  is a free abelian group. Therefore, the short exact sequence (2.3) must split. The formal statement of this finding, together with the result for  $k = 1$ , is stated in the following theorem.

**Theorem 7.1.3.** *Let  $k, n \in \mathbb{N}$ . Then*

$$[\#^k \mathbb{CP}^n, F/O] \cong \begin{cases} \pi_s^0(\#^k \mathbb{CP}^n) \oplus \mathbb{Z}^{k[\frac{n}{2}]} & \text{if } n \text{ odd,} \\ \pi_s^0(\#^k \mathbb{CP}^n) \oplus \mathbb{Z}^{k[\frac{n-1}{2}]+1} & \text{if } n \text{ even.} \end{cases}$$

Recall that, the set  $\{\eta^j \mid j = 1, 2, \dots, n\}$  forms an integral basis for the free abelian group  $\widetilde{K}^0(\mathbb{CP}^n)$ , for all  $n \in \mathbb{N}$ . Let  $\bar{\eta}$  be the realification of  $\eta$ . Then we have the following result using [36, §4].

**Theorem 7.1.4.** *Let  $\xi_1 = 24\bar{\eta} + 98\bar{\eta}^2 + 111\bar{\eta}^3$ ,  $\xi_2 = 240\bar{\eta}^2 + 380\bar{\eta}^3$ , and  $\xi_3 = 504\bar{\eta}^3$ . Then*



the image of  $c_* : [\#^k \mathbb{CP}^n, F/O] \rightarrow [\#^k \mathbb{CP}^n, BSO]$ , for  $n \leq 7$  and  $k \in \mathbb{Z}$ , is generated by  $\xi_i$  and  $q^*(\xi_i)$  for  $i = 1, 2, 3$ , where  $q : \#^k \mathbb{CP}^n \rightarrow \mathbb{CP}^n$  is the quotient map.

## 7.2 Smooth tangential structure set

In the concluding section, we compute the smooth tangential structure set of  $\#^k \mathbb{CP}^n$ . Recall from [11, §6] that, a *smooth tangential structure set* of a closed, smooth  $n$ -manifold  $M$  with stable normal bundle  $\nu_M$  of rank  $k \gg n$ , denoted by  $S_t^{\text{Diff}}(M)$ , is a triple  $(N, f, b)$  such that  $f : N \rightarrow M$  is a homotopy equivalence and  $b : \nu_N \rightarrow \nu_M$  is a map of stable bundles. Two structures  $(N_1, f_1, b_1)$ ,  $(N_2, f_2, b_2)$  are said to be equivalent if there is an  $s$ -cobordism  $(U; N_1, N_2)$  with  $F : U \rightarrow M$  is a simple homotopy equivalence,  $F : \nu_U \rightarrow \nu_M$  is a bundle map and these data restrict to  $(N_1, f_1, b_1)$  and  $(N_2, f_2, b_2)$  at the boundary of  $U$ .

In the smooth  $PL$  category, there exists an analogous notion of the tangential  $PL$  smoothings of a manifold  $M$  with a tangent bundle  $\tau_M$ , denoted by  $S_t^{PL}(M)$  [49, Page 194]. It is a collection of triples  $(V, t, \phi)$  with concordance relation, where  $V$  is a smooth manifold,  $t : V \rightarrow M$  is a piecewise differential homeomorphism, and  $\phi : E(\tau_V \oplus k) \rightarrow E(\tau_M \oplus k)$  is a vector bundle isomorphism covering  $t$ . Notably, [49, Theorem 3.8] establishes a one-to-one correspondence between  $S_t^{PL}(M)$  and the group  $[M, PL]$ , where  $PL$  is colimit of groups  $PL_n$ , the self  $PL$ -homeomorphisms of  $\mathbb{R}^n$  over  $n$ . Additionally, there exists a forgetful map  $H : S_t^{PL}(M) \rightarrow S_t^{\text{Diff}}(M)$ , which maps tangential  $PL$  smoothing to tangential homotopy smoothing.

To compute these groups and understand the behavior of the map  $H$  specifically for  $\#^k \mathbb{CP}^n$ , we explore some results. For that, consider the following homotopy commutative diagram

$$\begin{array}{ccccccccc}
 SO & \longrightarrow & SF & \xrightarrow{b} & F/O & \xrightarrow{c} & BSO & \xrightarrow{f} & BSF \\
 \parallel & & \uparrow l & & \uparrow & & \parallel & & \uparrow \\
 SO & \longrightarrow & PL & \xrightarrow{v} & PL/O & \xrightarrow{w} & BSO & \xrightarrow{g} & BPL \\
 & & \uparrow & & & & & & \\
 & & \Omega(F/PL) & & & & & & 
 \end{array} \tag{7.4}$$

where the rows and columns are homotopy fiber sequences and all maps are natural maps. Here,  $PL/O$  is a homotopy fiber of the map  $BO \rightarrow BPL$ , and  $F/PL$  is a homotopy fiber of the map

$BPL \rightarrow BSF$ , where  $BPL$  is the classifying space of  $PL$ .

Given a  $PL$  self-homeomorphism  $h$  of a smooth  $n$ -manifold  $M$  and a homotopy of  $h$  to the identity, there is an associated element of  $[M, \Omega(F/PL)]$ , which vanishes if the homotopy can be deformed to a pseudo-isotopy of  $h$  to the identity. In particular, for  $\#^k \mathbb{CP}^n$ , we have the following result:

**Lemma 7.2.1.** *Every  $PL$  self-homeomorphism of  $\#^k \mathbb{CP}^n$  homotopic to identity is pseudo-isotopic to the identity.*

*Proof.* The [7, Corollary 2.3] states that  $[\mathbb{CP}^n, \Omega(F/PL)] = 0$  for all  $n \in \mathbb{N}$ . Thus, using Lemma 6.2.1, we get  $[\#^k \mathbb{CP}^n, \Omega(F/PL)] = 0$  for all  $k, n \in \mathbb{N}$ , implying the statement.  $\square$

**Theorem 7.2.2.** *The forgetful map  $H : S_t^{PL}(\#^k \mathbb{CP}^n) \rightarrow S_t^{Diff}(\#^k \mathbb{CP}^n)$  is injective, for all  $k, n \in \mathbb{N}$ .*

*Proof.* The statement follows immediately by using Lemma 7.2.1 in the following commutative diagram mentioned in [49, Theorem 3.9] with  $M$  substituted by  $\#^k \mathbb{CP}^n$ .

$$\begin{array}{ccccc} [\Sigma \#^k \mathbb{CP}^n, F/PL] & \longrightarrow & S_t^{PL}(\#^k \mathbb{CP}^n) & \longrightarrow & [\#^k \mathbb{CP}^n, SF] \\ \downarrow & & \downarrow H & & \downarrow \cong \\ 0 & \longrightarrow & S_t^{Diff}(\#^k \mathbb{CP}^n) & \longrightarrow & \pi_s^0(\#^k \mathbb{CP}^n) \end{array} \quad (7.5)$$

$\square$

In the category of  $PL$ -manifolds, the notion of  $PL$ -normal invariant set  $\mathcal{N}_{PL}(M)$ , which is in bijection with  $[M, F/PL]$ , is analogous to the smooth normal invariant set. For the next theorem, we compute the group  $[\#^k \mathbb{CP}^n, F/PL]$ , using the sequence (5.1), and the fact that  $[\mathbb{CP}^m, F/PL] = \prod_{k=2}^m P_{2k}$  where  $P_i = \mathbb{Z}, 0, \mathbb{Z}_2, 0$  for  $i = 0, 1, 2, 3 \pmod{4}$ , respectively,

$$[\#^k \mathbb{CP}^n, F/PL] \cong \begin{cases} \mathbb{Z}^{k \lfloor \frac{n-1}{2} \rfloor + 1} \oplus \mathbb{Z}_2^{k \lfloor \frac{n-1}{2} \rfloor} & \text{if } n \text{ is even,} \\ \mathbb{Z}^{k \lfloor \frac{n}{2} \rfloor} \oplus \mathbb{Z}_2^{k(\lfloor \frac{n}{2} \rfloor - 1) + 1} & \text{if } n \text{ is odd.} \end{cases} \quad (7.6)$$

**Theorem 7.2.3.** *For all  $k, n \in \mathbb{N}$  we have the following.*

1. *The map  $l_* : [\#^k \mathbb{CP}^n, PL] \rightarrow [\#^k \mathbb{CP}^n, SF]$  is injective, and is an isomorphism on odd torsion.*

2. The map  $w_* : [\#^k \mathbb{CP}^n, PL/O] \rightarrow [\#^k \mathbb{CP}^n, BSO]$  is zero.
3. The natural induced map  $v_* : [\#^k \mathbb{CP}^n, PL] \rightarrow [\#^k \mathbb{CP}^n, PL/O]$  is an isomorphism.

*Proof.* 1. Using Lemma 7.2.1 we have  $[\#^k \mathbb{CP}^n, \Omega(F/PL)] = 0$ . Therefore, from the diagram (7.4) the induced map  $l_* : [\#^k \mathbb{CP}^n, PL] \rightarrow [\#^k \mathbb{CP}^n, SF]$  is injective for all  $k, n \in \mathbb{N}$ .

Now, from (7.6) it is clear that  $[\#^k \mathbb{CP}^n, F/PL]$  has no odd torsion subgroup. This implies  $l_*$  is an isomorphism after localizing at odd primes.

2. It is known from [30] that, the group  $[\#^k \mathbb{CP}^n, PL/O]$  is finite abelian for all  $k, n \in \mathbb{N}$ . Now, consider the following homotopy commutative diagram obtained using (7.4)

$$\begin{array}{ccc} [\#^k \mathbb{CP}^n, BSO] & \xrightarrow{g_*} & [\#^k \mathbb{CP}^n, BPL] \\ \parallel & & \downarrow \\ [\#^k \mathbb{CP}^n, BSO] & \xrightarrow{f_*} & [\#^k \mathbb{CP}^n, BSF] \end{array}$$

In Theorem 7.1.1, we proved that the torsion part of  $[\#^k \mathbb{CP}^n, BSO]$  gets mapped injectively under  $f_*$ , and consequently, under  $g_*$  as well. Hence, in (7.4), the  $\text{Im}(w_*)$  is a free abelian group. Since  $[\#^k \mathbb{CP}^n, PL/O]$  is a finite abelian group, the map  $w_*$  must be zero.

3. Theorem 6.2.2(2) gives  $[\#^k \mathbb{CP}^n, SO] = 0$  for all  $k, n \in \mathbb{N}$ . Thus, the map  $v_* : [\#^k \mathbb{CP}^n, PL] \rightarrow [\#^k \mathbb{CP}^n, PL/O]$  is injective, and its surjectivity follows from (2) of this theorem and by from diagram (7.4).

□

**Remark 7.2.4.** Recall that, the set  $[\#^k \mathbb{CP}^n, PL/O]$  can be identified with the concordance structure set  $\mathcal{C}(\#^k \mathbb{CP}^n)$  (Chapter 2). From Theorem 7.2.3 (3),  $\mathcal{C}(\#^k \mathbb{CP}^n)$  and the  $S_t^{PL}(\#^k \mathbb{CP}^n)$  are isomorphic, for all  $k, n \in \mathbb{N}$ .

Now, we use Theorem 7.2.3, and compute  $\pi_0^s(\#^k \mathbb{CP}^5)$ .

**Theorem 7.2.5.**  $\pi_s^0(\#^k \mathbb{CP}^5) \cong \mathbb{Z}_2^{2k} \oplus \mathbb{Z}_3$

*Proof.* From Theorem 5.4.4 (iii), we obtained  $[\#^k \mathbb{CP}^5, SF] \cong \mathbb{Z}_2^{2k} \oplus \mathbb{Z}_3$  or  $\mathbb{Z}_2^{2k-2} \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_3$  depending upon the splitting or non-splitting of the corresponding short exact sequence. To

conclude this, let us consider the following commutative diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & [\#^k \mathbb{CP}^n, PL/O] & \xrightarrow{b'_*} & [\#^k \mathbb{CP}^n, F/O] & \xrightarrow{c'_*} & [\#^k \mathbb{CP}^n, F/PL] \xrightarrow{f'_*} \dots \\
& & \downarrow & & \parallel & & \downarrow \\
0 & \longrightarrow & [\#^k \mathbb{CP}^n, SF] & \xrightarrow{b_*} & [\#^k \mathbb{CP}^n, F/O] & \xrightarrow{c_*} & [\#^k \mathbb{CP}^n, BSO] \xrightarrow{f_*} \dots
\end{array} \tag{7.7}$$

The injectivity of the map from  $[\#^k \mathbb{CP}^n, PL/O] \rightarrow [\#^k \mathbb{CP}^n, SF]$  follows from Theorem 7.2.3(1) and (3), which maps a homeomorphism to a tangential homotopy equivalence. We showed the splitting of the following short exact sequence while proving Theorem 7.1.3

$$0 \longrightarrow [\#^k \mathbb{CP}^n, SF] \xrightarrow{b_*} [\#^k \mathbb{CP}^n, F/O] \xrightarrow{c_*} \text{Im}(c_*) \longrightarrow 0.$$

This shows that the short exact sequence

$$0 \longrightarrow [\#^k \mathbb{CP}^n, PL/O] \xrightarrow{b'_*} [\#^k \mathbb{CP}^n, F/O] \xrightarrow{c'_*} \text{Im}(c'_*) \longrightarrow 0,$$

obtained from the top row of (7.7) is a split short exact sequence as well.

Furthermore, according to Theorem 5.3.2 (i), we have  $[\#^k \mathbb{CP}^5, PL/O] \cong \mathbb{Z}_2^{k+1} \oplus \mathbb{Z}_3$ , and from (7.6), we find that  $[\#^k \mathbb{CP}^5, F/PL] \cong \mathbb{Z}^{2k} \oplus \mathbb{Z}_2^{k+1}$ . Consequently, we conclude that the torsion part of  $[\#^k \mathbb{CP}^n, F/O]$  does not contain  $\mathbb{Z}_4$ . Hence,  $[\#^k \mathbb{CP}^n, SF]$  does not contain  $\mathbb{Z}_4$ , thereby completing the proof.  $\square$

Recall that, the smooth tangential structure set of a simply-connected manifold  $M$  is a part of the following smooth tangential surgery exact sequence for  $M$

$$L_{2n+1}(\mathbb{Z}\pi) \longrightarrow S_t^{\text{Diff}}(M) \xrightarrow{\eta} \mathcal{N}_t^{\text{Diff}}(M) \xrightarrow{s} L_{2n}(\mathbb{Z}\pi), \tag{7.8}$$

where  $\mathcal{N}_t^{\text{Diff}}(M)$  is the set of smooth tangential normal invariants of  $M$ , and  $L_i(\mathbb{Z}\pi) = \mathbb{Z}, 0, \mathbb{Z}_2, 0$  for  $i \equiv 0, 1, 2, 3 \pmod{4}$  respectively. The surgery obstruction map  $s$  is a group homomorphism if  $n$  is odd; otherwise, it is not a group homomorphism. Further, note that, there exists a bijection

$$\mathcal{N}_t^{\text{Diff}}(M) \cong [M, SF].$$

We have computed  $\mathcal{N}_t^{\text{Diff}}(\#^k \mathbb{CP}^n)$  as detailed in Theorems 5.4.4, 5.4.5 and 7.2.5. Since  $L_{2n+1} = 0$ , the map  $\eta : S_t^{\text{Diff}}(\#^k \mathbb{CP}^n) \rightarrow \mathcal{N}_t^{\text{Diff}}(\#^k \mathbb{CP}^n)$  is injective for all  $k, n \in \mathbb{N}$ . More precisely, we have the following:

**Theorem 7.2.6.** *Let  $k, n \in \mathbb{N}$ . Then*

1. *The map  $\eta : S_t^{\text{Diff}}(\#^k \mathbb{CP}^n) \rightarrow \mathcal{N}_t^{\text{Diff}}(\#^k \mathbb{CP}^n)$  is an isomorphism for  $n = 3, 4$  and 6.*
2. *The map  $\eta : S_t^{\text{Diff}}(\#^k \mathbb{CP}^5) \rightarrow \mathcal{N}_t^{\text{Diff}}(\#^k \mathbb{CP}^5)$  has image isomorphic to  $\mathbb{Z}_2^{2k-1} \oplus \mathbb{Z}_3$ .*
3. *The map  $\eta : S_t^{\text{Diff}}(\#^k \mathbb{CP}^7) \rightarrow \mathcal{N}_t^{\text{Diff}}(\#^k \mathbb{CP}^7)$  has image isomorphic to  $S_t^{\text{PL}}(\#^k \mathbb{CP}^7)$ .*

*Proof.* Consider the commutative diagram (7.5). Then using Theorems 7.2.1 and 7.2.3(3) together, we obtain that  $[\#^k \mathbb{CP}^n, PL/O]$  embeds injectively in  $S_t^{\text{Diff}}(\#^k \mathbb{CP}^n)$ . In addition to this, Proposition 7.2.3(1) implies that the map from  $[\#^k \mathbb{CP}^n, PL/O]$  to  $\pi_s^0(\#^k \mathbb{CP}^n)$  is injective.

1. For  $n = 6$ , note that groups  $\pi_s^0(\#^k \mathbb{CP}^6)$  and  $[\#^k \mathbb{CP}^6, PL/O]$  are isomorphism Theorem 5.4.4 and Corollary 5.3.1. This confirms statement (1) for this case.

For the remaining case, consider the following commutative diagram

$$\begin{array}{ccccccc}
 0 & \longrightarrow & S_t^{\text{Diff}}(\#^k \mathbb{CP}^n) & \longrightarrow & [\#^k \mathbb{CP}^n, SF] & \xrightarrow{s} & L_{2n} \\
 & & & & \uparrow q^* & & \parallel \\
 & & & & \oplus_k [\mathbb{CP}^n, SF] & \xrightarrow{s} & L_{2n}
 \end{array} \tag{7.9}$$

In case of  $n = 3$  and 4, the surgery obstruction map  $s : [\mathbb{CP}^n, SF] \rightarrow L_{2n}$  is zero, and so is the map  $s : \oplus_k [\mathbb{CP}^n, SF] \rightarrow L_{2n}$ . Also, note that, the map  $q^* : \oplus_k [\mathbb{CP}^n, SF] \rightarrow [\#^k \mathbb{CP}^n, SF]$  is surjective for these case (see the proof of Theorem 5.4.5 (i) for  $n = 4$ ). Therefore, the map  $s : [\#^k \mathbb{CP}^3, SF] \rightarrow L_{2n}$  is zero, implying the statement (1).

2. For  $n = 5$ , the following commutative square

$$\begin{array}{ccc}
 [\mathbb{S}^{10}, F/PL] & \xrightarrow{d^*} & [\#^k \mathbb{CP}^5, F/PL] \\
 \cong \downarrow & & \downarrow \sigma \\
 L_{10} & \xlongequal{\quad} & L_{10}
 \end{array}$$

dictates that the surgery obstruction homomorphism  $\sigma$  is non-zero. Therefore, the statement (2) follows immediately by applying this in the commutative diagram (7.5), as the map  $s : [\#^k \mathbb{CP}^5, SF] \rightarrow L_{10}$  becomes non-zero.

3. For  $n = 7$ , the map  $s : [\mathbb{CP}^7, SF] \rightarrow L_{14}$  is a non-zero homomorphism. From the proof of Theorem 5.4.5 (ii), note that  $q^* : \oplus_k [\mathbb{CP}^7, SF] \rightarrow [\#^k \mathbb{CP}^7, SF]$  is surjective. Hence in (7.9) the map  $s : [\#^k \mathbb{CP}^7, SF] \rightarrow L_{14}$  is a non-zero homomorphism.

This completes the proof in all cases. □

Combining Theorems 7.2.2, 7.2.3 and 7.2.6, we have the following.

**Theorem 7.2.7.** *Let  $k \in \mathbb{N}$ . Then*

1.  $S_t^{\text{Diff}}(\#^k \mathbb{CP}^3) \cong \mathbb{Z}_2$
2.  $S_t^{\text{Diff}}(\#^k \mathbb{CP}^4) \cong \mathbb{Z}_2^{k+1}$
3.  $S_t^{\text{Diff}}(\#^k \mathbb{CP}^5) \cong \mathbb{Z}_2^{2k-1} \oplus \mathbb{Z}_3$
4.  $S_t^{\text{Diff}}(\#^k \mathbb{CP}^6) \cong \mathbb{Z}_2^{2k-1} \oplus \mathbb{Z}_3^k$
5.  $S_t^{\text{Diff}}(\#^k \mathbb{CP}^7) \cong \mathbb{Z}_2^{k+1} \oplus \mathbb{Z}_3^{k-1}$

The following corollary summarizes this section.

**Corollary 7.2.8.** *Let  $M$  be a closed, smooth  $2n$ -manifold, and  $k \in \mathbb{N}$ .*

1. *For any  $k, n \in \mathbb{N}$ , every homeomorphism from  $M$  to  $\#^k \mathbb{CP}^n$  is homotopic to a tangential homotopy equivalence.*
2. *For  $n = 3, 6$  and  $7$ , every tangential homotopy equivalence from  $M$  to  $\#^k \mathbb{CP}^n$  is homotopic to a homeomorphism.*

# Chapter 8

## Summary and Conclusions

The thesis presents the concordance classes of smooth structures on  $PL$ -manifolds with dimensions between 8 and 10 in terms of cohomologies and Steenrod operations with the help of cohomology operations and the Postnikov decomposition of the space  $PL/O$ . Additionally, these findings lead to the discussion of another important factor involved in the classification problem: the homotopy inertia groups which are determined for smooth manifolds of dimensions 8, 9, and 10. As a consequence of these results, the classification of smooth structures up to diffeomorphism in real projective spaces of dimensions between 8 and 10, as well as lens spaces in 9 dimensions, is addressed.

The thesis proceeds with computations of various structure sets of the  $k$ -fold connected sum of complex projective spaces, denoted by  $\#^k \mathbb{CP}^n$ . This involves explicit computations of concordance structure sets and reduced 0<sup>th</sup> stable cohomotopy groups of  $\#^k \mathbb{CP}^n$  in real dimensions between 8 and 16. More generally, it is demonstrated that the concordance inertia group of a connected sum of manifolds is the sum of the individual manifold concordance inertia groups, i.e.  $I_c(\#_i M_i) = \Sigma_i I_c(M_i)$ .

Furthermore, the thesis determines the reduced complex  $K$ -groups  $\widetilde{K}^i(\#^k \mathbb{CP}^n)$  for  $i = -1, 0$ , and the reduced real  $K$ -groups  $\widetilde{KO}^i(\#^k \mathbb{CP}^n)$  for  $-7 \leq i \leq 0$ , along with their genera-

tors for  $k, n \in \mathbb{N}$  and  $k \geq 2$ . Moreover, it proves that the smooth normal invariant set, which also plays a crucial role in the manifold classification problem, of  $\#^k \mathbb{CP}^n$  is obtained as a direct sum of reduced  $0^{\text{th}}$  stable cohomotopy groups of  $\#^k \mathbb{CP}^n$  and the free part as the reduced  $KO$ -groups of  $\#^k \mathbb{CP}^n$  for  $k, n \in \mathbb{N}$  and  $k \geq 2$ . Finally, it computes the smooth and  $PL$ -tangential structure sets of  $\#^k \mathbb{CP}^n$  using several results proved in Chapter 7.

## 8.1 Scope for future research

Our primary focus here was computing the concordance structure set  $\mathcal{C}(M)$  for 8, 9, and 10-dimensional manifolds. Future research aimed at classifying smooth structures on  $M$  up to diffeomorphism could concentrate on computing the group of self-homeomorphisms of  $M$ , denoted by  $\mathcal{H}(M)$  and understanding its action on  $\mathcal{C}(M)$ .

**Problem 8.1.1.** *Let  $M$  be a manifold of dimension 8, 9 or 10.*

1. *Compute the group  $\mathcal{H}(M)$ .*
2. *Find the orbit space  $\mathcal{C}(M)/\mathcal{H}(M)$ .*

This action results in an orbit space that establishes a bijection with the set of diffeomorphism classes, providing a classification of smooth structures up to diffeomorphism.

Another approach to classification involves determining the inertia groups of 8, 9, and 10-dimensional manifolds, as well as the  $k$ -fold connected sum of complex projective spaces.

**Problem 8.1.2.** *What is the inertia group of manifolds of dimension 8, 9 and 10?*

An answer to this would further contribute to achieving a complete classification up to diffeomorphism.

In addition to these manifolds, we have also extended our analysis to the  $k$ -fold connected sum of complex projective spaces for dimensions between 8 and 16. Therefore, similar problems can be asked for these manifolds.



# Appendix A

## A.1 Fiber bundles

**Definition A.1.1.** A *fiber bundle* is a structure  $(E, B, p, F)$ , where  $E$ ,  $B$  and  $F$  are topological spaces and  $p : E \rightarrow B$  is a continuous surjection satisfying a local triviality condition.

The local triviality condition says that; for every  $x \in B$ , there is an open neighborhood  $U \subseteq B$ , which is called a trivializing neighborhood, such that there is a homeomorphism  $\phi : p^{-1}(U) \rightarrow U \times F$  with the subspace topology on  $p^{-1}(U)$  and the product topology on  $U \times F$ , such that the following diagram commutes

$$\begin{array}{ccc} & p^{-1}(U) & \\ \phi \swarrow & \downarrow p & \\ U \times F & \xrightarrow{Pr} & U \end{array}$$

for the projection map  $Pr : U \times F \rightarrow U$ .

There are two kinds of maps that are of fundamental importance in algebraic topology:

fibrations and cofibrations.

### A.1.1 Fibrations

Consider a map  $p : E \rightarrow B$ . For a space  $X$  if we are given a homotopy  $F : X \times \mathbb{I} \rightarrow B$  and a map  $g : X \rightarrow E$  such that,  $p \circ g(x) = F(x, 0)$ ,  $\forall x \in X$  then this is called a *homotopy lifting data* for  $p$ . We say  $p$  satisfies the *homotopy lifting property* if for each homotopy lifting data, there exists a homotopy  $G : X \times \mathbb{I} \rightarrow E$  with the property  $p \circ G = F$  and  $G(x, 0) = g(x)$ ,  $\forall x \in X$ .

$$\begin{array}{ccc}
 X \times 0 & \xrightarrow{g} & E \\
 \downarrow & \nearrow G & \downarrow p \\
 X \times \mathbb{I} & \xrightarrow{F} & B
 \end{array}$$

Homotopy lifting property

If a map  $p : E \rightarrow B$  has the *homotopy lifting property* for any space  $X$  then it is called a *Hurewicz fibration* or simply a **fibration**.

#### Remark A.1.2.

- *Pullback of a fibration is a fibration.*
- *If  $B$  is paracompact then any locally trivial map  $p : E \rightarrow B$  is a fibration.*
- *So for a fiber bundle  $p : E \rightarrow B$  over a paracompact space  $B$  is a fibration.*
- *Let  $p : E \rightarrow B$  be a fibration. If  $B$  is path connected then all fibers are homotopy equivalent.*

Let  $f : A \rightarrow X$  be any continuous map. For the space  $X$  let  $X^{\mathbb{I}}$  be the space of all paths in  $X$  endowed with the compact-open topology. Consider the space  $E_f = \{(a, \gamma) \in A \times X^{\mathbb{I}} \mid \gamma(0) = f(a)\}$ . Then the map  $p : E_f \rightarrow X$ ,  $(a, \gamma) \mapsto \gamma(1)$  is a fibration. Also  $A$  can be regarded as the subspace of  $E_f$  consisting of pairs  $(a, \gamma)$  with  $\gamma$  the constant path at  $f(a)$ . So  $A$  is a deformation retract of  $E_f$ . Therefore, we can factor an arbitrary map  $f : A \rightarrow X$  as the composition of a

homotopy equivalence and a fibration.

$$\begin{array}{ccc} & E_f & \\ \nearrow & & \searrow p \\ A & \xrightarrow{f} & X \end{array}$$

It means we can replace any map by a fibration up to homotopy.

For a given map  $f : A \rightarrow X$  we get a fibration sequence up to homotopy  $F \rightarrow A \rightarrow X$  where  $F$  is a homotopy fiber. Considering the fibration up to homotopy of the map  $F \rightarrow A$  then we get its homotopy fiber to be the loop space  $\Omega X$ . Hence the fibration will be  $\Omega X \rightarrow F \rightarrow A$ . Continuing this way we get the fiber sequence, up to homotopy, as

$$\begin{array}{ccccccc} \cdots & \longrightarrow & \Omega^n A & \longrightarrow & \Omega^n X & \longrightarrow & \cdots \longrightarrow \Omega A \\ & & & & & & \searrow \\ & & & & & & \Omega X \longrightarrow F \longrightarrow A \longrightarrow X \end{array}$$

Applying a covariant functor  $[Y, \_]$  for any compactly generated space  $Y$ , we get the following long exact sequence of sets for  $n \geq 0$ , of groups for  $n \geq 1$  and of abelian groups for  $n \geq 2$

$$\begin{array}{ccccccc} \cdots & \longrightarrow & [Y, \Omega^n A] & \longrightarrow & [Y, \Omega^n X] & \longrightarrow & \cdots \longrightarrow [Y, \Omega A] \\ & & & & & & \searrow \\ & & & & & & [Y, \Omega X] \longrightarrow [Y, F] \longrightarrow [Y, A] \longrightarrow [Y, X] \end{array} \quad (\text{A.1})$$

### A.1.2 Cofibrations

Given a map  $\eta : A \rightarrow X$ , consider the following data: A homotopy  $F : A \times \mathbb{I} \rightarrow Y$  and a map  $g : X \rightarrow Y$  such that  $g(\eta(a)) = F(a, 0)$  for all  $a \in A$ . Such a data is called a *homotopy extension data* for  $\eta$ . We say  $\eta$  has *homotopy extension property* with respect to  $Y$  if for each such data,

there exists a homotopy  $H : X \times \mathbb{I} \rightarrow Y$  such that  $H \circ (\eta \times Id) = F$ , and  $H(x, 0) = g(x)$ ,  $\forall x \in X$ .

$$\begin{array}{ccc}
 A \times 0 & \hookrightarrow & A \times \mathbb{I} \\
 \eta \times 0 \downarrow & \nearrow F & \downarrow \eta \times id \\
 & Y & \\
 X \times 0 & \hookrightarrow & X \times \mathbb{I} \\
 & \nwarrow g & \nearrow H \\
 & & 
 \end{array}$$

Homotopy extension property

If  $\eta$  has homotopy extension property with respect to all spaces  $Y$  then it is called a **cofibration**.

A *pushout* of maps  $(f, g)$  for  $f : A \rightarrow X$  and  $g : A \rightarrow Y$  is a triple  $(D, i_1, i_2)$ ,

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & X & & \\
 g \downarrow & & \downarrow i_2 & \nearrow j_2 & \\
 Y & \xrightarrow{i_1} & D & & \\
 & \searrow j_1 & & \nearrow & \\
 & & & E & 
 \end{array}$$

where  $D = f_*(Y)$  is the pushout of  $g$  along  $f$ . It is universal in the sense that for any triple  $(E, j_1, j_2)$  satisfying the outer commutative diagram, there exists a unique map  $D \rightarrow E$  making the whole diagram commute. Note that the pushout is unique up to homeomorphism.

**Remark A.1.3.**

- If  $g : A \rightarrow Y$  is a cofibration and  $f : A \rightarrow X$  is a map then the pushout  $X \rightarrow f_*(Y)$  is a cofibration.

$$\begin{array}{ccc}
 A & \xrightarrow{f} & X \\
 g \downarrow & & \downarrow \\
 Y & \longrightarrow & f_*(Y)
 \end{array}$$

- $A \hookrightarrow X$  is a cofibration if and only if  $X \times 0 \cup A \times \mathbb{I}$  is a retract of  $X \times \mathbb{I}$

Let  $f : A \rightarrow X$  be a continuous map. The *mapping cylinder* of a map  $f$  is the space

$$M_f = \frac{(A \times \mathbb{I}) \sqcup X}{(a, 1) \sim f(a)}$$

The *mapping cone* is the space

$$C_f = \frac{M_f}{A \times \{0\}}$$

Then  $f$  can be written as a composition  $A \xrightarrow{r} M_f \xrightarrow{p} X$ , where  $r$  is a cofibration and  $p$  is a homotopy equivalence. Hence we can replace any map by a cofibration up to homotopy. If  $f : A \rightarrow X$  is a map then we get a cofibration sequence  $A \rightarrow X \rightarrow C_f$  where  $C_f$  denotes the cofiber up to homotopy.

Similarly suppose the cofibration up to homotopy of the map  $X \rightarrow C_f$  is the cofibration sequence  $X \rightarrow C_f \rightarrow W$ . Then  $W$  is homotopy equivalent to the suspension of  $A$ ,  $\Sigma A$ . Continuing in this way we will get the following cofiber sequence, up to homotopy

$$\begin{array}{c} A \xrightarrow{f} X \longrightarrow C_f \longrightarrow \Sigma A \\ \searrow \hspace{10em} \nearrow \\ \Sigma X \longrightarrow \cdots \longrightarrow \Sigma^n A \longrightarrow \Sigma^n X \longrightarrow \cdots \end{array}$$

This induces a long exact sequence of sets for  $n \geq 0$ , of groups for  $n \geq 1$  and of abelian groups for  $n \geq 2$  after applying a contravariant functor  $[-, Y]$  for any compactly generated space  $Y$ ,

$$\begin{array}{c} \cdots \longrightarrow [\Sigma^n X, Y] \longrightarrow [\Sigma^n A, Y] \longrightarrow \cdots \rightarrow [\Sigma X, Y] \\ \searrow \hspace{10em} \nearrow \\ [\Sigma A, Y] \longrightarrow [C_f, Y] \longrightarrow [X, Y] \longrightarrow [A, Y]. \end{array} \tag{A.2}$$

The equations (A.1) and (A.2) are known as *Puppe sequences*.

## A.2 Spin manifolds

A smooth manifold  $M$  admits a spin structure if and only if its second Stiefel-Whitney class  $w_2(M)$  is zero:

$$w_2(M) = 0.$$

This criterion means that the manifold must be orientable and that the associated principal  $SO(n)$ -bundle has a trivial second Stiefel-Whitney class.

Examples of spin structures include  $S^n$ ,  $\mathbb{RP}^n$  for  $n \equiv 3(\text{mod } 4)$ , all complex projective spaces with an odd dimension, such as  $\mathbb{CP}^{2n+1}$ , are spin. Additionally, the  $n$ -dimensional torus  $T^n$  is a spin manifold for all  $n$ , as it is simply connected and orientable with  $w_2(T^n) = 0$ . Cartesian products of spin manifolds, such as  $S^2 \times S^2$ , retain spin structures, that is the product of spin manifolds is also spin.

On the other hand, non-examples of spin structures include  $\mathbb{RP}^n$  for  $n \not\equiv 3(\text{mod } 4)$ . Non-orientable manifolds like the Möbius strip and the Klein bottle cannot possess spin structures due to their non-orientability. Certain lens spaces, specifically  $L(p, 1)$  for specific values of  $p$ , may also fail to have spin structures depending on the properties of  $w_2$ .

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# List of Publications

## Preprints

1. Basu. S, Kasilingam. R, **Magar-Sawant. P**, 2024, "Smooth structures on pl-manifolds of dimensions between 8 and 10". Positive review with Minor revisions in: *Proceedings of the Edinburgh Mathematical Society*. <https://arxiv.org/abs/2302.02301v2>
2. **Magar-Sawant. P**, 2024, "Concordance structure set of connected sum of projective spaces". In: *Proceedings-Mathematical Sciences*,  
<https://doi.org/10.1007/s12044-024-00800-7>
3. **Magar-Sawant. P**, 2024, "Tangential smoothings of k-fold connected sum of complex projective spaces". Submitted. <https://arxiv.org/abs/2306.16911>